

4-2-2b

1. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	4	8	12	16
$p(t)$ (feet)	25	21	16	12	7.5

The position of a particle, A , moving along the x -axis is modeled by a twice-differentiable function p , where time t is measured in seconds and $p(t)$ is measured in feet. Selected values of $p(t)$ are shown in the table above.

(a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate the value of $\int_0^{16} p(t) \, dt$. Show the computations that lead to your answer.

(b) Use the data in the table to approximate $p'(2)$ using the average rate of change of $p(t)$ over the interval $0 \leq t \leq 4$. Show the computations that lead to your answer. Indicate units of measure.

(c) Interpret the meaning of $p'(2)$ in the context of the problem.

(d) Must there exist a value of d , for $0 < d < 16$, such that $p'(d)$ is equal to the average rate of change of p over the interval $0 \leq x \leq 16$? Justify your answer.

(e) Find $\int_0^4 p'(3t + 4) \, dt$. Show the computations that lead to your answer.

(f) Let $h(x) = \int_1^{2x} p(3t) \, dt$. Find $h'(2)$. Show the computations that lead to your answer.

(g) The velocity of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(3) = 2$, $g'(3) = -4$, and $g''(3) = -8$. Is the speed of particle Q increasing or decreasing at time $t = 3$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -2x^2 + y^2$ with initial condition $f(3) = 5$. Write an equation for the line tangent to the graph of f at the point $(3, 5)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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First Point

Philosophy: Earned for correct Midpoint Riemann sum, with appropriate values taken from the table.

Need to see sum of two terms, each as a product of two factors, i.e. $\text{base}_1 \cdot \text{height}_1 + \text{base}_2 \cdot \text{height}_2$

$p(4) \cdot (8 - 0) + p(12) \cdot (16 - 8)$ or $p(4) \cdot (8) + p(12) \cdot (8)$ earns first point.

If at least three of the four base/height values are correct in the sum of products, the expression earns the first point.

Second Point

264 or equivalent

Can be earned without first point with Riemann sum setup demonstration. With credible evidence that a midpoint Riemann sum attempt was made without not enough work shown, this point can be earned in the presence of the correct answer. For example, a sum of two terms with no evidence of products ($168 + 96$) or a graph with areas labeled and summed to 264 or equivalent, can earn this point.

Examples:

- $264 = 168 + 96$ by itself earns second point, not first point
- 264 declared, with $21 \cdot 8$ and $12 \cdot 8$ identified (perhaps graphically) earns second point, not first point
- $k(21 \cdot 8 + 12 \cdot 8)$, nonzero constant $k \neq 1$ by itself earns first point, not second point
- $p(4) \cdot (8) + p(12) \cdot (16 - 8)$ by itself earns first point, not second point



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint Riemann sum
- ☐ Answer

Solution:

$$\int_0^{16} p(t) dt \approx p(4)(8 - 0) + p(12)(16 - 8)$$

$$= 21 \cdot 8 + 12 \cdot 8 = 264$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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First Point

Philosophy: Earned for correct difference quotient, with appropriate values from the table.

Need to see both a difference and a quotient

$\frac{p(4)-p(0)}{4-0}$ by itself doesn't earn first point; however $\frac{21-25}{4-0}$ or $\frac{21-25}{4}$ by itself earns first point

Second Point

Earned for correct units of ft per sec or ft/sec or f/s.

Correct units must be tied to a number to earn second point.

Correct units must appear in part (b); correct units in part (c) only doesn't earn second point.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$p'(2) \approx \frac{21-25}{4-0} = \frac{-4}{4} = -1 \text{ foot per second}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct $p'(2)$ interpretation.

Need two components in response:

- “rate of change of position” or “velocity” of the particle
- at $t = 2$

Correct ways to describe rate at time at $t = 2$ include:

- Rate at which position is changing/decreasing
- Rate at which particle is moving left
- Instantaneous velocity or velocity of particle

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Ambiguous/incorrect ways to describe rate include:

- Rate at which particle is changing/decreasing
- Rate at which particle is moving

Units are not required in part (c). If units are presented for t , they must be in seconds.



0	1
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The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$p'(2)$ is the rate at which the position of the particle is changing, in feet per second, at time $t = 2$ seconds.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for both Mean Value Theorem (MVT) conditions with the appropriate implication that p is continuous because we know that p is differentiable.

Ways to earn first point include:

- p differentiable $\Rightarrow p$ continuous
- p differentiable, hence p continuous
- p differentiable, so p continuous

p differentiable and p continuous does not earn first point.

Second Point

Philosophy: Earned for concluding “yes” and invoking Mean Value Theorem or its equivalent formulation, with consideration of the continuity of p .

MVT title does not have to be stated to earn second point.

This point is not earned if other theorems (IVT, etc.) are invoked.

If presented, the value for $p'(d) = -\frac{17.5}{16}$ must be correct to earn second point.

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Examples:

- “By MVT, yes since p differentiable” earns neither point
- “By MVT, yes since p continuous” earns second point only
- “By MVT, yes since p differentiable and continuous” earns second point only
- “By MVT, yes since p differentiable hence continuous” earns both points
- “By MVT, yes since p twice differentiable hence continuous” earns both points
- “By MVT, yes since function differentiable hence continuous” earns both points



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

p is differentiable. $\Rightarrow p$ is continuous.

Therefore, the Mean Value Theorem can be applied to p on the interval $0 \leq x \leq 16$ to guarantee that there exists a value of d , for $0 < d < 16$, such that $p'(d)$ equals the average rate of change of p over the interval $0 \leq x \leq 16$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct substitution application.

Need three elements in response (with consistent use of u or another substitution variable):

- $u = 3t + 4$ (can be explicit or implicit)
- $du = 3 dt$ (yielding integral coefficient of $\frac{1}{3}$)
- Correct integral limits throughout, with t from 0 to 4 and u from 4 to 16

Omitting the coefficient of $\frac{1}{3}$ or using some other value will not earn first point; response could earn second point, but not third point.

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In the third element above, if response shows an otherwise correct substitution with unchanged limits of integration on a definite integral from 0 to 4, then point is not earned.

Alternately, the integral can be worked with as an indefinite integral with u returned to $3t + 4$ in the antiderivative before evaluating as a means to eliminate the need to change limits. Such a response can earn the first point if handled correctly.

For example: $\int_0^4 p'(3t + 4) dt = \frac{1}{3}p(3t + 4) \Big|_0^4$ earns the substitution point.

Second Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, with evidence of connection between $\int p'(u)$ and $p(u)$.

Ways to earn second point include:

· $k(p(16) - p(4))$ or $k(p(3 \cdot 4 + 4) - p(3 \cdot 0 + 4))$ for k a nonzero constant

· $kp(3t + 4) \Big|_0^4$ or $kp(u) \Big|_4^{16}$ for k a nonzero constant

· $\int p'(3t + 4) dt = kp(3t + 4)$ or $\int p'(u) du = kp(u)$ for any nonzero constant k , with or without appropriate limits of integration on either integral

First and second points can be earned simultaneously by responses such as $\int_0^4 p'(3t + 4) dt = \frac{1}{2}p(3t + 4) \Big|_0^4$

Third Point

Earned only for an answer of -4.5 or equivalent value.

Value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring Examples:

· $\frac{1}{3} \int p'(u) du = \frac{1}{3}p(u)$ earns 0 – 1 – 0

· $\frac{1}{3} \int p'(u) du = \frac{1}{3}p(u) \Big|_0^4 = \frac{1}{3}(p(4) - p(0))$ earns 0 – 1 – 0

· $\int p'(u) du = p(u) \Big|_4^{16} = (p(16) - p(4)) = -13.5$ earns 0 – 1 – 0

· $3 \int p'(u) du = 3p(u) \Big|_4^{16} = 3[7.5 - 21]$ earns 0 – 1 – 0

· $\int_0^4 p'(3t + 4) dt = p(1.5t^2 + 4t) \Big|_0^4 = p(40) - p(0)$ earns 0 – 0 – 0

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$$\cdot \frac{1}{3} \int_0^4 p'(u) \, du = \frac{1}{3}(p(16) - p(4)) = -4.5 \text{ earns } 0 - 1 - 1$$

$$\cdot \int_0^4 p'(3t + 4) \, dt = \frac{1}{3}(p(16) - p(4)) = -4.5 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^4 p'(3t + 4) \, dt = \frac{1}{3}p(u) \Big|_4^{16} = \frac{1}{3}(p(16) - p(4)) = -4.5 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^4 p'(3t + 4) \, dt = \frac{1}{3}p(3t + 4) \Big|_0^4 = \frac{1}{3}(p(16) - p(4)) \text{ earns } 1 - 1 - 0$$

$$\cdot \frac{1}{3} \int p'(u) = \frac{1}{3}p(u); \text{ and } \frac{1}{3}p(3t + 4) \Big|_0^4 = \frac{1}{3}(7.5 - 21) \text{ earns } 1 - 1 - 1$$



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t + 4$. Then $du = 3 \, dt$.

When $t = 0$, $u = 4$.

When $t = 4$, $u = 16$.

$$\int_0^4 p'(3t + 4) \, dt = \frac{1}{3} \int_4^{16} p'(u) \, du$$

$$= \frac{1}{3}(p(16) - p(4))$$

$$= \frac{1}{3}(7.5 - 21) = -4.5$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

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Philosophy: Earned for correct Fundamental Theorem of Calculus application, resulting in $a \cdot p(6x)$ for some nonzero constant a .

Ways to earn first point include responses of $p(3 \cdot 2x)$ or $p(6x)$ (Note $p(6t)$ would also earn this point.)

The expression $p(3 \cdot 2x) - p(3 \cdot 1)$ or $(p(3 \cdot 2x) - p(3 \cdot 1)) \cdot 2$ earns this point. The $p(3 \cdot 1)$ is viewed as a chain rule error.

Any response correctly presenting and evaluating the derivative of the integral in terms of the substitution $u = 3t$ earns first point. Part of this response may include $\int_1^{2x} p(3t) dt = \frac{1}{3} \int_3^{6x} p(u) du$.

Second Point

Philosophy: This point is earned for correctly applying the chain rule to find $h'(x)$ or $h'(2)$. That is accomplished by multiplying the term involving $p(x)$ by 2.

Can only be earned if response earned first (FTC) point.

The first two points can be, and often are, earned simultaneously.

The point is not earned if the presented derivative is $a \cdot p(6x)$ for any $a \neq 2$.

The point is not earned for responses of $2p(3t) \big|_1^{2x}$ or $2(p(6x) - p(b))$, for any value of b .

Third Point

Can only be earned if response earned first (FTC) and second (chain rule) points.

Earned only for answer of 24 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring examples:

$\cdot p(3 \cdot 2x) - p(3)$; $p(12) - p(3)$ earns $1 - 0 - 0$

$\cdot p(3 \cdot 2x) \cdot 2 - p(3) \cdot 0$; 24 earns $1 - 1 - 1$

$\cdot 2p(3t) \big|_1^{2x} = 2(p(12) - p(3))$ earns $1 - 0 - 0$

$\cdot p(3x) \cdot 2$; $p(6) \cdot 2$ earns $0 - 0 - 0$

$\cdot p(6x) \cdot 2$; $p(12) \cdot 2$ earns $1 - 1 - 0$



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

4-2-2b

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_1^{2x} p(3t) dt = p(3 \cdot 2x) \cdot 2$$

$$h'(2) = p(12) \cdot 2 = 12 \cdot 2 = 24$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for narrowing consideration of given information to g and g' .

g and g' need not be evaluated to earn the first point.

Considering velocity and acceleration not sufficient to earn first point, unless acceleration is linked to g' .

If response considers g and g' and g'' , then eventually g'' must be eliminated to earn first point.

Second Point

Philosophy: Earned for “decreasing” with reason, using the opposite signs of g and g' .

Ways to earn second point include:

- “decreasing since $g < 0$ and $g' > 0$ ”
- “particle slowing down since g and g' have opposite signs”

Can only be earned if response earned first point.

Examples:

- “decreasing since velocity and acceleration have opposite signs” earns neither point
- “decreasing since velocity < 0 and acceleration > 0 ” earns neither point
- “decreasing since $(2) \cdot (-4) = -8 < 0$ ” earns both points



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g and g'
- ☐ Answer with reason

Solution:

Because $g(3) = 2 > 0$, the velocity of the particle is positive. Because $g'(3) = -4 < 0$, the acceleration of the particle is negative. Because the velocity and acceleration have opposite signs, the speed of the particle is decreasing at time $t = 3$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct tangent line equation.

Correct responses include:

$$y = 7(x - 3) + 5 \text{ or } y - 5 = 7(x - 3)$$

$$y = 7x - 16$$

$$y = (-2 \cdot 3^2 + 5^2)(x - 3) + 5$$

Incorrect responses include:

$$y = (-2x^2 + y^2)(x - 3) + 5$$

$$y = -7x - 16$$

$$f(x) = 7x - 16$$



0	1
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The student response accurately includes the criteria below.

- ☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(3,5)} = -2 \cdot 3^2 + 5^2 = 7$$

$$y = 7(x - 3) + 5$$

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2. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	5	10	15	20
$p(t)$ (feet)	-25	-20	-14	-8	-2.5

The position of a particle, A , moving along the x -axis is modeled by a twice-differentiable function p , where time t is measured in seconds and $p(t)$ is measured in feet. Selected values of $p(t)$ are shown in the table above.

(a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate the value of $\int_0^{20} p(t) \, dt$. Show the computations that lead to your answer.

(b) Use the data in the table to approximate $p'(7)$ using the average rate of change of $p(t)$ over the interval $5 \leq t \leq 10$. Show the computations that lead to your answer. Indicate units of measure.

(c) Interpret the meaning of $p'(7)$ in the context of the problem.

(d) Must there exist a value of b , for $0 < b < 20$, such that $p'(b)$ is equal to the average rate of change of p over the interval $0 \leq x \leq 20$? Justify your answer.

(e) Find $\int_0^5 p'(2t + 5) \, dt$. Show the computations that lead to your answer.

(f) Let $h(x) = \int_3^{5x} p(4t) \, dt$. Find $h'(1)$. Show the computations that lead to your answer.

(g) The velocity of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(1) = -2$, $g'(1) = 4$, and $g''(1) = 7$. Is the speed of particle Q increasing or decreasing at time $t = 1$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -2x^3y^2 - 5$ with initial condition $f(-1) = 2$. Write an equation for the line tangent to the graph of f at the point $(-1, 2)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-2-2b**First Point**

Philosophy: Earned for correct Midpoint Riemann sum, with appropriate values taken from the table.

Need to see sum of two terms, each as a product of two factors, i.e. $\text{base}_1 \cdot \text{height}_1 + \text{base}_2 \cdot \text{height}_2$

$p(5) \cdot (10 - 0) + p(15) \cdot (20 - 10)$ or $p(5) \cdot (10) + p(15) \cdot (10)$ earns first point.

If at least three of the four base/height values are correct in the sum of products, the expression earns the first point.

Second Point

−280 or equivalent

Can be earned without first point with Riemann sum setup demonstration. With credible evidence that a midpoint Riemann sum attempt was made without not enough work shown, this point can be earned in the presence of the correct answer. For example, a sum of two terms with no evidence of products $(-200 - 80)$ or a graph with areas labeled and summed to −280 or equivalent, can earn this point.

Examples:

- $-280 = -200 - 80$ by itself earns second point, not first point
- -280 declared, with $-20 \cdot 10$ and $-8 \cdot 10$ identified (perhaps graphically) earns second point, not first point
- $k(-20 \cdot 10 - 8 \cdot 10)$, nonzero constant $k \neq 1$ by itself earns first point, not second point
- $p(3) \cdot (10) + p(15) \cdot (20 - 10)$ by itself earns first point, not second point



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint Riemann sum
- ☐ Answer

Solution:

$$\int_0^{20} p(t) dt \approx p(5)(10 - 0) + p(15)(20 - 10)$$

$$= -20 \cdot 10 + (-8) \cdot 10 = -280$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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First Point

Philosophy: Earned for correct difference quotient, with appropriate values from the table.

Need to see both a difference and a quotient

$\frac{p(10)-p(5)}{10-5}$ by itself doesn't earn first point; however $\frac{-14-(-20)}{10-5}$ or $\frac{-14-(-20)}{5}$ by itself earns first point

Second Point

Earned for correct units of ft per sec or ft/sec or f/s.

Correct units must be tied to a number to earn second point.

Correct units must appear in part (b); correct units in part (c) only doesn't earn second point.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$p'(7) \approx \frac{-14-(-20)}{10-5} = \frac{6}{5} \text{ feet per second}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct $p'(7)$ interpretation.

Need two components in response:

- “rate of change of position” or “velocity” of the particle
- at $t = 7$

Correct ways to describe rate at time at $t = 7$ include:

- Rate at which position is changing/increasing
- Rate at which particle is moving right
- Instantaneous velocity or velocity of particle

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Ambiguous/incorrect ways to describe rate include:

- Rate at which particle is changing/increasing
- Rate at which particle is moving

Units are not required in part (c). If units are presented for t , they must be in seconds.



0	1
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The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$p'(7)$ is the rate at which the position of the particle is changing, in feet per second, at time $t = 7$ seconds.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for both Mean Value Theorem (MVT) conditions with the appropriate implication that p is continuous because we know that p is differentiable.

Ways to earn first point include:

- p differentiable $\Rightarrow p$ continuous
- p differentiable, hence p continuous
- p differentiable, so p continuous

p differentiable and p continuous does not earn first point.

Second Point

Philosophy: Earned for concluding “yes” and invoking Mean Value Theorem or its equivalent formulation, with consideration of the continuity of p .

MVT title does not have to be stated to earn second point.

This point is not earned if other theorems (IVT, etc.) are invoked.

If presented, the value for $p'(b) = \frac{22.5}{20}$ must be correct to earn second point.

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Examples:

- “By MVT , yes since p differentiable” earns neither point
- “By MVT , yes since p continuous” earns second point only
- “By MVT , yes since p differentiable and continuous” earns second point only
- “By MVT , yes since p differentiable hence continuous” earns both points
- “By MVT , yes since p twice differentiable hence continuous” earns both points
- “By MVT , yes since function differentiable hence continuous” earns both points



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

p is differentiable. $\Rightarrow p$ is continuous.

Therefore, the Mean Value Theorem can be applied to p on the interval $0 \leq x \leq 20$ to guarantee that there exists a value of b , for $0 < b < 20$, such that $p'(b)$ equals the average rate of change of p over the interval $0 \leq x \leq 20$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct substitution application.

Need three elements in response (with consistent use of u or another substitution variable):

- $u = 2t + 5$ (can be explicit or implicit)
- $du = 2dt$ (yielding integral coefficient of $\frac{1}{2}$)
- Correct integral limits throughout, with t from 0 to 5 and u from 5 to 15

Omitting the coefficient of $\frac{1}{2}$ or using some other value will not earn first point; response could earn second point, but not third point.

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In the third element above, if response shows an otherwise correct substitution with unchanged limits of integration on a definite integral from 0 to 5, then point is not earned.

Alternately, the integral can be worked with as an indefinite integral with u returned to $2t + 5$ in the antiderivative before evaluating as a means to eliminate the need to change limits. Such a response can earn the first point if handled correctly.

For example: $\int_0^5 p'(2t + 5)dt = \frac{1}{2}p(2t + 5)\big|_0^5$ earns the substitution point.

Second Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, with evidence of connection between $\int p'(u)$ and $p(u)$.

Ways to earn second point include:

· $k(p(15) - p(5))$ or $k(p(2 \cdot 5 + 5) - p(2 \cdot 0 + 5))$ for k a nonzero constant

· $kp(2t + 5)\big|_0^5$ or $kp(u)\big|_5^{15}$ for k a nonzero constant

· $\int p'(2t + 5)dt = kp(2t + 5)$ or $\int p'(u)du = kp(u)$ for any nonzero constant k , with or without appropriate limits of integration on either integral

First and second points can be earned simultaneously by responses such as $\int_0^5 p'(2t + 5)dt = \frac{1}{2}p(2t + 5)\big|_0^5$

Third Point

Earned only for an answer of 6 or equivalent value.

Value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring Examples:

· $\frac{1}{2} \int p'(u)du = \frac{1}{2}p(u)$ earns 0 – 1 – 0

· $\frac{1}{2} \int p'(u)du = \frac{1}{2}p(u)\big|_0^5 = \frac{1}{2}(p(5) - p(0))$ earns 0 – 1 – 0

· $\int p'(u)du = p(u)\big|_5^{15} = (p(15) - p(5)) = 12$ earns 0 – 1 – 0

· $2 \int p'(u)du = 2p(u)\big|_0^5 = 2[-8 - (-20)]$ earns 0 – 1 – 0

· $\int_0^5 p'(2t + 5)dt = p(t^2 + 5t)\big|_0^5 = p(50) - p(0)$ earns 0 – 0 – 0

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$$\cdot \frac{1}{2} \int_0^5 p'(u) du = \frac{1}{2}(p(15) - p(5)) = 6 \text{ earns } 0 - 1 - 1$$

$$\cdot \int_0^5 p'(2t + 5) dt = \frac{1}{2}(p(15) - p(5)) = 6 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^5 p'(2t + 5) dt = \frac{1}{2} p(u) \Big|_5^{15} = \frac{1}{2}(p(15) - p(5)) = 6 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^5 p'(2t + 5) dt = \frac{1}{2} p(2t + 5) \Big|_0^5 = \frac{1}{2}(p(15) - p(5)) \text{ earns } 1 - 1 - 0$$

$$\cdot \frac{1}{2} \int p'(u) = \frac{1}{2} p(u); \text{ and } \frac{1}{2} p(2t + 5) \Big|_0^5 = \frac{1}{2}(-8 - (-20)) \text{ earns } 1 - 1 - 1$$



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t + 5$. Then $du = 2 \, dt$.

When $t = 0$, $u = 5$.

When $t = 5$, $u = 15$.

$$\int_0^5 p'(2t + 5) \, dt = \frac{1}{2} \int_5^{15} p'(u) \, du$$

$$= \frac{1}{2}(p(15) - p(5))$$

$$= \frac{1}{2}(-8 - (-20)) = 6$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

4-2-2b

Philosophy: Earned for correct Fundamental Theorem of Calculus application, resulting in $a \cdot p(20x)$ for some nonzero constant a .

Ways to earn first point include responses of $p(4 \cdot 5x)$ or $p(20x)$ (Note $p(20t)$ would also earn this point.)

The expression $p(4 \cdot 5x) - p(4 \cdot 3)$ or $(p(4 \cdot 5x) - p(4 \cdot 3)) \cdot 5$ earns this point. The $p(4 \cdot 3)$ is viewed as a chain rule error.

Any response correctly presenting and evaluating the derivative of the integral in terms of the substitution $u = 4t$ earns first point. Part of this response may include $\int_3^{5x} p(4t) dt = \frac{1}{4} \int_{12}^{20x} p(u) du$.

Second Point

Philosophy: This point is earned for correctly applying the chain rule to find $h'(x)$ or $h'(1)$. That is accomplished by multiplying the term involving $p(x)$ by 5.

Can only be earned if response earned first (FTC) point.

The first two points can be, and often are, earned simultaneously.

The point is not earned if the presented derivative is $a \cdot p(20x)$ for any $a \neq 5$.

The point is not earned for responses of $5p(4t) \big|_3^{5x}$ or $5(p(20x) - p(b))$, for any value of b .

Third Point

Can only be earned if response earned first (FTC) and second (chain rule) points.

Earned only for answer of -12.5 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring examples:

$\cdot p(4 \cdot 5x) - p(12)$; $p(20) - p(12)$ earns $1 - 0 - 0$

$\cdot p(4 \cdot 5x) \cdot 5 - p(12) \cdot 0$; -12.5 earns $1 - 1 - 1$

$\cdot 5p(4t) \big|_3^{5x} = 5(p(20) - p(12))$ earns $1 - 0 - 0$

$\cdot p(4x) \cdot 5$; $p(4) \cdot 5$ earns $0 - 0 - 0$

$\cdot p(4 \cdot 5x) \cdot 5$; $p(20) \cdot 5$ earns $1 - 1 - 0$



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

4-2-2b

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_3^{5x} p(4t) dt = p(4 \cdot 5x) \cdot 5$$

$$h'(1) = p(20) \cdot 5 = -2.5 \cdot 5 = -12.5$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for narrowing consideration of given information to g and g' .

g and g' need not be evaluated to earn the first point.

Considering velocity and acceleration not sufficient to earn first point, unless acceleration is linked to g' .

If response considers g and g' and g'' , then eventually g'' must be eliminated to earn first point.

Second Point

Philosophy: Earned for “decreasing” with reason, using the opposite signs of g and g' .

Ways to earn second point include:

- “decreasing since $g < 0$ and $g' > 0$ ”
- “particle slowing down since g and g' have opposite signs”

Can only be earned if response earned first point.

Examples:

- “decreasing since velocity and acceleration have opposite signs” earns neither point
- “decreasing since velocity < 0 and acceleration > 0 ” earns neither point
- “decreasing since $(-2) \cdot 4 = -8 < 0$ ” earns both points



0	1	2
---	---	---

4-2-2b

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g and g'
- ☐ Answer with reason

Solution:

Because $g(1) = -2 < 0$, the velocity of the particle is negative. Because $g'(1) = 4 > 0$, the acceleration of the particle is positive. Because the velocity and acceleration have opposite signs, the speed of the particle is decreasing at time $t = 1$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct tangent line equation.

Correct responses include:

- $y = 3(x + 1) + 2$ or $y - 2 = 3(x + 1)$
- $y = 3x + 5$
- $y = (-2 \cdot (-1)^3 \cdot 2^2 - 5)(x + 1) + 2$

Incorrect responses include:

- $y = (-2x^3y^2 - 5)(x + 1) + 2$
- $y = -3x + 5$
- $f(x) = 3x + 5$



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(-1, 2)} = -2(-1)^3 \cdot 2^2 - 5 = 3$$

$$y = 3(x + 1) + 2$$

4-2-2b

3. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	3	6	9	12
$p(t)$ (feet)	-5	-8.5	-12.5	-15	-20

The position of a particle, A , moving along the x -axis is modeled by a twice-differentiable function p , where time t is measured in seconds and $p(t)$ is measured in feet. Selected values of $p(t)$ are shown in the table above.

(a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate the value of $\int_0^{12} p(t) \, dt$. Show the computations that lead to your answer.

(b) Use the data in the table to approximate $p'(5)$ using the average rate of change of $p(t)$ over the interval $3 \leq t \leq 6$. Show the computations that lead to your answer. Indicate units of measure.

(c) Interpret the meaning of $p'(5)$ in the context of the problem.

(d) Must there exist a value of k , for $0 < k < 12$, such that $p'(k)$ is equal to the average rate of change of p over the interval $0 \leq x \leq 12$? Justify your answer.

(e) Find $\int_0^3 p'(2t + 6) \, dt$. Show the computations that lead to your answer.

(f) Let $h(x) = \int_1^{2x} p(3t) \, dt$. Find $h'(2)$. Show the computations that lead to your answer.

(g) The velocity of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(3) = -1$, $g'(3) = -5$, and $g''(3) = 7$. Is the speed of particle Q increasing or decreasing at time $t = 3$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = 12x^2 - y^3$ with initial condition $f\left(\frac{1}{2}\right) = -2$. Write an equation for the line tangent to the graph of f at the point $\left(\frac{1}{2}, -2\right)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-2-2b

First Point

Philosophy: Earned for correct midpoint Riemann sum, with appropriate values taken from the table.

Need to see sum of two terms, each as a product of two factors, i.e. $\text{base}_1 \cdot \text{height}_1 + \text{base}_2 \cdot \text{height}_2$

$$p(3) \cdot (6 - 0) + p(9) \cdot (12 - 6) \text{ or } p(3) \cdot (6) + p(9) \cdot (6)$$

If at least three of the four base/height values are correct in the sum of products, the expression earns the first point. However, if one of these values is incorrect, the response does not earn the second point.

Second Point

Earned for an answer of -141 or equivalent.

Can be earned without the first point for the Riemann sum setup demonstration. With credible evidence that a midpoint Riemann sum attempt was made, but not enough work shown, this point can be earned in the presence of the correct answer. For example, a sum of two terms with no evidence of products ($-51 - 90$), or a graph with areas labeled and summed to -141 , can earn this point.

Examples:

- $-141 = -51 - 90$ by itself earns second point, not first point
- -141 declared, with $-8.5 \cdot 6$ and $-15 \cdot 6$ identified earns second point, not first point (perhaps graphically)
- $k(-8.5 \cdot 6 - 15 \cdot 6)$, nonzero constant $k \neq 1$ by itself earns first point, not second point
- $p(3) \cdot (6 - 0) + p(9) \cdot (12 - 6)$ by itself earns first point, not second point



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint Riemann sum
- ☐ Answer

Solution:

$$\int_0^{12} p(t) dt \approx p(3)(6 - 0) + p(9)(12 - 6)$$

$$= -8.5 \cdot 6 + (-15) \cdot 6 = -141$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-2-2b

First Point

Philosophy: Earned for a correct difference quotient, with appropriate values from the table.

The response must show both a difference and a quotient.

$\frac{p(6)-p(3)}{6-3}$ by itself doesn't earn first point; however $\frac{-12.5-(-8.5)}{6-3}$ or $A \frac{-12.5-(-8.5)}{3}$ by itself earns the first point

Second Point

Earned for correct units of ft per sec or ft/sec or f/s.

Correct units must be tied to a number to earn second point.

Correct units must appear in part (b); correct units in part (c) only do not earn the second point.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$p'(5) \approx \frac{-12.5-(-8.5)}{6-3} = \frac{-4}{3} \text{ feet per second}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct $p'(5)$ interpretation.

Need two components in the response:

- “rate of change of position” or “velocity” of the particle
- at $t = 5$

Correct ways to describe the rate of change include:

- rate at which position (of the particle) is changing/decreasing
- rate at which particle is moving left

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- instantaneous velocity or velocity of the particle

Ambiguous/incorrect ways to describe rate include:

- rate at which particle is changing/decreasing
- rate at which particle is moving

The response does not need to include the value of the approximation found in part (b), but will not be penalized for doing so (e.g. "the rate at which the particle's position is decreasing is $\frac{4}{3}$ at time $t = 5$ " earns the point).

If the response includes the approximation value from part (b) and makes an incorrect statement such as "the rate at which the position of the particle is decreasing is $-\frac{4}{3}$ at time $t = 5$ ", the point is not earned.

Units are not required in part (c). If units are presented for t , they must be in seconds.



0	1
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The student response accurately includes the criteria below.

- ☐ Interpretation

Solution:

$p'(5)$ is the rate at which the position of the particle is changing, in feet per second, at time $t = 5$ seconds.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for both Mean Value Theorem (MVT) conditions with the appropriate implication that p is continuous because we know that p is differentiable.

Ways to earn first point include:

- p differentiable $\Rightarrow p$ continuous
- p differentiable, hence p continuous
- p differentiable, so p continuous

p differentiable and p continuous does not earn first point.

Second Point

4-2-2b

Philosophy: Earned for concluding “yes” and invoking Mean Value Theorem or its equivalent formulation, with consideration of the continuity of p .

MVT title does not have to be stated to earn second point.

This point is not earned if other theorems (IVT, etc.) are invoked.

If presented, the value for $p'(c) = -\frac{5}{4}$ must be correct to earn second point.

Examples:

- “By MVT, yes since p differentiable” earns neither point
- “By MVT, yes since p continuous” earns second point only
- “By MVT, yes since p differentiable and continuous” earns second point only
- “By MVT, yes since p differentiable hence continuous” earns both points
- “By MVT, yes since p twice differentiable hence continuous” earns both points
- “By MVT, yes since function differentiable hence continuous” earns both points



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

p is differentiable. $\Rightarrow p$ is continuous.

Therefore, the Mean Value Theorem can be applied to p on the interval $0 \leq x \leq 12$ to guarantee that there exists a value of k , for $0 < k < 12$, such that $p'(k)$ equals the average rate of change of p over the interval $0 \leq x \leq 12$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct substitution.

A correct substitution requires three elements (with consistent use of u or another substitution variable):

4-2-2b

- $u = 2t + 6$ (can be explicit or implicit)
- $du = 2 dt$ (yielding integral coefficient of $\frac{1}{2}$)
- Correct integral limits throughout, with t from 0 to 3 and u from 6 to 12

The substitution could be implicit: $\int_0^3 p'(2t + 6) dt = \frac{1}{2}p(2t + 6) \Big|_0^3$

Omitting the coefficient of $\frac{1}{2}$ or using some other constant will not earn the substitution or answer points; the response could still earn the second point.

In the third element above, if response shows an otherwise correct substitution with unchanged limits of integration on a definite integral from 0 to 3, then point is not earned.

Alternately, the integral can be worked with as an indefinite integral with u returned to $2t + 6$ in the antiderivative before evaluating as a means to eliminate the need to change limits. Such a response can earn the first point, if handled correctly.

Second Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, with evidence of connection between $\int p'(u)$ and $p(u)$.

Ways to earn second point include:

- $k(p(12) - p(6))$ or $k(p(2 \cdot 3 + 6) - p(2 \cdot 0 + 6))$ for k a nonzero constant
- $k p(2t + 6) \Big|_0^3$ or $k p(u) \Big|_6^{12}$ for k a nonzero constant
- $\int p'(2t + 6) dt = k p(2t + 6)$ or $\int p'(u) du = k p(u)$ for any nonzero constant k , with or without appropriate limits of integration on either integral

The first two points can be earned simultaneously by responses such as $\int_0^3 p'(2t + 6) dt = \frac{1}{2}p(2t + 6) \Big|_0^3$

Third Point

Earned only for an answer of -3.75 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring Examples:

- $\frac{1}{2} \int p'(u) du = \frac{1}{2}p(u)$ earns $0 - 1 - 0$

4-2-2b

$$\cdot \frac{1}{2} \int p'(u) \, du = \frac{1}{2} p(u) \Big|_0^3 = \frac{1}{2} (p(3) - p(0)) \text{ earns } 0 - 1 - 0$$

$$\cdot \int p'(u) \, du = p(u) \Big|_6^{12} = (p(12) - p(6)) = -7.5 \text{ earns } 0 - 1 - 0$$

$$\cdot 2 \int p'(u) \, du = 2p(u) \Big|_0^3 = 2[-8.5 + 5] \text{ earns } 0 - 1 - 0$$

$$\cdot \int_0^3 p'(2t+6) \, dt = p(t^2+6t) \Big|_0^3 = p(27) - p(0) \text{ earns } 0 - 0 - 0$$

$$\cdot \frac{1}{2} \int_0^3 p'(u) \, du = \frac{1}{2} (p(12) - p(6)) = -3.75 \text{ earns } 0 - 1 - 1$$

$$\cdot \int_0^3 p'(2t+6) \, dt = \frac{1}{2} (p(12) - p(6)) = -3.75 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^3 p'(2t+6) \, dt = \frac{1}{2} p(u) \Big|_6^{12} = \frac{1}{2} (p(12) - p(6)) = -3.75 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^3 p'(2t+6) \, dt = \frac{1}{2} p(2t+6) \Big|_0^3 = \frac{1}{2} (p(12) - p(6)) \text{ earns } 1 - 1 - 0$$

$$\cdot \frac{1}{2} \int p'(u) = \frac{1}{2} p(u); \text{ and } \frac{1}{2} p(2t+6) \Big|_0^3 = \frac{1}{2} (-20 - (12.5)) \text{ earns } 1 - 1 - 1$$



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t + 6$. Then $du = 2 \, dt$.

When $t = 0$, $u = 6$.

When $t = 3$, $u = 12$.

$$\begin{aligned} \int_0^3 p'(2t+6) \, dt &= \frac{1}{2} \int_6^{12} p'(u) \, du \\ &= \frac{1}{2} (p(12) - p(6)) \end{aligned}$$

4-2-2b

$$= \frac{1}{2}(-20 - (-12.5)) = -3.75$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, resulting in $a \cdot p(6x)$ for some nonzero constant a .

Ways to earn first point include responses of $p(3 \cdot 2x)$ or $p(6x)$ (Note $p(6t)$ would also earn this point.)

The expression $p(3 \cdot 2x) - p(3 \cdot 1)$ or $(p(3 \cdot 2x) - p(3 \cdot 1)) \cdot 3$ earns this point. The $p(3 \cdot 1)$ is viewed as a chain rule error.

Any response correctly presenting and evaluating the derivative of the integral in terms of the substitution $u = 3t$ earns first point. Part of this response may include $\int_1^{2x} p(3t) dt = \frac{1}{3} \int_3^{6x} p(u) du$.

Second Point

Philosophy: This point is earned for correctly applying the chain rule to find $h'(x)$ or $h'(2)$. That is accomplished by multiplying the term involving $p(x)$ by 2.

Can only be earned if response earned first (FTC) point.

The first two points can be, and often are, earned simultaneously.

The point is not earned if the presented derivative is $a \cdot p(6x)$ for any $a \neq 2$.

The point is not earned for responses of $2p(3t) \big|_1^{2x}$ or $2(p(6x) - p(b))$, for any value of b .

Third Point

Can only be earned if response earned first (FTC) and second (chain rule) points.

Earned only for answer of -40 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring examples:

$$\cdot p(3 \cdot 2x) - p(3); p(12) - p(3) = -20 + 8.5 \text{ earns } 1 - 0 - 0$$

$$\cdot p(3 \cdot 2x) \cdot 2 - p(3) \cdot 0; -40 \text{ earns } 1 - 1 - 1$$

$$\cdot 2p(3t) \big|_1^{2x} = 2(p(12) - p(3)) \text{ earns } 1 - 0 - 0$$

$$\cdot p(3x) \cdot 2; p(6) \cdot 2 = -12.5 \cdot 2 \text{ earns } 0 - 0 - 0$$

4-2-2b

$\cdot p(6x) \cdot 2$; $p(12) \cdot 2$ earns $1 - 1 - 0$



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_1^{2x} p(3t) dt = p(3 \cdot 2x) \cdot 2$$

$$h'(2) = p(12) \cdot 2 = -20 \cdot 2 = -40$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for narrowing consideration of given information to g and g' .

g and g' need not be evaluated to earn the first point.

Considering velocity and acceleration not sufficient to earn first point, unless acceleration is linked to g' .

If response considers g and g' and g'' , then eventually g'' must be eliminated to earn first point.

Second Point

Philosophy: Earned for “increasing” with reason, using the opposite signs of g and g' .

Ways to earn second point include:

- $\cdot$ “increasing since $g < 0$ and $g' < 0$ ”
- $\cdot$ “particle speeding up since g and g' have the same signs”

Can only be earned if response earned first point.

A response that calls out all three given values ($g(3)$, $g'(3)$, and $g''(3)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g and g' only, both points can be earned.

4-2-2b

Examples:

- “increasing since velocity and acceleration have the same signs” earns neither point
- “increasing since velocity < 0 and acceleration < 0 ” earns neither point
- “increasing since $(-1) \cdot (-5) = 5 > 0$ ” earns both points



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g and g'
- ☐ Answer with reason

Solution:

Because $g(3) = -1 < 0$, the velocity of the particle is negative. Because $g'(3) = -5 < 0$, the acceleration of the particle is negative. Because the velocity and acceleration have the same signs, the speed of the particle is increasing at time $t = 3$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct tangent line equation, written in any form.

Correct responses include:

- $y = 11\left(x - \frac{1}{2}\right) - 2$ or $y + 2 = 11\left(x - \frac{1}{2}\right)$
- $y = 11x - \frac{15}{2}$
- $y = \left(12 \cdot \left(\frac{1}{2}\right)^2 - (-2)^3\right)\left(x - \frac{1}{2}\right) - 2$

Incorrect responses include:

- $y = (12x^2 - y^3)\left(x - \frac{1}{2}\right) - 2$
- $y = 11x - 2$
- $f(x) = 11x - \frac{15}{2}$. ($f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

4-2-2b



0

1

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{\left(\frac{1}{2}, -2\right)} = 12 \cdot \left(\frac{1}{2}\right)^2 - (-2)^3 = 11$$

$$y = 11 \left(x - \frac{1}{2}\right) - 2$$

4-2-2b

4. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	4	8	12	16
$p(t)$ (feet)	3	9	16.5	26	36

The position of a particle, A , moving along the x -axis is modeled by a twice-differentiable function p , where time t is measured in seconds and $p(t)$ is measured in feet. Selected values of $p(t)$ are shown in the table above.

(a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate the value of $\int_0^{16} p(t) \, dt$. Show the computations that lead to your answer.

(b) Use the data in the table to approximate $p'(11)$ using the average rate of change of $p(t)$ over the interval $8 \leq t \leq 12$. Show the computations that lead to your answer. Indicate units of measure.

(c) Interpret the meaning of $p'(11)$ in the context of the problem.

(d) Must there exist a value of k , for $0 < k < 16$, such that $p'(k)$ is equal to the average rate of change of p over the interval $0 \leq x \leq 16$? Justify your answer.

(e) Find $\int_0^4 p'(2t + 4) \, dt$. Show the computations that lead to your answer.

(f) Let $h(x) = \int_1^{4x} p(2t) \, dt$. Find $h'(1)$. Show the computations that lead to your answer.

(g) The velocity of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(3) = 2$, $g'(3) = -5$, and $g''(3) = 11$. Is the speed of particle Q increasing or decreasing at time $t = 3$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -xy^2 + 3$ with initial condition $f(5) = 2$. Write an equation for the line tangent to the graph of f at the point $(5, 2)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-2-2b

First Point

Philosophy: Earned for correct Midpoint Riemann sum, with appropriate values taken from the table.

Need to see sum of two terms, each as a product of two factors, i.e. $\text{base}_1 \cdot \text{height}_1 + \text{base}_2 \cdot \text{height}_2$

$p(4) \cdot (8 - 0) + p(12) \cdot (16 - 8)$ or $p(3) \cdot (6) + p(9) \cdot (6)$ earns first point.

If at least three of the four base/height values are correct in the sum of products, the expression earns the first point.

Second Point

280 or equivalent

Can be earned without first point with Riemann sum setup demonstration. With credible evidence that a midpoint Riemann sum attempt was made without not enough work shown, this point can be earned in the presence of the correct answer. For example, a sum of two terms with no evidence of products $(72 + 208)$ or a graph with areas labeled and summed to 280 or equivalent, can earn this point.

Examples:

- $280 = 72 + 208$ by itself earns second point, not first point
- 280 declared, with $9 \cdot 8$ and $26 \cdot 8$ identified (perhaps graphically) earns second point, not first point
- $k(9 \cdot 8 + 26 \cdot 8)$, nonzero constant $k \neq 1$ by itself earns first point, not second point
- $p(4) \cdot (8) + p(12) \cdot (16 - 8)$ by itself earns first point, not second point



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint Riemann sum
- ☐ Answer

Solution:

$$\int_0^{16} p(t) dt \approx p(4)(8 - 0) + p(12)(16 - 8)$$

$$= 9 \cdot 8 + 26 \cdot 8 = 280$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-2-2b

First Point

Philosophy: Earned for correct difference quotient, with appropriate values from the table.

Need to see both a difference and a quotient

$\frac{p(12)-p(8)}{12-8}$ by itself doesn't earn first point; however $\frac{26-16.5}{12-8}$ or $\frac{26-16.5}{4}$ by itself earns first point

Second Point

Earned for correct units of ft per sec or ft/sec or f/s.

Correct units must be tied to a number to earn second point.

Correct units must appear in part (b); correct units in part (c) only doesn't earn second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$p'(11) \approx \frac{26-16.5}{12-8} = \frac{9.5}{4} = 2.375 \text{ feet per second}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct $p'(11)$ interpretation.

Need two components in response:

- “rate of change of position” or “velocity” of the particle
- at $t = 11$

Correct ways to describe rate at time at $t = 11$ include:

- Rate at which position is changing/increasing
- Rate at which particle is moving right
- Instantaneous velocity or velocity of particle

4-2-2b

Ambiguous/incorrect ways to describe rate include:

- Rate at which particle is changing/increasing
- Rate at which particle is moving

Units are not required in part (c). If units are presented for t , they must be in seconds.



0	1
---	---

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$p'(11)$ is the rate at which the position of the particle is changing, in feet per second, at time $t = 11$ seconds.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for both Mean Value Theorem (MVT) conditions with the appropriate implication that p is continuous because we know that p is differentiable.

Ways to earn first point include:

- p differentiable $\Rightarrow p$ continuous
- p differentiable, hence p continuous
- p differentiable, so p continuous

p differentiable and p continuous does not earn first point.

Second Point

Philosophy: Earned for concluding “yes” and invoking Mean Value Theorem or its equivalent formulation, with consideration of the continuity of p .

MVT title does not have to be stated to earn second point.

This point is not earned if other theorems (IVT, etc.) are invoked.

If presented, the value for $p'(k) = \frac{33.5}{16}$ must be correct to earn second point.

4-2-2b

Examples:

- “By MVT, yes since p differentiable” earns neither point
- “By MVT, yes since p continuous” earns second point only
- “By MVT, yes since p differentiable and continuous” earns second point only
- “By MVT, yes since p differentiable hence continuous” earns both points
- “By MVT, yes since p twice differentiable hence continuous” earns both points
- “By MVT, yes since function differentiable hence continuous” earns both points



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

p is differentiable. $\Rightarrow p$ is continuous.

Therefore, the Mean Value Theorem can be applied to p on the interval $0 \leq x \leq 16$ to guarantee that there exists a value of k , for $0 < k < 16$, such that $p'(k)$ equals the average rate of change of p over the interval $0 \leq x \leq 16$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct substitution application.

Need three elements in response (with consistent use of u or another substitution variable):

- $u = 2t + 4$ (can be explicit or implicit)
- $du = 2 dt$ (yielding integral coefficient of $\frac{1}{2}$)
- Correct integral limits throughout, with t from 0 to 4 and u from 4 to 12

Omitting the coefficient of $\frac{1}{2}$ or using some other value will not earn first point; response could earn second point, but not third point.

4-2-2b

In the third element above, if response shows an otherwise correct substitution with unchanged limits of integration on a definite integral from 0 to 4, then point is not earned.

Alternately, the integral can be worked with as an indefinite integral with u returned to $2t + 4$ in the antiderivative before evaluating as a means to eliminate the need to change limits. Such a response can earn the first point if handled correctly.

For example: $\int_0^4 p'(2t + 4) dt = \frac{1}{2}p(2t + 4) \Big|_0^4$ earns the substitution point.

Second Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, with evidence of connection between $\int p'(u)$ and $p(u)$.

Ways to earn second point include:

· $k(p(12) - p(4))$ or $k(p(2 \cdot 4 + 4) - p(2 \cdot 0 + 4))$ for k a nonzero constant

· $k p(2t + 4) \Big|_0^3$ or $k p(u) \Big|_4^{12}$ for k a nonzero constant

· $\int p'(2t + 4) dt = k p(2t + 4)$ or $\int p'(u) du = k p(u)$ for any nonzero constant k , with or without appropriate limits of integration on either integral

First and second points can be earned simultaneously by responses such as $\int_0^4 p'(2t + 4) dt = \frac{1}{2}p(2t + 4) \Big|_0^4$

Third Point

Earned only for an answer of 8.5 or equivalent value.

Value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring Examples:

· $\frac{1}{2} \int p'(u) du = \frac{1}{2}p(u)$ earns 0 – 1 – 0

· $\frac{1}{2} \int p'(u) du = \frac{1}{2}p(u) \Big|_0^4 = \frac{1}{2}(p(4) - p(0))$ earns 0 – 1 – 0

· $\int p'(u) du = p(u) \Big|_4^{12} = (p(12) - p(4)) = 17$ earns 0 – 1 – 0

· $2 \int p'(u) du = 2p(u) \Big|_0^4 = 2[26 - 9]$ earns 0 – 1 – 0

· $\int_0^4 p'(2t + 4) dt = p(t^2 + 4t) \Big|_0^4 = p(32) - p(0)$ earns 0 – 0 – 0

4-2-2b

$$\cdot \frac{1}{2} \int_0^4 p'(u) \, du = \frac{1}{2}(p(12) - p(4)) = 8.5 \text{ earns } 0 - 1 - 1$$

$$\cdot \int_0^4 p'(2t + 4) \, dt = \frac{1}{2}(p(12) - p(4)) = 8.5 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^4 p'(2t + 4) \, dt = \frac{1}{2}p(u)\Big|_4^{12} = \frac{1}{2}(p(12) - p(4)) = 8.5 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^4 p'(2t + 4) \, dt = \frac{1}{2}p(2t + 4)\Big|_0^4 = \frac{1}{2}(p(12) - p(4)) \text{ earns } 1 - 1 - 0$$

$$\cdot \frac{1}{2} \int p'(u) = \frac{1}{2}p(u); \text{ and } \frac{1}{2}p(2t + 4)\Big|_0^4 = \frac{1}{2}(26 - 9) \text{ earns } 1 - 1 - 1$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t + 4$. Then $du = 2 \, dt$.

When $t = 0$, $u = 4$.

When $t = 4$, $u = 12$.

$$\int_0^4 p'(2t + 4) \, dt = \frac{1}{2} \int_4^{12} p'(u) \, du$$

$$= \frac{1}{2}(p(12) - p(4))$$

$$= \frac{1}{2}(26 - 9) = 8.5$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

4-2-2b

Philosophy: Earned for correct Fundamental Theorem of Calculus application, resulting in $a \cdot p(8x)$ for some nonzero constant a .

Ways to earn first point include responses of $p(2 \cdot 4x)$ or $p(8x)$ (Note $p(8t)$ would also earn this point.)

The expression $p(2 \cdot 4x) - p(2 \cdot 1)$ or $(p(2 \cdot 4x) - p(2 \cdot 1)) \cdot 4$ earns this point. The $p(2 \cdot 1)$ is viewed as a chain rule error.

Any response correctly presenting and evaluating the derivative of the integral in terms of the substitution $u = 2t$ earns first point. Part of this response may include $\int_1^{4x} p(2t) dt = \frac{1}{3} \int_2^{8x} p(u) du$.

Second Point

Philosophy: This point is earned for correctly applying the chain rule to find $h'(x)$ or $h'(1)$. That is accomplished by multiplying the term involving $p(x)$ by 4.

Can only be earned if response earned first (FTC) point.

The first two points can be, and often are, earned simultaneously.

The point is not earned if the presented derivative is $a \cdot p(8x)$ for any $a \neq 4$.

The point is not earned for responses of $4p(2t) \big|_1^{4x}$ or $4(p(8x) - p(b))$, for any value of b .

Third Point

Can only be earned if response earned first (FTC) and second (chain rule) points.

Earned only for answer of 66 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring examples:

$\cdot p(2 \cdot 4x) - p(2)$; $p(8) - p(2)$ earns $1 - 0 - 0$

$\cdot p(2 \cdot 4x) \cdot 4 - p(2) \cdot 0$; 66 earns $1 - 1 - 1$

$\cdot 4p(2t) \big|_1^{4x} = 4(p(8) - p(2))$ earns $1 - 0 - 0$

$\cdot p(2x) \cdot 2$; $p(2) \cdot 2$ earns $0 - 0 - 0$

$\cdot p(2 \cdot 4x) \cdot 4$; $p(8) \cdot 4$ earns $1 - 1 - 0$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

4-2-2b

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_1^{4x} p(2t) dt = p(2 \cdot 4x) \cdot 4$$

$$h'(1) = p(8) \cdot 4 = 16.5 \cdot 4 = 66$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for narrowing consideration of given information to g and g' .

g and g' need not be evaluated to earn the first point.

Considering velocity and acceleration not sufficient to earn first point, unless acceleration is linked to g' .

If response considers g and g' and g'' , then eventually g'' must be eliminated to earn first point.

Second Point

Philosophy: Earned for “decreasing” with reason, using the opposite signs of g and g' .

Ways to earn second point include:

- “decreasing since $g > 0$ and ” $g' < 0$ ”
- “particle slowing down since g and g' have opposite signs”

Can only be earned if response earned first point.

Examples:

- “decreasing since velocity and acceleration have opposite signs” earns neither point
- “decreasing since velocity > 0 and acceleration < 0 ” earns neither point
- “decreasing since $(2) \cdot (-5) = -10 < 0$ ” earns both points



0	1	2
---	---	---

4-2-2b

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g and g'
- ☐ Answer with reason

Solution:

Because $g(3) = 2 > 0$, the velocity of the particle is positive. Because $g'(3) = -5 < 0$, the acceleration of the particle is negative. Because the velocity and acceleration have opposite signs, the speed of the particle is decreasing at time $t = 3$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct tangent line equation.

Correct responses include:

· $y = -17(x - 5) + 2$ or $y - 2 = -17(x - 5)$

· $y = -17x + 87$

· $y = (-5 \cdot 2^2 + 3)(x - 5) + 2$

Incorrect responses include:

· $y = (-xy^2 + 3)(x - 5) + 2$

· $y = 17x + 87$

· $f(x) = -17x + 87$



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(5,2)} = -5 \cdot 2^2 + 3 = -17$$

$$y = -17(x - 5) + 2$$

4-2-2b

An object moves along the x -axis with initial position $x(0) = 2$. The velocity of the object at time $t \geq 0$ is given by $v(t) = \sin\left(\frac{\pi}{3}t\right)$.

5.  Consider the following two statements.

Statement I: For $3 < t < 4.5$, the velocity of the object is decreasing.

Statement II: For $3 < t < 4.5$, the speed of the object is increasing.

Are either or both of these statements correct? For each statement provide a reason why it is correct or not correct.

Part B

1 point is earned for I correct, with reason

1 point is earned for II correct

1 point is earned for the reason for II

On $3 < t < 4.5$:

$$a(t) = v'(t) = \frac{\pi}{3} \cos\left(\frac{\pi}{3}t\right) < 0$$

Statement I is correct since $a(t) < 0$.

Statement II is correct since $v(t) < 0$ and $a(t) < 0$.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for I correct, with reason

1 point is earned for II correct

1 point is earned for the reason for II

On $3 < t < 4.5$:

$$a(t) = v'(t) = \frac{\pi}{3} \cos\left(\frac{\pi}{3}t\right) < 0$$

Statement I is correct since $a(t) < 0$.

4-2-2b

Statement II is correct since $v(t) < 0$ and $a(t) < 0$.

6. What is the acceleration of the object time $t = 4$?
-

Part A

1 point is earned for the answer

$$\begin{aligned} a(4) &= v'(4) = \frac{\pi}{3} \cos\left(\frac{4\pi}{3}\right) \\ &= -\frac{\pi}{6} \text{ or } -0.523 \text{ or } -0.524 \end{aligned}$$



0

1

The student response earns all of the following points:

1 point is earned for the answer

$$\begin{aligned} a(4) &= v'(4) = \frac{\pi}{3} \cos\left(\frac{4\pi}{3}\right) \\ &= -\frac{\pi}{6} \text{ or } -0.523 \text{ or } -0.524 \end{aligned}$$

4-2-2b

7.  An object moves along the x -axis with initial position $x(0) = 2$. The velocity of the object at time $t \geq 0$ is given by $v(t) = \sin\left(\frac{\pi}{3}t\right)$.

- (a) What is the acceleration of the object at time $t = 4$?
 (b) Consider the following two statements.

Statement I: For $3 < t < 4.5$, the velocity of the object is decreasing.

Statement II: For $3 < t < 4.5$, the speed of the object is increasing.

Are either or both of these statements correct? For each statement provide a reason why it is correct or not correct.

- (c) What is the total distance traveled by the object over the time interval $0 \leq t \leq 4$?
 (d) What is the position of the object at time $t = 4$?

Part A

1 point is earned for the answer

$$\begin{aligned} a(4) &= v'(4) = \frac{\pi}{3} \cos\left(\frac{4\pi}{3}\right) \\ &= -\frac{\pi}{6} \text{ or } -0.523 \text{ or } -0.524 \end{aligned}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$\begin{aligned} a(4) &= v'(4) = \frac{\pi}{3} \cos\left(\frac{4\pi}{3}\right) \\ &= -\frac{\pi}{6} \text{ or } -0.523 \text{ or } -0.524 \end{aligned}$$

Part B

1 point is earned for I correct, with reason

1 point is earned for II correct

1 point is earned for the reason for II

On $3 < t < 4.5$:

4-2-2b

$$a(t) = v'(t) = \frac{\pi}{3} \cos\left(\frac{\pi}{3}t\right) < 0$$

Statement I is correct since $a(t) < 0$.

Statement II is correct since $v(t) < 0$ and $a(t) < 0$.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for I correct, with reason

1 point is earned for II correct

1 point is earned for the reason for II

On $3 < t < 4.5$:

$$a(t) = v'(t) = \frac{\pi}{3} \cos\left(\frac{\pi}{3}t\right) < 0$$

Statement I is correct since $a(t) < 0$.

Statement II is correct since $v(t) < 0$ and $a(t) < 0$.

Part C

1 point is earned for limits of 0 and 4 on an integral of $v(t)$ or $|v(t)|$

or

uses $x(0)$ and $x(4)$ to compute distance

1 point is earned for handle change of direction at student's turning point answer

1 point is earned for the answer

$$\text{Distance} = \int_0^4 |v(t)| dt = 2.387$$

OR

4-2-2b

$$x(t) = -\frac{3}{\pi} \cos\left(\frac{\pi}{3}t\right) + \frac{3}{\pi} + 2$$

$$x(0) = 2$$

$$x(4) = 2 + \frac{9}{2\pi} = 3.43239$$

$$v(t) = 0 \text{ When } t = 3$$

$$x(3) = \frac{6}{\pi} + 2 = 3.90986$$

$$|x(3) - x(0)| + |x(4) - x(3)| = \frac{15}{2\pi} = 2.387$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for limits of 0 and 4 on an integral of $v(t)$ or $|v(t)|$

or

uses $x(0)$ and $x(4)$ to compute distance

1 point is earned for handles change of direction at student's turning point answer

1 point is earned for the answer

$$\text{Distance} = \int_0^4 |v(t)| dt = 2.387$$

OR

$$x(t) = -\frac{3}{\pi} \cos\left(\frac{\pi}{3}t\right) + \frac{3}{\pi} + 2$$

$$x(0) = 2$$

$$x(4) = 2 + \frac{9}{2\pi} = 3.43239$$

$$v(t) = 0 \text{ When } t = 3$$

$$x(3) = \frac{6}{\pi} + 2 = 3.90986$$

$$|x(3) - x(0)| + |x(4) - x(3)| = \frac{15}{2\pi} = 2.387$$

Part D

1 point is earned for integral

4-2-2b

1 point is earned for the answer

OR

1 point is earned for $x(t) = -\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right) + C$

1 point is earned for the answer

$$x(4) = x(0) + \int_0^4 v(t)dt = 3.432$$

OR

$$x(t) = -\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right) + \frac{3}{\pi} + 2$$

$$x(4) = 2 + \frac{9}{2\pi} = 3.432$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

OR

1 point is earned for $x(t) = -\frac{3}{\pi}\cos\left(\frac{\pi}{3}t\right) + C$

1 point is earned for the answer

$$x(4) = x(0) + \int_0^4 v(t)dt = 3.432$$

OR

4-2-2b

$$x(t) = -\frac{3}{\pi} \cos\left(\frac{\pi}{3}t\right) + \frac{3}{\pi} + 2$$

$$x(4) = 2 + \frac{9}{2\pi} = 3.432$$

4-2-2b

A particle starts at time $t=0$ and moves along the x -axis so that its position at any time $t \geq 0$ is given by $x(t) = (t-1)^3(2t-3)$.

8. Find the value of t when the particle is moving and the acceleration is zero.

Part C

1 point is earned for derivative of their v from (a) or (b)

1 point is earned for solving for $v'(t)=0$ and solves for at least 2 roots

1 point is earned for answer

$$\begin{aligned} a(t) &= v'(t) \\ &= 2(t-1)(8t-11) + 8(t-1)^2 \\ &= 6(t-1)(4t-5) \end{aligned}$$

$$a(t) = 0 \text{ when } t = 1, t = \frac{5}{4}$$

but particle not moving at $t = 1$ so $t = \frac{5}{4}$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for derivative of their v from (a) or (b)

1 point is earned for solving for $v'(t)=0$ and solves for at least 2 roots

1 point is earned for answer

$$\begin{aligned} a(t) &= v'(t) \\ &= 2(t-1)(8t-11) + 8(t-1)^2 \\ &= 6(t-1)(4t-5) \end{aligned}$$

$$a(t) = 0 \text{ when } t = 1, t = \frac{5}{4}$$

but particle not moving at $t = 1$ so $t = \frac{5}{4}$

4-2-2b

9. For what values of t is the velocity of the particle less than zero?

Part B

1 point is earned for setting their $v(t) < 0$

3 points are earned for answer

1 point is **deducted** for allowing $t=1$

1 point is **deducted** for allowing $t < 0$

1 point is **deducted** for including $t = \frac{11}{8}$

1 point is **deducted** for algebra error

$$v(t) < 0 \text{ when } (t - 1)^2 (8t - 11) < 0$$

Therefore $8t - 11 < 0$ and $t \neq 1$

or $t < \frac{11}{8}$ and $t \neq 1$

Since $t \geq 0$, answer is $0 \leq t < \frac{11}{8}$, except $t = 1$



0	1	2	3	4
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The student response earns four of the following points:

1 point is earned for setting their $v(t) < 0$

3 points are earned for answer

1 point is **deducted** for allowing $t=1$

1 point is **deducted** for allowing $t < 0$

1 point is **deducted** for including $t = \frac{11}{8}$

1 point is **deducted** for algebra error

4-2-2b

$$v(t) < 0 \text{ when } (t-1)^2(8t-11) < 0$$

Therefore $8t - 11 < 0$ and $t \neq 1$

$$\text{or } t < \frac{11}{8} \text{ and } t \neq 1$$

Since $t \geq 0$, answer is $0 \leq t < \frac{11}{8}$, except $t = 1$

10. Find the velocity of the particle at any time $t \geq 0$.

Part A

2 points are earned for the derivative

1 point is **deducted** for algebra error

$$v(t) = x'(t)$$

$$= 3(t-1)^2(2t-3) + 2(t-1)^3$$

$$= (t-1)^2(8t-11)$$



0	1	2
---	---	---

The student response earns two of the following points:

2 points are earned for the derivative

1 point is **deducted** for algebra error

$$v(t) = x'(t)$$

$$= 3(t-1)^2(2t-3) + 2(t-1)^3$$

$$= (t-1)^2(8t-11)$$

4-2-2b

11. A particle starts at time $t = 0$ and moves along the x -axis so that its position at any time $t \geq 0$ is given by $x(t) = (t - 1)^3(2t - 3)$.
- (a) Find the velocity of the particle at any time $t \geq 0$.
- (b) For what values of t is the velocity of the particle less than zero?
- (c) Find the value of t when the particle is moving and the acceleration is zero.

Part A

2 points are earned for the derivative

1 point is **deducted** for algebra error

$$\begin{aligned} v(t) &= x'(t) \\ &= 3(t - 1)^2(2t - 3) + 2(t - 1)^3 \\ &= (t - 1)^2(8t - 11) \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for the derivative

1 point is **deducted** for algebra error

$$\begin{aligned} v(t) &= x'(t) \\ &= 3(t - 1)^2(2t - 3) + 2(t - 1)^3 \\ &= (t - 1)^2(8t - 11) \end{aligned}$$

Part B

1 point is earned for setting their $v(t) < 0$

3 points are earned for answer

1 point is **deducted** for allowing $t = 1$

1 point is **deducted** for allowing $t < 0$

1 point is **deducted** for including $t = \frac{11}{8}$

4-2-2b

1 point is **deducted** for algebra error

$$v(t) < 0 \text{ when } (t - 1)^2 (8t - 11) < 0$$

$$\therefore 8t - 11 < 0 \text{ and } t \neq 1$$

$$\text{or } t < \frac{11}{8} \text{ and } t \neq 1$$

Since $t \geq 0$, answer is

$$0 \leq t < \frac{11}{8}, \text{ except } t = 1$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for setting their $v(t) < 0$

3 points are earned for answer

1 point is **deducted** for allowing $t = 1$

1 point is **deducted** for allowing $t < 0$

1 point is **deducted** for including $t = \frac{11}{8}$

1 point is **deducted** for algebra error

$$v(t) < 0 \text{ when } (t - 1)^2 (8t - 11) < 0$$

$$\therefore 8t - 11 < 0 \text{ and } t \neq 1$$

$$\text{or } t < \frac{11}{8} \text{ and } t \neq 1$$

Since $t \geq 0$, answer is

4-2-2b

$$0 \leq t < \frac{11}{8}, \text{ except } t = 1$$

Part C

1 point is earned for derivative of their v from (a) or (b)

1 point is earned for solving for $v'(t) = 0$ and solves for at least 2 roots

1 point is earned for answer

$$a(t) = v'(t)$$

$$= 2(t - 1)(8t - 11) + 8(t - 1)^2$$

$$= 6(t - 1)(4t - 5)$$

$$a(t) = 0 \text{ when } t = 1, t = \frac{5}{4}$$

but particle not moving at $t = 1$ so $t = \frac{5}{4}$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for derivative of their v from (a) or (b)

1 point is earned for solving for $v'(t) = 0$ and solves for at least 2 roots

1 point is earned for answer

$$a(t) = v'(t)$$

$$= 2(t - 1)(8t - 11) + 8(t - 1)^2$$

$$= 6(t - 1)(4t - 5)$$

$$a(t) = 0 \text{ when } t = 1, t = \frac{5}{4}$$

but particle not moving at $t = 1$ so $t = \frac{5}{4}$

4-2-2b

A particle moves along the x -axis so that its velocity at any time $t \geq 0$ is given by $v(t) = 1 - \sin(2\pi t)$.

12. Find all values of t , $0 \leq t \leq 2$, for which the particle is at rest.

Part B

1 point is earned for $1 - \sin(2\pi t) = 0$

1 point is earned for $t = \frac{1}{4}$

1 point is earned for $t = \frac{5}{4}$

$$v(t) = 0$$

$$1 - \sin(2\pi t) = 0 \text{ or } 1 = \sin(2\pi t)$$

$$2\pi t = \frac{\pi}{2} + 2k\pi,$$

where $k = 0, \pm 1, \pm 2, \dots$

$$\text{and } 0 \leq t \leq 2$$

$$t = \frac{1}{4} \text{ and } t = \frac{5}{4}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $1 - \sin(2\pi t) = 0$

1 point is earned for $t = \frac{1}{4}$

1 point is earned for $t = \frac{5}{4}$

$$v(t) = 0$$

4-2-2b

$$1 - \sin(2\pi t) = 0 \text{ or } 1 = \sin(2\pi t)$$

$$2\pi t = \frac{\pi}{2} + 2k\pi,$$

where $k = 0, \pm 1, \pm 2, \dots$

and $0 \leq t \leq 2$

$$t = \frac{1}{4} \text{ and } t = \frac{5}{4}$$

13. Find the acceleration $a(t)$ of the particle at any time t .
-

Part A

2 points are earned for correct differentiation of velocity

$$\begin{aligned} a(t) &= v'(t) \\ &= -2\pi \cos(2\pi t) \end{aligned}$$



0	1	2
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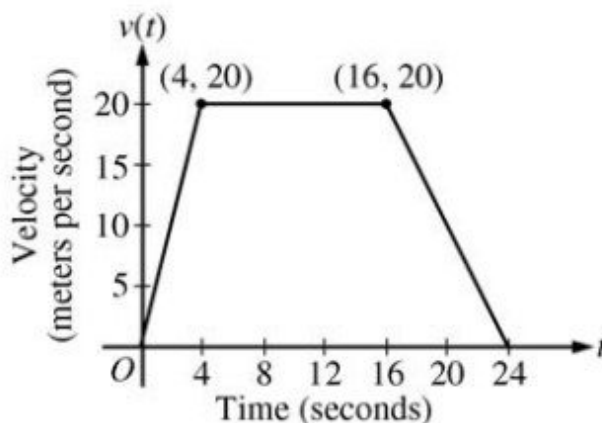
The student response earns all of the following points:

2 points are earned for correct differentiation of velocity

$$\begin{aligned} a(t) &= v'(t) \\ &= 2\pi \cos(2\pi t) \end{aligned}$$

4-2-2b

14.



A car is traveling on a straight road. For $0 \leq t \leq 24$ seconds, the car's velocity $v(t)$, in meters per second, is modeled by the piecewise-linear function defined by the graph above.

Let $a(t)$ be the car's acceleration at time t , in meters per second per second. For $0 < t < 24$, write a piecewise-defined function for $a(t)$.

Part C

1 point is earned for finds the values $5, 0, -\frac{5}{2}$

1 point is earned for identifies constants with correct intervals

$$a(t) = \begin{cases} 5 & \text{if } 0 < t < 4 \\ 0 & \text{if } 4 < t < 16 \\ -\frac{5}{2} & \text{if } 16 < t < 24 \end{cases}$$

$a(t)$ does not exist at $t = 4$ and $t = 16$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for finds the values $5, 0, -\frac{5}{2}$

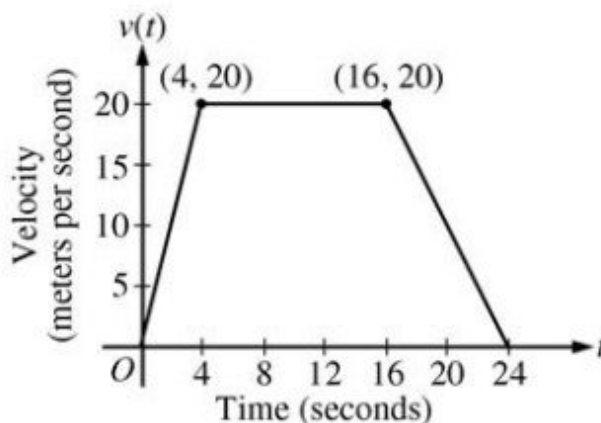
1 point is earned for identifies constants with correct intervals

$$a(t) = \begin{cases} 5 & \text{if } 0 < t < 4 \\ 0 & \text{if } 4 < t < 16 \\ -\frac{5}{2} & \text{if } 16 < t < 24 \end{cases}$$

$a(t)$ does not exist at $t = 4$ and $t = 16$.

4-2-2b

15.



A car is traveling on a straight road. For $0 \leq t \leq 24$ seconds, the car's velocity $v(t)$, in meters per second, is modeled by the piecewise-linear function defined by the graph above.

- (a) Find $\int_0^{24} v(t) dt$. Using correct units, explain the meaning of $\int_0^{24} v(t) dt$.
- (b) For each of $v'(4)$ and $v'(20)$, find the value or explain why it does not exist. Indicate units of measure.
- (c) Let $a(t)$ be the car's acceleration at time t , in meters per second per second. For $0 < t < 24$, write a piecewise-defined function for $a(t)$.
- (d) Find the average rate of change of v over the interval $8 \leq t \leq 20$. Does the Mean Value Theorem guarantee a value of c , for $8 < c < 20$, such that $v'(c)$ is equal to this average rate of change? Why or why not?

Part A

1 point is earned for the value

1 point is earned for the meaning with units

$$\int_0^{24} v(t) dt = \frac{1}{2}(4)(20) + (12)(20) + \frac{1}{2}(8)(20) = 360$$

The car travels 360 meters in these 24 seconds.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value

1 point is earned for the meaning with units

4-2-2b

$$\int_0^{24} v(t) dt = \frac{1}{2}(4)(20) + (12)(20) + \frac{1}{2}(8)(20) = 360$$

The car travels 360 meters in these 24 seconds.

Part B

1 point is earned for: $v'(4)$ does not exist, with explanation

1 point is earned for: $v'(20)$

1 point is earned for the units

$v'(4)$ does not exist because

$$\lim_{t \rightarrow 4^-} \left(\frac{v(t) - v(4)}{t - 4} \right) = 5 \neq 0 = \lim_{t \rightarrow 4^+} \left(\frac{v(t) - v(4)}{t - 4} \right)$$

$$v'(20) = \frac{20 - 0}{16 - 24} = -\frac{5}{2} \text{ m/sec}^2$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $v'(4)$ does not exist, with explanation

1 point is earned for: $v'(20)$

1 point is earned for the units

$v'(4)$ does not exist because

$$\lim_{t \rightarrow 4^-} \left(\frac{v(t) - v(4)}{t - 4} \right) = 5 \neq 0 = \lim_{t \rightarrow 4^+} \left(\frac{v(t) - v(4)}{t - 4} \right)$$

$$v'(20) = \frac{20 - 0}{16 - 24} = -\frac{5}{2} \text{ m/sec}^2$$

Part C

1 point is earned for finds the values 5, 0, $-\frac{5}{2}$

1 point is earned for identifies constants with correct intervals

4-2-2b

$$a(t) = \begin{cases} 5 & \text{if } 0 < t < 4 \\ 0 & \text{if } 4 < t < 16 \\ -\frac{5}{2} & \text{if } 16 < t < 24 \end{cases}$$

$a(t)$ does not exist at $t = 4$ and $t = 16$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for finds the values 5 , 0 , $-\frac{5}{2}$

1 point is earned for identifies constants with correct intervals

$$a(t) = \begin{cases} 5 & \text{if } 0 < t < 4 \\ 0 & \text{if } 4 < t < 16 \\ -\frac{5}{2} & \text{if } 16 < t < 24 \end{cases}$$

$a(t)$ does not exist at $t = 4$ and $t = 16$.

Part D

1 point is earned for the average rate of change of v on $[8, 20]$

1 point is earned for the answer with explanation

The average rate of change of v on $[8, 20]$ is

$$\frac{v(20) - v(8)}{20 - 8} = -\frac{5}{6} \text{ m/sec}^2.$$

No, the Mean Value Theorem does not apply to v on $[8, 20]$ because v is not differentiable at

$t = 16$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the average rate of change of v on $[8, 20]$

1 point is earned for the answer with explanation

4-2-2b

The average rate of change of v on $[8, 20]$ is

$$\frac{v(20) - v(8)}{20 - 8} = -\frac{5}{6} \text{ m/sec}^2.$$

No, the Mean Value Theorem does not apply to v on $[8, 20]$ because v is not differentiable at

$t = 16$.

4-2-2b

A particle moves along the x -axis so that its velocity at time t is given by

$$v(t) = -(t + 1) \sin\left(\frac{t^2}{2}\right).$$

At time $t = 0$, the particle is at position $x = 1$.

16.  Find all times t in the open interval $0 < t < 3$ when the particle changes direction. Justify your answer.

Part B

1 point is earned for the correct answer

$$t = \sqrt{2\pi} \text{ only}$$

1 point is earned for the justification

$$v(t) = 0 \text{ when } \frac{t^2}{2} = \pi$$

$$t = \sqrt{2\pi} \text{ or } 2.506 \text{ or } 2.507$$

Since $v(t) < 0$ for $0 < t < \sqrt{2\pi}$ and $v(t) > 0$ for $\sqrt{2\pi} < t < 3$, the particle changes directions at $t = \sqrt{2\pi}$.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the correct answer

$$t = \sqrt{2\pi} \text{ only}$$

1 point is earned for the justification

$$v(t) = 0 \text{ when } \frac{t^2}{2} = \pi$$

$$t = \sqrt{2\pi} \text{ or } 2.506 \text{ or } 2.507$$

Since $v(t) < 0$ for $0 < t < \sqrt{2\pi}$ and $v(t) > 0$ for $\sqrt{2\pi} < t < 3$, the particle changes directions at $t = \sqrt{2\pi}$.

4-2-2b

17.  Find the acceleration of the particle at time $t = 2$. Is the speed of the particle increasing at $t = 2$? Why or why not?

Part A

1 point is earned for correctly solving for $a(2)$

$$a(2) = v'(2) = 1.587 \text{ or } 1.588$$

$$v(2) = -3 \sin(2) < 0$$

1 point is earned for the reason for the speed decreasing

Speed is decreasing since $a(2) > 0$ and $v(2) < 0$.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for correctly solving for $a(2)$

$$a(2) = v'(2) = 1.587 \text{ or } 1.588$$

$$v(2) = -3 \sin(2) < 0$$

1 point is earned for the reason for the speed decreasing

Speed is decreasing since $a(2) > 0$ and $v(2) < 0$.

4-2-2b

18.  A particle moves along the x -axis so that its velocity at time t is given by

$$v(t) = -(t + 1) \sin\left(\frac{t^2}{2}\right).$$

At time $t = 0$, the particle is at position $x = 1$.

- (a) Find the acceleration of the particle at time $t = 2$. Is the speed of the particle increasing at $t = 2$? Why or why not?
- (b) Find all times t in the open interval $0 < t < 3$ when the particle changes direction. Justify your answer.
- (c) Find the total distance traveled by the particle from time $t = 0$ until time $t = 3$.
- (d) During the time interval $0 \leq t \leq 3$, what is the greatest distance between the particle and the origin? Show the work that leads to your answer.

Part A

1 point is earned for correctly solving for $a(2)$

$$a(2) = v'(2) = 1.587 \text{ or } 1.588$$

$$v(2) = -3 \sin(2) < 0$$

1 point is earned for the reason for the speed decreasing

Speed is decreasing since $a(2) > 0$ and $v(2) < 0$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correctly solving for $a(2)$

$$a(2) = v'(2) = 1.587 \text{ or } 1.588$$

$$v(2) = -3 \sin(2) < 0$$

1 point is earned for the reason for the speed decreasing

Speed is decreasing since $a(2) > 0$ and $v(2) < 0$.

Part B

1 point is earned for the correct answer $t = \sqrt{2\pi}$ only

1 point is earned for the justification

4-2-2b

$$v(t) = 0 \text{ when } \frac{t^2}{2} = \pi$$

$$t = \sqrt{2\pi} \text{ or } 2.506 \text{ or } 2.507$$

Since $v(t) < 0$ for $0 < t < \sqrt{2\pi}$ and $v(t) > 0$ for $\sqrt{2\pi} < t < 3$, the particle changes directions at $t = \sqrt{2\pi}$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct answer $t = \sqrt{2\pi}$ only

1 point is earned for the justification

$$v(t) = 0 \text{ when } \frac{t^2}{2} = \pi$$

$$t = \sqrt{2\pi} \text{ or } 2.506 \text{ or } 2.507$$

Since $v(t) < 0$ for $0 < t < \sqrt{2\pi}$ and $v(t) > 0$ for $\sqrt{2\pi} < t < 3$, the particle changes directions at $t = \sqrt{2\pi}$.

Part C

1 point is earned for the correct limits

1 point is earned for the correct integrand

1 point is earned for the correct answer

$$\text{Distance} = \int_0^3 |v(t)| dt = 4.333 \text{ or } 4.334$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the correct limits

1 point is earned for the correct integrand

1 point is earned for the correct answer

4-2-2b

$$\text{Distance} = \int_0^3 |v(t)| dt = 4.333 \text{ or } 4.334$$

Part D

1 point is earned for the $\pm$ distance that the particle travels while velocity is negative

$$\int_0^{\sqrt{2\pi}} v(t) dt = -3.265$$

$$x(\sqrt{2\pi}) = x(0) + \int_0^{\sqrt{2\pi}} v(t) dt = -2.265$$

1 point is earned for the correct answer

Since the total distance from $t = 0$ to $t = 3$ is 4.334, the particle is still to the left of the origin at $t = 3$. Hence the greatest distance from the origin is 2.265.



0	1	2
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The student response earns all of the following points:

1 point is earned for the $\pm$ distance that the particle travels while velocity is negative

$$\int_0^{\sqrt{2\pi}} v(t) dt = -3.265$$

$$x(\sqrt{2\pi}) = x(0) + \int_0^{\sqrt{2\pi}} v(t) dt = -2.265$$

1 point is earned for the correct answer

Since the total distance from $t = 0$ to $t = 3$ is 4.334, the particle is still to the left of the origin at $t = 3$. Hence the greatest distance from the origin is 2.265.

4-2-2b

19. A particle moves along the x -axis so that its velocity at any time $t \geq 0$ is given by $v(t) = 1 - \sin(2\pi t)$.

(a) Find the acceleration $a(t)$ of the particle at any time t .

(b) Find all values of t , $0 \leq t \leq 2$, for which the particle is at rest.

(c) Find the position $x(t)$ of the particle at any time t if $x(0) = 0$.

Part A

2 points are earned for correct differentiation of velocity

$$\begin{aligned} a(t) &= v'(t) \\ &= -2\pi \cos(2\pi t) \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for correct differentiation of velocity

$$\begin{aligned} a(t) &= v'(t) \\ &= -2\pi \cos(2\pi t) \end{aligned}$$

Part B

1 point is earned for $1 - \sin(2\pi t) = 0$

1 point is earned for $t = \frac{1}{4}$

1 point is earned for $t = \frac{5}{4}$

$$v(t) = 0$$

$$1 - \sin(2\pi t) = 0 \text{ or } 1 = \sin(2\pi t)$$

$$2\pi t = \frac{\pi}{2} + 2k\pi,$$

where $k = 0, \pm 1, \pm 2, \dots$

4-2-2band $0 \leq t \leq 2$

$$\therefore t = \frac{1}{4} \text{ and } t = \frac{5}{4}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $1 - \sin(2\pi t) = 0$ 1 point is earned for $t = \frac{1}{4}$ 1 point is earned for $t = \frac{5}{4}$

$$v(t) = 0$$

$$1 - \sin(2\pi t) = 0 \text{ or } 1 = \sin(2\pi t)$$

$$2\pi t = \frac{\pi}{2} + 2k\pi,$$

where $k = 0, \pm 1, \pm 2, \dots$ and $0 \leq t \leq 2$

$$\therefore t = \frac{1}{4} \text{ and } t = \frac{5}{4}$$

Part C2 points are earned for the correct antiderivative of $v(t)$ 1 point is earned for $x(0) = 0$ 1 point is earned for finding value of C

4-2-2b

$$\begin{aligned}
 x(t) &= \int v(t) dt \\
 &= \int [1 - \sin(2\pi t)] dt \\
 &= t + \frac{1}{2\pi} \cos(2\pi t) + C
 \end{aligned}$$

$$x(0) = 0 = 0 + \frac{\cos(0)}{2\pi} + C$$

$$\therefore x(t) = t + \frac{1}{2\pi} \cos(2\pi t) - \frac{1}{2\pi}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for the correct antiderivative of $v(t)$

1 point is earned for $x(0) = 0$

1 point is earned for finding value of C

$$\begin{aligned}
 x(t) &= \int v(t) dt \\
 &= \int [1 - \sin(2\pi t)] dt \\
 &= t + \frac{1}{2\pi} \cos(2\pi t) + C
 \end{aligned}$$

$$x(0) = 0 = 0 + \frac{\cos(0)}{2\pi} + C$$

$$\therefore x(t) = t + \frac{1}{2\pi} \cos(2\pi t) - \frac{1}{2\pi}$$

4-2-2b

20. A particle moves along the y -axis so that its velocity v at time $t \geq 0$ is given by $v(t) = 1 - \tan^{-1}(e^t)$. At time $t = 0$, the particle is at $y = -1$. (Note: $\tan^{-1}x = \arctan x$)



Is the speed of the particle increasing or decreasing at time $t = 2$. Give a reason for your answer.

Part B

1 point is earned for the answer with reason

$$v(2) = -0.436$$

Speed is increasing since $a(2) < 0$ and $v(2) < 0$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer with reason

$$v(2) = -0.436$$

Speed is increasing since $a(2) < 0$ and $v(2) < 0$.

4-2-2b

21.  A particle moves along the y -axis so that its velocity v at time $t \geq 0$ is given by $v(t) = 1 - \tan^{-1}(e^t)$. At time $t = 0$, the particle is at $y = -1$. (Note: $\tan^{-1}x = \arctan x$)
- Find the acceleration of the particle at time $t = 2$.
 - Is the speed of the particle increasing or decreasing at time $t = 2$? Give a reason for your answer.
 - Find the time $t \geq 0$ at which the particle reaches its highest point. Justify your answer.
 - Find the position of the particle at time $t = 2$. Is the particle moving toward the origin or away from the origin at time $t = 2$? Justify your answer.

Part A

1 point is earned for the answer

$$a(2) = v'(2) = -0.132 \text{ or } -0.133$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$a(2) = v'(2) = -0.132 \text{ or } -0.133$$

Part B

1 point is earned for the answer with reason

$$v(2) = -0.436$$

Speed is increasing since $a(2) < 0$ and $v(2) < 0$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer with reason

$$v(2) = -0.436$$

Speed is increasing since $a(2) < 0$ and $v(2) < 0$.

Part C

4-2-2b

1 point is earned for: set $v(t) = 0$

1 point is earned for identifying $t = 0.433$ as a candidate

1 point is earned for justifying absolute maximum

$$v(t) = 0 \text{ when } \tan^{-1}(e^t) = 1$$

$t = \ln(\tan(1)) = 0.433$ is the only critical value for y .

$$v(t) > 0 \text{ for } 0 < t < \ln(\tan(1))$$

$$v(t) < 0 \text{ for } t > \ln(\tan(1))$$

$y(t)$ has an absolute maximum at $t = 0.433$.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for: set $v(t) = 0$

1 point is earned for identifying $t = 0.433$ as a candidate

1 point is earned for justifying absolute maximum

$$v(t) = 0 \text{ when } \tan^{-1}(e^t) = 1$$

$t = \ln(\tan(1)) = 0.433$ is the only critical value for y .

$$v(t) > 0 \text{ for } 0 < t < \ln(\tan(1))$$

$$v(t) < 0 \text{ for } t > \ln(\tan(1))$$

$y(t)$ has an absolute maximum at $t = 0.433$.

Part D

1 point is earned for: $\int_0^2 v(t) dt$

1 point is earned for the handles initial condition

1 point is earned for the value of $y(2)$

1 point is earned for the answer with reason

4-2-2b

$$y(2) = -1 + \int_0^2 v(t) dt = -1.360 \text{ or } -1.361$$

The particle is moving away from the origin since $v(2) < 0$ and $y(2) < 0$.



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for: $\int_0^2 v(t) dt$

1 point is earned for the handles initial condition

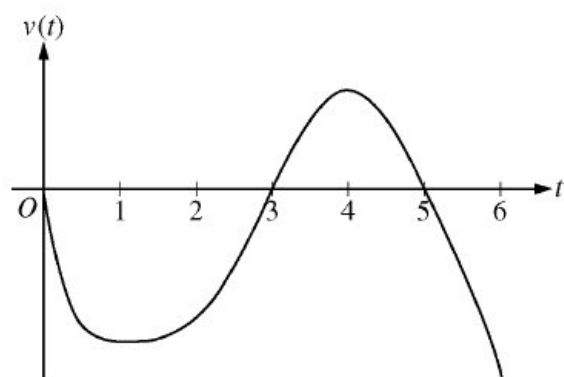
1 point is earned for the value of $y(2)$

1 point is earned for the answer with reason

$$y(2) = -1 + \int_0^2 v(t) dt = -1.360 \text{ or } -1.361$$

The particle is moving away from the origin since $v(2) < 0$ and $y(2) < 0$.

4-2-2b

Graph of v

A particle moves along the x -axis so that its velocity at time t , for $0 \leq t \leq 6$, is given by a differentiable function v whose graph is shown above. The velocity is 0 at $t = 0$, $t = 3$, and $t = 5$, and the graph has horizontal tangents at $t = 1$ and $t = 4$. The areas of the regions bounded by the t -axis and the graph of v on the intervals $[0,3]$, $[3,5]$, and $[5,6]$ are 8, 3, and 2 respectively. At time $t = 0$, the particle is $x = -2$.

22. During what time intervals, if any, is the acceleration of the particle negative? Justify your answer.

Part D

2 points can be earned.

1 point for answer.

1 point for justification.

The acceleration is negative on the intervals $0 < t < 1$ and $4 < t < 6$ since velocity is decreasing on these intervals.



0	1	2
---	---	---

The student response earns two of the following points:

2 points can be earned.

1 point for answer.

1 point for justification.

The acceleration is negative on the intervals $0 < t < 1$ and $4 < t < 6$ since velocity is decreasing on these intervals.

4-2-2b

23. On the interval $2 < t < 3$, is the speed of the particle increasing or decreasing? Give a reason for your answer.
-

Part C

1 point can be earned.

1 point for answer with reason.

The speed is decreasing on the interval $2 < t < 3$ since on the interval $v < 0$ and v is increasing .



0	1
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The student response earns one of the following points:

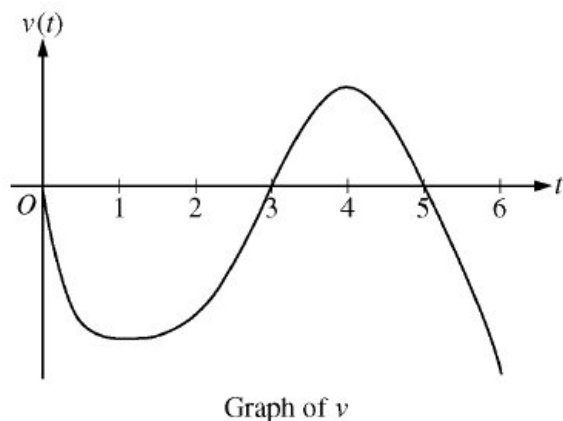
1 point can be earned.

1 point for answer with reason.

The speed is decreasing on the interval $2 < t < 3$ since on the interval $v < 0$ and v is increasing .

4-2-2b

24.



A particle moves along the x -axis so that its velocity at time t , for $0 \leq t \leq 6$, is given by a differentiable function v whose graph is shown above. The velocity is 0 at $t = 0$, $t = 3$, and $t = 5$, and the graph has horizontal tangents at $t = 1$ and $t = 4$. The areas of the regions bounded by the t -axis and the graph of v on the intervals $[0,3]$, $[3,5]$, and $[5,6]$ are 8, 3, and 2 respectively. At time $t = 0$, the particle is $x = -2$.

- (a) For $0 \leq t \leq 6$, find *both the time and the position of the particle when the particle is farthest to the left. Justify your answer.*
- (b) For how many values of t , where $0 \leq t \leq 6$, is the particle at $x = -8$? Explain your reasoning.
- (c) On the interval $2 < t < 3$, is the speed of the particle increasing or decreasing? Give a reason for your answer.
- (d) During what time intervals, if any, is the acceleration of the particle negative? Justify your answer.

Part A

1 point is earned for identifying $t = 3$ as a candidate

1 point is earned for considering $\int_0^6 v(t) dt$

1 point is earned for conclusion

Since $v(t) < 0$ for $0 < t < 3$ and $5 < t < 6$, and $v(t) > 0$ for $3 < t < 5$, we consider $t = 3$ and $t = 6$.

$$x(3) = -2 + \int_0^3 v(t) dt = -2 - 8 = -10$$

$$x(6) = -2 + \int_0^6 v(t) dt = -2 - 8 + 3 - 2 = -9$$

Therefore, the particle is farthest left at time $t = 3$ when its position is $x(3) = -10$.

4-2-2b



0	1	2	3
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The student response earns all of the following points:

1 point is earned for identifying $t = 3$ as a candidate

1 point is earned for considering $\int_0^6 v(t)dt$

1 point is earned for conclusion

Since $v(t) < 0$ for $0 < t < 3$ and $5 < t < 6$, and $v(t) > 0$ for $3 < t < 5$, we consider $t = 3$ and $t = 6$.

$$x(3) = -2 + \int_0^3 v(t)dt = -2 - 8 = -10$$

$$x(6) = -2 + \int_0^6 v(t)dt = -2 - 8 + 3 - 2 = -9$$

Therefore, the particle is farthest left at time $t = 3$ when its position is $x(3) = -10$.

Part B

1 point for positioning at $t = 3$, $t = 5$, and $t = 6$

1 point for description of motion

1 point for conclusion

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.

By the Intermediate Value Theorem, there are three values of t for which the particle is at $x(t) = -8$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point for positioning at $t = 3$, $t = 5$, and $t = 6$

1 point for description of motion

1 point for conclusion

4-2-2b

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.

By the Intermediate Value Theorem, there are three values of t for which the particle is at $x(t) = -8$.

Part C

1 point is earned for answer with reason

The speed is decreasing on the interval $2 < t < 3$ since on this interval $v < 0$ and v is increasing.



0	1
---	---

The student response earns all of the following points:

1 point is earned for answer with reason

The speed is decreasing on the interval $2 < t < 3$ since on this interval $v < 0$ and v is increasing.

Part D

1 point for answer

1 point for justification

The acceleration is negative on the intervals $0 < t < 1$ and $4 < t < 6$ since velocity is decreasing on these intervals.



0	1	2
---	---	---

The student response earns all of the following points:

1 point for answer

1 point for justification

The acceleration is negative on the intervals $0 < t < 1$ and $4 < t < 6$ since velocity is decreasing on these intervals.

4-2-2b

25. A particle moves along the x -axis in such a way that its acceleration at time t for $t \geq 0$ is given by $a(t) = 4\cos(2t)$. At time $t = 0$, the velocity of the particle is $v(0) = 1$ and its position is $x(0) = 0$.

For what values of t , $0 \leq t \leq \pi$, is the particle at rest?

Part C

1 point is earned if sets $v(t) = 0$

2 point are earned for solving for t

$$2\sin 2t + 1 = 0$$

$$\sin 2t = -\frac{1}{2}$$

$$2t = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$t = \frac{7\pi}{12}, \frac{11\pi}{12}$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned if sets $v(t) = 0$

2 point are earned for solving for t

$$2\sin 2t + 1 = 0$$

$$\sin 2t = -\frac{1}{2}$$

$$2t = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$t = \frac{7\pi}{12}, \frac{11\pi}{12}$$

4-2-2b

26. A particle moves along the x -axis in such a way that its acceleration at time t for $t \geq 0$ is given by $a(t) = 4\cos(2t)$. At time $t = 0$, the velocity of the particle is $v(0) = 1$ and its position is $x(0) = 0$.

- (a) Write an equation for the velocity $v(t)$ of the particle.
 (b) Write an equation for the position $x(t)$ of the particle.
 (c) For what values of t , $0 \leq t \leq \pi$, is the particle at rest?

Part A

1 point is earned for integrand

1 point is earned for $v(t) = 2\sin 2t + C$ or $v(t) = 2\sin 2t$

1 point is earned for answer

$$\begin{aligned} v(t) &= \int 4 \cos 2t \, dt \\ \therefore v(t) &= 2 \sin 2t + C \\ v(0) &= 1 \Rightarrow C = 1 \\ \therefore v(t) &= 2 \sin 2t + 1 \end{aligned}$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for $v(t) = 2\sin 2t + C$ or $v(t) = 2\sin 2t$

1 point is earned for answer

$$\begin{aligned} v(t) &= \int 4 \cos 2t \, dt \\ \therefore v(t) &= 2 \sin 2t + C \\ v(0) &= 1 \Rightarrow C = 1 \\ \therefore v(t) &= 2 \sin 2t + 1 \end{aligned}$$

Part B

1 point is earned for integrand

1 point is earned for $x(t) = -\cos 2t + t + C$ or $x(t) = -\cos 2t + t$

1 point is earned for answer

4-2-2b

$$x(t) = \int 2\sin 2t + 1 \, dt$$

$$\therefore x(t) = -\cos 2t + t + C$$

$$x(0) = 0 \Rightarrow C = 1$$

$$\therefore x(t) = -\cos 2t + t + 1$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for $x(t) = -\cos 2t + t + C$ or $x(t) = -\cos 2t + t$

1 point is earned for answer

$$x(t) = \int 2\sin 2t + 1 \, dt$$

$$\therefore x(t) = -\cos 2t + t + C$$

$$x(0) = 0 \Rightarrow C = 1$$

$$\therefore x(t) = -\cos 2t + t + 1$$

Part C

1 point is earned if sets $v(t) = 0$

2 point are earned for solving for t

$$2 \sin 2t + 1 = 0$$

$$\sin 2t = -\frac{1}{2}$$

$$\therefore 2t = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$\therefore t = \frac{7\pi}{12}, \frac{11\pi}{12}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

4-2-2b

1 point is earned if sets $v(t) = 0$

2 point are earned for solving for t

$$2 \sin 2t + 1 = 0$$

$$\sin 2t = -\frac{1}{2}$$

$$\therefore 2t = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$\therefore t = \frac{7\pi}{12}, \frac{11\pi}{12}$$

27.  A particle moves on the x -axis with velocity given by $v(t) = 3t^4 - 11t^2 + 9t - 2$ for $-3 \leq t \leq 3$. How many times does the particle change direction as t increases from -3 to 3 ?

(A) Zero

(B) One

(C) Two



(D) Three

(E) Four

28. A particle moves along the x -axis with its position at time t given by $x(t) = (t - a)(t - b)$ where a and b are constants and $a \neq b$. For which of the following values of t is the particle at rest?

(A) $t = ab$

(B) $t = \frac{a+b}{2}$



(C) $t = a + b$

(D) $t = 2(a + b)$

(E) $t = a$ and $t = b$

29. A particle moves on the x -axis so that at any time $t, 0 \leq t \leq 1$ its position is given by $x(t) = \sin(2\pi t) + 2\pi t$. For what value of t is the particle at rest?

(A) 0

(B) $\frac{1}{8}$

(C) $\frac{1}{4}$

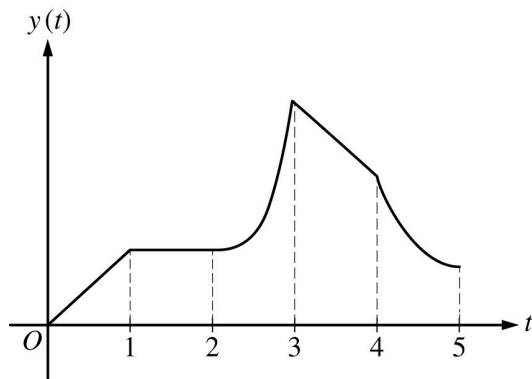
(D) $\frac{1}{2}$



(E) 1

4-2-2b

30.



A particle moves along the y -axis. The graph of the particle's position $y(t)$ at time t is shown above for $0 \leq t \leq 5$. For what values of t is the velocity of the particle negative and the acceleration positive?

- (A) $0 < t < 1$
(B) $1 < t < 2$
(C) $2 < t < 3$
(D) $3 < t < 4$
(E) $4 < t < 5$



31. A particle moves along the x -axis so that at time $t \geq 0$ its position is given by $x(t) = 2t^3 - 9t^2 - 60t + 4$. What is the total distance traveled by the particle over the time interval $0 \leq t \leq 7$?

- (A) 168
(B) 171
(C) 175
(D) 375

**Answer D**

Correct. The velocity of the particle is given by $x'(t) = 6t^2 - 18t - 60 = 6(t - 5)(t + 2)$. This is negative for $0 \leq t < 5$ and positive for $5 < t \leq 7$. So the particle is moving left over the time interval $0 \leq t < 5$ and moving right over the time interval $5 < t \leq 7$. Therefore, the total distance traveled by the particle over the time interval $0 \leq t \leq 7$ is given by $(x(0) - x(5)) + (x(7) - x(5)) = (4 - (-271)) + (-171 - (-271)) = 375$.

4-2-2b

32. A particle moves along the y -axis so that at time $t \geq 0$ its position is given by $y(t) = \frac{2}{3}t^3 - 5t^2 + 8t$. Over the time interval $0 < t < 5$, for what values of t is the speed of the particle increasing?

- (A) $2.5 < t < 5$
(B) $4 < t < 5$ only
(C) $0 < t < 1$ and $4 < t < 5$
(D) $1 < t < 2.5$ and $4 < t < 5$

**Answer D**

Correct. The velocity of the particle is given by

$$y'(t) = 2t^2 - 10t + 8 = 2(t - 1)(t - 4),$$

and the acceleration of the particle is given by $y''(t) = 4t - 10 = 4(t - 2.5)$. The speed of the particle is increasing when the velocity and acceleration have the same sign, which occurs for $1 < t < 2.5$ and $4 < t < 5$.

33. An object moves along a straight line so that at any time t , for $0 \leq t \leq 8$, its position is given by $x(t) = 5 + 4t - t^2$. For what value of t is the object at rest?

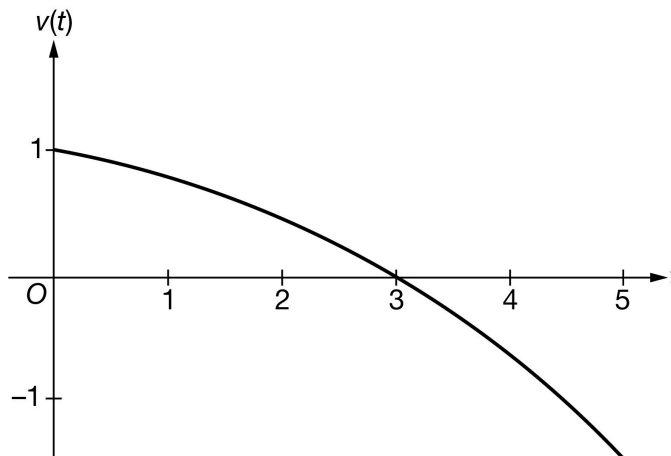
- (A) $t = 2$
(B) $t = 4$
(C) $t = \frac{9}{2}$
(D) $t = 5$

**Answer A**

Correct. The object is at rest when its velocity equals 0. Because the object's velocity is given by $x'(t) = 4 - 2t$, it is at rest at $t = 2$.

4-2-2b

34.



A particle traveling on the x -axis has position $x(t)$ at time t . The graph of the particle's velocity $v(t)$ is shown above for $0 \leq t \leq 5$. Which of the following expressions gives the total distance traveled by the particle over the time interval $0 \leq t \leq 5$?

- (A) $x(0) - x(5)$
(B) $x(5) - x(0)$
(C) $(x(3) - x(0)) + (x(3) - x(5))$ ✓
(D) $(x(0) - x(3)) + (x(5) - x(3))$

Answer C

Correct. Since the particle's velocity is positive for $0 \leq t < 3$ and negative for $3 < t \leq 5$, the particle moves to the right for $0 \leq t < 3$ and to the left for $3 < t \leq 5$. Therefore, the particle moves $x(3) - x(0)$ units to the right and then $x(3) - x(5)$ units to the left.

35. A particle moves along the x -axis so that at any time $t \geq 0$ its position is given by $x(t) = \frac{1}{2}(a - t)^2$, where a is a positive constant. For what values of t is the particle moving to the right?
- (A) The particle is moving to the right only if $0 < t < a$.
(B) The particle is moving to the right only if $a < t$. ✓
(C) The particle is moving to the right only if $t = a$.
(D) The particle is not moving to the right.

Answer B

Correct. The velocity of the particle is given by $x'(t) = t - a$. Since $x'(t)$ is positive for $a < t$ and negative for $0 < t < a$, the particle is moving to the right for $a < t$ and moving to the left for $0 < t < a$. Therefore, the particle is moving to the right only if $a < t$.

4-2-2b

36.  The velocity of a particle moving along a straight line is given by $v(t) = 1.3t \ln(0.2t + 0.4)$ for time $t \geq 0$. What is the acceleration of the particle at time $t = 1.2$?
- (A) -0.580
(B) -0.548
(C) -0.093 ✓
(D) 0.660

Answer C

$$v'(1.2)$$

37. For time $t \geq 0$, the position of an object moving along the x -axis is given by $x(t) = \frac{1}{3}t^3 - 2t^2 + 3t + 1$. Which of the following statements about the object is true at time $t = 0.5$?
- (A) The object is moving to the right and slowing down. ✓
(B) The object is moving to the right and speeding up.
(C) The object is moving to the left and slowing down.
(D) The object is moving to the left and speeding up.

Answer A

Correct. This statement is true. The object's velocity is $v(t) = x'(t) = t^2 - 4t + 3$ and the object's acceleration is $a(t) = v'(t) = 2t - 4$. At time $t = 0.5$, $v(0.5) = 0.25 - 2 + 3 = 1.25$ and $a(0.5) = 1 - 4 = -3$. Because $v(0.5)$ is positive, the object is moving to the right, and because $v(0.5)$ and $a(0.5)$ have opposite signs, the object is slowing down.

4-2-2b

38.  Two particles move along the x -axis with velocities given by $v_1(t) = -3t^2 + 8t + 5$ and $v_2(t) = \sin t$ for time $t \geq 0$. At what time t do the two particles have equal acceleration?

(A) 1.287



(B) 3.194

(C) 5.010

(D) There is no such time t .

Answer A

Correct. The accelerations of the two particles are $a_1(t) = v_1'(t) = -6t + 8$ and $a_2(t) = v_2'(t) = \cos t$. Using the calculator to solve the equation $-6t + 8 = \cos t$ yields the solution $t = 1.287$.

4-2-2b

39. A particle moves along the x -axis so that its velocity v at any time t , for $0 \leq t \leq 16$ is given by $v(t) = e^{2\sin t} - 1$. At time $t = 0$, the particle is at the origin.



During what intervals of time is the particle moving to the left? Give a reason for your answer.

Part B

2 points are earned for the intervals

< -1 > each missing or incorrect interval

1 point is earned for the reason

Particle is moving to the left when

$$v(t) < 0, \text{ i.e., } e^{2\sin t} < 1.$$

$$(\pi, 2\pi), (3\pi, 4\pi) \text{ and } (5\pi, 16)$$



0	1	2	3
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The student response earns all of the following points:

2 points are earned for the intervals

< -1 > each missing or incorrect interval

1 point is earned for the reason

Particle is moving to the left when

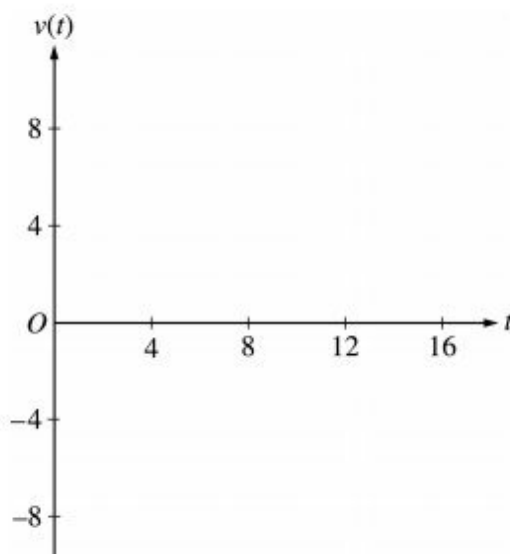
$$v(t) < 0, \text{ i.e., } e^{2\sin t} < 1.$$

$$(\pi, 2\pi), (3\pi, 4\pi) \text{ and } (5\pi, 16)$$

4-2-2b

40. A particle moves along the x -axis so that its velocity v at any time t , for $0 \leq t \leq 16$, is given by $v(t) = e^2 \sin t - 1$. At time $t = 0$, the particle is at the origin.

 (a) On the axes provided, sketch the graph of $v(t)$ for $0 \leq t \leq 16$.



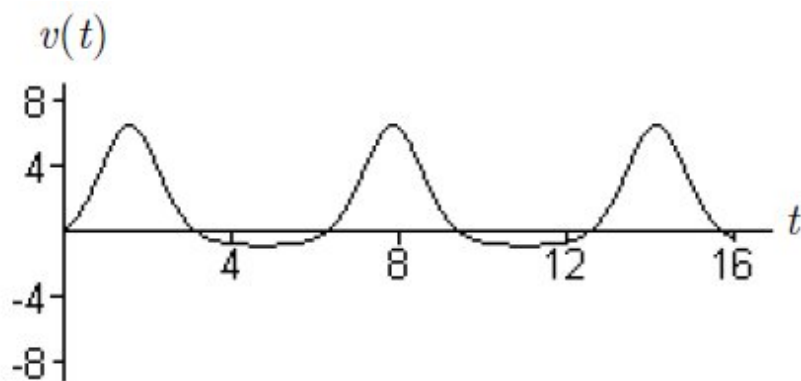
(b) During what intervals of time is the particle moving to the left? Give a reason for your answer.

(c) Find the total distance traveled by the particle from $t = 0$ to $t = 4$.

(d) Is there any time t , $0 < t \leq 16$, at which the particle returns to the origin? Justify your answer.

Part A

1 point is earned for the graph three "humps", periodic behavior, starts at origin, reasonable relative max and min values



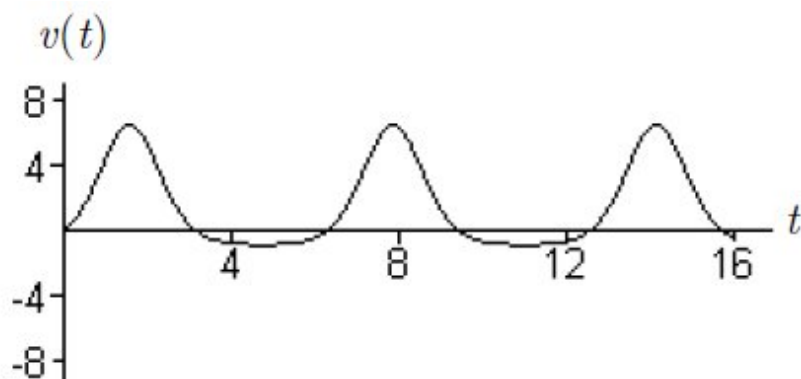
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The student response earns all of the following points:

4-2-2b

1 point is earned for the graph three "humps", periodic behavior, starts at origin, reasonable relative max and min values



Part B

2 points are earned for the intervals

<-1> each missing or incorrect interval

1 point is earned for the reason

Particle is moving to the left when

$$v(t) < 0, \text{ i.e. } e^{2\sin t} < 1.$$

$$(\pi, 2\pi), (3\pi, 4\pi) \text{ and } (5\pi, 16]$$



0	1	2	3
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The student response earns all of the following points:

2 points are earned for the intervals

<-1> each missing or incorrect interval

1 point is earned for the reason

Particle is moving to the left when

$$v(t) < 0, \text{ i.e. } e^{2\sin t} < 1.$$

$$(\pi, 2\pi), (3\pi, 4\pi) \text{ and } (5\pi, 16]$$

4-2-2b

Part C

1 point is earned for the limits of 0 and 4 on an integral of $v(t)$ or $|v(t)|$

or

uses $x(0)$ and $x(4)$ to compute distance

1 point is earned for the handles change of direction at student's turning point

1 point is earned for the answer

$$\int_0^4 |v(t)| dt = 10.542$$

or

$$v(t) = e^{2 \sin t} - 1 = 0$$

$$t = 0 \text{ or } t = \pi$$

$$x(\pi) = \int_0^{\pi} v(t) dt = 10.10656$$

$$x(4) = \int_0^4 v(t) dt = 9.67066$$

$$|x(\pi) - x(0)| + |x(4) - x(\pi)| = 10.542$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the limits of 0 and 4 on an integral of $v(t)$ or $|v(t)|$

or

uses $x(0)$ and $x(4)$ to compute distance

1 point is earned for the handles change of direction at student's turning point

1 point is earned for the answer

4-2-2b

$$\int_0^4 |v(t)| dt = 10.542$$

or

$$v(t) = e^{2 \sin t} - 1 = 0$$

$$t = 0 \text{ or } t = \pi$$

$$x(\pi) = \int_0^{\pi} v(t) dt = 10.10656$$

$$x(4) = \int_0^4 v(t) dt = 9.67066$$

$$|x(\pi) - x(0)| + |x(4) - x(\pi)| = 10.542$$

Part D

1 point is earned for no such time

1 point is earned for the reason

There is no such time because $\int_0^T v(t) dt > 0$ for all $T > 0$.



0	1	2
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The student response earns all of the following points:

1 point is earned for no such time

1 point is earned for the reason

There is no such time because $\int_0^T v(t) dt > 0$ for all $T > 0$.

4-2-2b

41.  A particle is moving along the x -axis with a velocity at time t seconds given by $v(t) = 0.85t - 0.3e^{(4-t)}$. What is the acceleration of the particle when it is at rest?

- (A) -16.379
(B) -12.508
(C) 2.703
(D) 17.229

**Answer C**

Correct. The particle is at rest when $v(t) = 0$. Since $v'(t) = 0.85 + 0.3e^{4-t} > 0$ for all t , the graph of $v(t)$ is increasing and therefore the particle is at rest for at most one value of t , which is found with the graphing calculator to be at $t = 2.179466$. The acceleration at this moment is $a(2.179466) = v'(2.179466) = 2.703$. The graphing calculator is used to compute the derivative value.

4-2-2b

42. A particle moves along the x -axis with position at time t given by $x(t) = e^{-t} \sin t$ for $0 \leq t \leq 2\pi$. Find the time t at which the particle is farthest to the left. Justify your answer.

Part A

2 points are earned for: $x'(t)$

1 point is earned for sets $x'(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

$$x'(t) = -e^{-t} \sin t + e^{-t} \cos t = -e^{-t} (\cos t - \sin t)$$

$x'(t) = 0$, when, $\cos t = \sin t$. Therefore, $x'(t) = 0$, on

$$0 \leq t \leq 2\pi, \text{ for, } t = \frac{\pi}{4} \text{ and, } t = \frac{5\pi}{4}$$

The candidates for the absolute minimum are at

$$t = 0, \frac{\pi}{4}, \frac{5\pi}{4} \text{ and, } 2\pi.$$

t	$x(t)$
0	$e^0 \sin(0) = 0$
$\frac{\pi}{4}$	$e^{-\frac{\pi}{4}} \sin\left(\frac{\pi}{4}\right) > 0$
$\frac{5\pi}{4}$	$e^{-\frac{5\pi}{4}} \sin\left(\frac{5\pi}{4}\right) < 0$
2π	$e^{-2\pi} \sin(2\pi) = 0$

The particle is farthest to the left when $t = \frac{5\pi}{4}$.



0	1	2	3	4	5
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The student response earns all of the following points:

2 points are earned for: $x'(t)$

1 point is earned for sets $x'(t) = 0$

4-2-2b

1 point is earned for the answer

1 point is earned for the justification

$$x'(t) = -e^{-t} \sin t + e^{-t} \cos t = -e^{-t} (\cos t - \sin t)$$

$$x'(t) = 0, \text{ when, } \cos t = \sin t. \text{ Therefore, } x'(t) = 0, \text{ on}$$

$$0 \leq t \leq 2\pi, \text{ for, } t = \frac{\pi}{4} \text{ and, } t = \frac{5\pi}{4}$$

The candidates for the absolute minimum are at

$$t = 0, \frac{\pi}{4}, \frac{5\pi}{4} \text{ and, } 2\pi.$$

t	$x(t)$
0	$e^0 \sin(0) = 0$
$\frac{\pi}{4}$	$e^{-\frac{\pi}{4}} \sin\left(\frac{\pi}{4}\right) > 0$
$\frac{5\pi}{4}$	$e^{-\frac{5\pi}{4}} \sin\left(\frac{5\pi}{4}\right) < 0$
2π	$e^{-2\pi} \sin(2\pi) = 0$

The particle is farthest to the left when $t = \frac{5\pi}{4}$.

4-2-2b

43. A particle moves along the x -axis with position at time t given by $x(t) = e^{-t} \sin t$ for $0 \leq t \leq 2\pi$.

(a) Find the time t at which the particle is farthest to the left. Justify your answer.

(b) Find the value of the constant A for which $x(t)$ satisfies the equation $Ax''(t) + x'(t) + x(t) = 0$ for $0 < t < 2\pi$.

Part A

2 points are earned for $x'(t)$

1 point is earned for setting $x'(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

$$x'(t) = -e^{-t} \sin t + e^{-t} \cos t = e^{-t} (\cos t - \sin t)$$

$x'(t) = 0$ when $\cos t = \sin t$. Therefore, $x'(t) = 0$ on

$$0 \leq t \leq 2\pi \text{ for } t = \frac{\pi}{4} \text{ and } t = \frac{5\pi}{4}.$$

The candidates for the absolute minimum are at $t = 0, \frac{\pi}{4}, \frac{5\pi}{4}$, and 2π .

t	$x(t)$
0	$e^0 \sin(0) = 0$
$\frac{\pi}{4}$	$e^{-\frac{\pi}{4}} \sin\left(\frac{\pi}{4}\right) > 0$
$\frac{5\pi}{4}$	$e^{-\frac{5\pi}{4}} \sin\left(\frac{5\pi}{4}\right) < 0$
2π	$e^{-2\pi} \sin(2\pi) = 0$

The particle is farthest to the left when $t = \frac{5\pi}{4}$.



0	1	2	3	4	5
---	---	---	---	---	---

4-2-2b

The student response earns all of the following points:

2 points are earned for $x'(t)$

1 point is earned for setting $x'(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

$$x'(t) = -e^{-t} \sin t + e^{-t} \cos t = e^{-t} (\cos t - \sin t)$$

$x'(t) = 0$ when $\cos t = \sin t$. Therefore, $x'(t) = 0$ on

$$0 \leq t \leq 2\pi \text{ for } t = \frac{\pi}{4} \text{ and } t = \frac{5\pi}{4}.$$

The candidates for the absolute minimum are at $t = 0, \frac{\pi}{4}, \frac{5\pi}{4}$, and 2π .

t	$x(t)$
0	$e^0 \sin(0) = 0$
$\frac{\pi}{4}$	$e^{-\frac{\pi}{4}} \sin\left(\frac{\pi}{4}\right) > 0$
$\frac{5\pi}{4}$	$e^{-\frac{5\pi}{4}} \sin\left(\frac{5\pi}{4}\right) < 0$
2π	$e^{-2\pi} \sin(2\pi) = 0$

The particle is farthest to the left when $t = \frac{5\pi}{4}$.

Part B

2 points are earned for $x''(t)$

1 point is earned for substituting $x''(t)$, $x'(t)$, and $x(t)$ into $Ax''(t) + x'(t) + x(t)$

1 point is earned for the answer

4-2-2b

$$\begin{aligned}
 x''(t) &= -e^{-t}(\cos t - \sin t) + e^{-t}(\sin t - \cos t) \\
 &= -2e^{-t} \cos t
 \end{aligned}$$

$$\begin{aligned}
 Ax''(t) + x'(t) + x(t) &= A(-2e^{-t} \cos t) + e^{-t}(\cos t - \sin t) + e^{-t} \sin t \\
 &= (-2A + 1)e^{-t} \cos t \\
 &= 0
 \end{aligned}$$

Therefore, $A = \frac{1}{2}$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for $x''(t)$

1 point is earned for substituting $x''(t)$, $x'(t)$, and $x(t)$ into $Ax''(t) + x'(t) + x(t)$

1 point is earned for the answer

$$\begin{aligned}
 x''(t) &= -e^{-t}(\cos t - \sin t) + e^{-t}(\sin t - \cos t) \\
 &= -2e^{-t} \cos t
 \end{aligned}$$

$$\begin{aligned}
 Ax''(t) + x'(t) + x(t) &= A(-2e^{-t} \cos t) + e^{-t}(\cos t - \sin t) + e^{-t} \sin t \\
 &= (-2A + 1)e^{-t} \cos t \\
 &= 0
 \end{aligned}$$

Therefore, $A = \frac{1}{2}$.

4-2-2b

44. A particle moves along the x -axis with velocity given by $v(t) = \frac{10 \sin(0.4t^2)}{t^2 - t + 3}$ for time $0 \leq t \leq 3.5$. The particle is at position $x = -5$ at time $t = 0$.



A second particle moves along the x -axis with position given by $x_2(t) = t^2 - t$ for $0 \leq t \leq 3.5$. At what time t are the two particles moving with the same velocity? -

Part D

1 point is earned for: sets $v(t) = x_2'(t)$

1 point is earned for: answer

$$v(t) = x_2'(t)$$

$$v(t) = 2t - 1 \Rightarrow t = 1.57054$$

The two particles are moving with the same velocity at time $t = 1.571$ (or 1570)



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for: sets $v(t) = x_2'(t)$

1 point is earned for: answer

$$v(t) = x_2'(t)$$

$$v(t) = 2t - 1 \Rightarrow t = 1.57054$$

The two particles are moving with the same velocity at time $t = 1.571$ (or 1570)

4-2-2b

45.  A particle moves along the x -axis with velocity given by $v(t) = \frac{10 \sin(0.4t^2)}{t^2 - t + 3}$ for time $0 \leq t \leq 3.5$. The particle is at position $x = -5$ at time $t = 0$.
- (a) Find the acceleration of the particle at time $t = 3$.
- (b) Find the position of the particle at time $t = 3$.
- (c) Evaluate $\int_0^{3.5} v(t) dt$, and evaluate $\int_0^{3.5} |v(t)| dt$. Interpret the meaning of each integral in the context of the problem.
- (d) A second particle moves along the x -axis with position given by $x_2(t) = t^2 - t$ for $0 \leq t \leq 3.5$. At what time t are the two particles moving with the same velocity?

Part A

1 point is earned for answer

$$v'(3) = -2.118$$

The acceleration of the particle at time $t = 3$ is -2.118 .



0	1
---	---

The student earns all of the following points:

1 point is earned for answer

$$v'(3) = -2.118$$

The acceleration of the particle at time $t = 3$ is -2.118 .

Part B

1 point is earned for $\int_0^3 v(t) dt$

1 point is earned for using initial condition

1 point is earned for the answer

$$x(3) = x(0) + \int_0^3 v(t) dt = -5 + \int_0^3 v(t) dt = -1.760213$$

The position of the particle at time $t = 3$ is -1.760 .

4-2-2b



0	1	2	3
---	---	---	---

The student earns all of the following point:

1 point is earned for $\int_0^3 v(t) dt$

1 point is earned for using initial condition

1 point is earned for the answer

$$x(3) = x(0) + \int_0^3 v(t) dt = -5 + \int_0^3 v(t) dt = -1.760213$$

The position of the particle at time $t = 3$ is -1.760 .

Part C

1 point is earned for the answers

2 points are earned for the interpretations of $\int_0^{3.5} v(t) dt$ and $\int_0^{3.5} |v(t)| dt$

$$\int_0^{3.5} v(t) dt = 2.844 \text{ (or } 2.843\text{)}$$

$$\int_0^{3.5} |v(t)| dt = 3.737$$

The integral $\int_0^{3.5} v(t) dt$ is the displacement of the particle over the time interval $0 \leq t \leq 3.5$.

The integral $\int_0^{3.5} |v(t)| dt$ is the total distance traveled by the particle over the time interval $0 \leq t \leq 3.5$.

4-2-2b



0	1	2	3
---	---	---	---

The student earns all of the following point:

1 point is earned for the answers

2 points are earned for the interpretations of $\int_0^{3.5} v(t) \, dt$ and $\int_0^{3.5} |v(t)| \, dt$

$$\int_0^{3.5} v(t) \, dt = 2.844 \text{ (or 2.843)}$$

$$\int_0^{3.5} |v(t)| \, dt = 3.737$$

The integral $\int_0^{3.5} v(t) \, dt$ is the displacement of the particle over the time interval $0 \leq t \leq 3.5$.

The integral $\int_0^{3.5} |v(t)| \, dt$ is the total distance traveled by the particle over the time interval $0 \leq t \leq 3.5$.

Part D

1 point is earned if sets $v(t) = x_2'(t)$

1 point is earned for the answer

$$v(t) = x_2'(t)$$

$$v(t) = 2t - 1 \Rightarrow t = 1.57054$$

The two particles are moving with the same velocity at time $t = 1.571$ (or 1570).

4-2-2b



0	1	2
---	---	---

The student earns all of the following point:

1 point is earned if sets $v(t) = x_2'(t)$

1 point is earned for the answer

$$v(t) = x_2'(t)$$

$$v(t) = 2t - 1 \Rightarrow t = 1.57054$$

The two particles are moving with the same velocity at time $t = 1.571$ (or 1570).

4-2-2b

A particle moves on the x -axis so that its velocity at any time $t \geq 0$ is given by $v(t) = 12t^2 - 36t + 15$. At $t = 1$, the particle is at the origin.

46. Find the maximum velocity of the particle for $0 \leq t \leq 2$.

Part C

1 point is earned for evaluating $v(t)$ at both endpoints

1 point is earned for correct velocity value

$$\frac{dv}{dt} = 24t - 36$$

$$\frac{dv}{dt} = 0 \text{ when } t = \frac{3}{2}$$

$$v(0) = 15$$

$$v\left(\frac{3}{2}\right) = -12$$

$$v(2) = -9$$

Maximum velocity is 15



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for evaluating $v(t)$ at both endpoints

1 point is earned for correct velocity value

$$\frac{dv}{dt} = 24t - 36$$

$$\frac{dv}{dt} = 0 \text{ when } t = \frac{3}{2}$$

$$v(0) = 15$$

$$v\left(\frac{3}{2}\right) = -12$$

$$v(2) = -9$$

Maximum velocity is 15

4-2-2b

47. Find all values of t for which the particle is at rest.
-

Part B

1 point is earned for setting $12t^2 - 36t + 15 = 0$

1 point is earned for answer

$$0 = v(t) = 12t^2 - 36t + 15 = 0$$

$$3(2t-1)(2t-5)=0$$

$$t = \frac{1}{2}, \frac{5}{2}$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for setting $12t^2 - 36t + 15 = 0$

1 point is earned for answer

$$0 = v(t) = 12t^2 - 36t + 15 = 0$$

$$3(2t-1)(2t-5)=0$$

$$t = \frac{1}{2}, \frac{5}{2}$$

4-2-2b

48. A particle moves on the x -axis so that its velocity at any time $t \geq 0$ is given by $v(t) = 12t^2 - 36t + 15$. At $t = 1$, the particle is at the origin.

- (a) Find the position $x(t)$ of the particle at any time $t \geq 0$.
 (b) Find all values of t for which the particle is at rest.
 (c) Find the maximum velocity of the particle for $0 \leq t \leq 2$.

Find the total distance traveled by the particle from $t = 0$ to $t = 2$.

Part A

2 points are earned for the antiderivative

1 point is earned for evaluating with constant

$$x(t) = 4t^3 - 18t^2 + 15t + C$$

$$0 = x(1) = 4 - 18 + 15 + C$$

$$\therefore C = -1$$

$$x(t) = 4t^3 - 18t^2 + 15t - 1$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the antiderivative

1 point is earned for evaluating with constant

$$x(t) = 4t^3 - 18t^2 + 15t + C$$

$$0 = x(1) = 4 - 18 + 15 + C$$

$$\therefore C = -1$$

$$x(t) = 4t^3 - 18t^2 + 15t - 1$$

Part B

4-2-2b

1 point is earned for setting $12t^2 - 36t + 15 = 0$

1 point is earned for answer

$$0 = v(t) = 12t^2 - 36t + 15$$

$$3(2t - 1)(2t - 5) = 0$$

$$t = \frac{1}{2}, \frac{5}{2}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for setting $12t^2 - 36t + 15 = 0$

1 point is earned for answer

$$0 = v(t) = 12t^2 - 36t + 15$$

$$3(2t - 1)(2t - 5) = 0$$

$$t = \frac{1}{2}, \frac{5}{2}$$

Part C

1 point is earned for evaluating $v(t)$ at both endpoints

1 point is earned for correct velocity value

4-2-2b

$$\frac{dv}{dt} = 24t - 36$$

$$\frac{dv}{dt} = 0 \text{ when } t = \frac{3}{2}$$

$$v(0) = 15$$

$$v\left(\frac{3}{2}\right) = -12$$

$$v(2) = -9$$

Maximum velocity is 15



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for evaluating $v(t)$ at both endpoints

1 point is earned for correct velocity value

$$\frac{dv}{dt} = 24t - 36$$

$$\frac{dv}{dt} = 0 \text{ when } t = \frac{3}{2}$$

$$v(0) = 15$$

$$v\left(\frac{3}{2}\right) = -12$$

$$v(2) = -9$$

Maximum velocity is 15

Part D

1 point is earned for student's $x(t)$ evaluated at endpoints and turning points

1 point is earned for answer

$$\begin{aligned}
 \text{Total Distance} &= \int_0^{1/2} v(t)dt - \int_{1/2}^2 v(t)dt \\
 &= \left(x\left(\frac{1}{2}\right) - x(0)\right) - \left(x(2) - x\left(\frac{1}{2}\right)\right) \\
 &= \frac{5}{2} - (-1) - \left(-11 - \frac{5}{2}\right) = \boxed{17}
 \end{aligned}$$

4-2-2b

✓

0	1	2
---	---	---

The student response earns all of the following points:

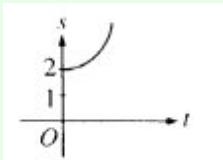
1 point is earned for student's $x(t)$ evaluated at endpoints and turning points

1 point is earned for answer

$$\begin{aligned}
 \text{Total Distance} &= \int_0^{1/2} v(t) dt - \int_{1/2}^2 v(t) dt \\
 &= \left(x\left(\frac{1}{2}\right) - x(0) \right) - \left(x(2) - x\left(\frac{1}{2}\right) \right) \\
 &= \frac{5}{2} - (-1) - \left(-11 - \frac{5}{2} \right) = \boxed{17}
 \end{aligned}$$

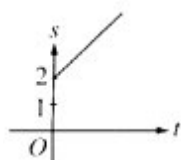
49. A particle starts from rest at the point $(2,0)$ and moves along the x -axis with a constant positive acceleration for time $t \geq 0$. Which of the following could be the graph of the distance $s(t)$ of the particle from the origin as a function of time t ?

(A)

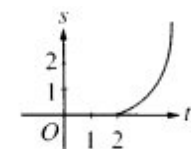


✓

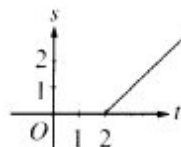
(B)



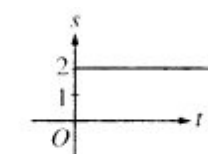
(C)



(D)



(E)



4-2-2b

50. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	6	12
$x'(t)$ (meters per second)	5.0	5.8	4.0

The position of a particle moving along a straight line is modeled by a twice-differentiable function $x(t)$, where time t is measured in seconds and $x(t)$ is measured in meters. Selected values of $x'(t)$, the derivative of $x(t)$, are shown in the table above. The position of the particle at time $t = 12$ is $x(12) = 8$.

(a) Approximate $x''(9)$ using the average rate of change of $x'(t)$ over the interval $6 \leq t \leq 12$. Show the computations that lead to your answer.

(b) Using correct units, explain the meaning of $x''(9)$ in the context of the problem.

(c) Use a right Riemann sum with the two subintervals indicated by the data in the table to approximate $\int_0^{12} |x'(t)| \, dt$. Show the computations that lead to your answer.

(d) Using correct units, explain the meaning of $\int_0^{12} |x'(t)| \, dt$ in the context of the problem.

(e) Use Euler's method, starting at $t = 12$ with two steps of equal size, to approximate the position of the particle at time $t = 0$. Show the computations that lead to your answer.

(f) Let s be a function such that the third derivative of s with respect to t is $(t - 3)^7$. Write the fourth-degree term of the fourth-degree Taylor polynomial for s about $t = 1$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The computations shown should include a quotient with at least one subtraction (numerator or denominator).

This point is earned only by a correct numerical value equivalent to -0.3 .

4-2-2b



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$x''(9) \approx \frac{x'(12) - x'(6)}{12 - 6} = \frac{4.0 - 5.8}{6} = -0.3$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Other acceptable interpretations include:

- $x''(9)$ is the acceleration of the particle at $t = 9$.
- $x''(9)$ is the rate of change of the particle's velocity at $t = 9$.
- The particle is decelerating at a rate of 0.3 meters per second per second at $t = 9$.
- The velocity of the particle is decreasing at a rate of 0.3 meters per second per second at $t = 9$.

Interpretations that will not earn the point include:

- $x''(9)$ is the rate of change of speed at $t = 9$.
- $x''(9)$ means the particle is slowing down at $t = 9$. (We cannot be certain the particle's velocity is positive at $t = 9$.)
- The particle is decelerating at a rate of -0.3 meters per second per second at $t = 9$. (Decelerating at a negative rate is not correct.)
- The velocity of the particle is decreasing at a rate of -0.3 meters per second per second at $t = 9$. (Decreasing at a negative rate is not correct.)

Units must include the rate of change units, but do not need to include the t (time) units. However, if t units are given, they must be correct.

Units appearing in part (a) do not need to be repeated in part (b).

Incorrect units written in part (b) will override any units presented in part (a).

We may read " $x = 9$ " to mean $t = 9$.

4-2-2b



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Interpretation
- ☐ Units

Solution:

$x''(9)$ is the rate of change of the rate of change of the position of the particle, in meters per second per second, at time $t = 9$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The computations shown must include sum of two explicit products. For example, $5.8 \cdot 6 + 4 \cdot 6$ or $(5.8 + 4) \cdot 6$.

Labeled areas on a graph are not enough without both the sum and explicit products.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$\int_0^{12} |x'(t)| dt \approx |x'(6)|(6-0) + |x'(12)|(12-6)$$

$$= 5.8 \cdot 6 + 4 \cdot 6 = 58.8$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The interpretation must clearly indicate the time interval $t = 0$ to $t = 12$. It cannot reference a specific point in time.

Other acceptable ways of indicating the time interval include:

- On the time interval $[0, 12]$

4-2-2b

$$0 \leq t \leq 12$$

- Over the first 12 seconds
- During the first 12 seconds
- Over the 12 second time interval
- During the 12 second time interval
- Over the 12 seconds
- During the 12 seconds
- In all of these, “seconds” may be replaced by any unit of time, correct or incorrect. Incorrect time units will cause the second point not to be awarded.

Interpretations that will not earn the point include:

- At time $t = 12$
- After $t = 12$ seconds

The interpretation must include “distance traveled.” “Displacement” does not demonstrate a correct interpretation of the integral given.

Other acceptable ways of indicating distance traveled include:

- The particle traveled a distance of 58.8 meters (or other unit of distance, correct or incorrect) over the time interval $t = 0$ to $t = 12$.

Interpretations that will not earn the point include:

- The particle moved a distance of 58.8 meters over the time interval $t = 0$ to $t = 12$.

Units must include the distance units but do not need to include the t (time) units. However, if the time units are given, they must be correct.

The distance units may have been written in part (c). They do not need to be written again in part (d).

If the distance units in part (d) are incorrect, but distance units are present (e.g., inches instead of meters), and the identical distance unit substitution was already penalized in part (b), the student will not be penalized again in part (d).



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

4-2-2b

- ☐ Interpretation
- ☐ Units

Solution:

$\int_0^{12} |x'(t)| dt$ is the total distance traveled by the particle, in meters, from $t = 0$ to $t = 12$ seconds.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for doing the first step of Euler's method with at most one error. This could be done in either equation or tabular form.

If presented in a table and the table is unlabeled, a correct answer will earn both points. An incorrect final answer from an unlabeled table will earn no points.

Some answers that might indicate one error include:

- Uses positive change in time: $8 + 4(6) = 32$, $32 + 5.8(6) = 66.8$
- Uses starting t -value instead of x -value: $12 + 4(-6) = -12$, $-12 + 5.8(-6) = -46.8$
- Uses value from part (c) as starting x -value: $58.8 + 4(-6) = 34.8$, $34.8 + 5.8(-6) = 0$

The second point is earned only by the correct answer. That answer does not need to be simplified.

A response of only $8 + 4(-6) + 5.8(-6)$ would earn both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$x(6) \approx x(12) + x'(12)(6 - 12) = 8 + 4 \cdot (-6) = -16$$

$$x(0) \approx x(6) + x'(6)(0 - 6) \approx -16 + 5.8 \cdot (-6) = -50.8$$

Part F

4-2-2b

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for recognition that it is necessary to find the fourth derivative.

This point can be earned by a declared attempt at the fourth derivative, even if the derivative is not correct.

If the derivative is not declared to be the fourth derivative, then it must be correct in order to earn the point.

This point can be earned by the correct derivative only embedded in an incorrect expression for the fourth-degree term of the Taylor polynomial, e.g., $\frac{7(t-3)^6}{4!}(t-1)^4$.

The second point is for using that derivative to write the coefficient of the fourth-degree term. This point would be earned for dividing an evaluation of the declared or correct fourth derivative at $t = 1$ by $4!$. This coefficient does not have to be attached to $(t-1)^4$ in order to earn this point.

The coefficient does not have to be simplified. You may see the coefficient in the answer correctly expressed in any of these forms (among others):

$$\cdot \frac{7(-2)^6}{4!}$$

$$\cdot \frac{7(1-3)^6}{4!}$$

$$\cdot \frac{448}{4!}$$

$$\cdot \frac{7(64)}{4!}$$

$$\cdot \frac{56}{3}$$

$$\cdot 18.666$$

$$\cdot 18.667 \text{ (Note: } 18.67 \text{ would not earn the point.)}$$

The third point is for putting it all together and presenting the correct term. The correct term must have a correct coefficient, attached to $(t-1)^4$. We will not read any presented terms of degree less than four. If there are terms of degree higher than four, the response must clearly indicate that they are not part of the final answer in order to earn this point.

A response of only $\frac{7(2^6)}{4!}(t-1)^4$ (or equivalent) would earn all three points.

If the fourth derivative is never declared and is incorrectly computed, the response can earn at most one point (the second point).

The variable used in the expression need not be t (for example, it might be x).

4-2-2b



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ $s^{(4)}(t)$
- ☐ $\frac{s^{(4)}(1)}{4!}$
- ☐ Term

Solution:

$$s^{(3)}(t) = (t - 3)^7$$

$$s^{(4)}(t) = 7(t - 3)^6$$

$$s^{(4)}(1) = 7(1 - 3)^6 = 7(-2)^6 = 7 \cdot 2^6$$

The fourth-degree term is $\frac{s^{(4)}(1)}{4!}(t - 1)^4 = \frac{7 \cdot 2^6}{4!}(t - 1)^4$.

4-2-2b

51. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	4	8
$x'(t)$ (feet per second)	7.0	7.6	6.0

The position of a particle moving along a straight line is modeled by a twice-differentiable function $x(t)$, where time t is measured in seconds and $x(t)$ is measured in feet. Selected values of $x'(t)$, the derivative of $x(t)$, are shown in the table above. The position of the particle at time $t = 8$ is $x(8) = 12$.

(a) Approximate $x''(6)$ using the average rate of change of $x'(t)$ over the interval $4 \leq t \leq 8$. Show the computations that lead to your answer.

(b) Using correct units, explain the meaning of $x''(6)$ in the context of the problem.

(c) Use a left Riemann sum with the two subintervals indicated by the data in the table to approximate $\int_0^8 |x'(t)| \, dt$. Show the computations that lead to your answer.

(d) Using correct units, explain the meaning of $\int_0^8 |x'(t)| \, dt$ in the context of the problem.

(e) Use Euler's method, starting at $t = 8$ with two steps of equal size, to approximate the position of the particle at time $t = 0$. Show the computations that lead to your answer.

(f) Let s be a function such that the fifth derivative of s with respect to t is $(t - 4)^9$. Write the sixth-degree term of the sixth-degree Taylor polynomial for s about $t = 2$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The computations shown should include a quotient with at least one subtraction (numerator or denominator).

This point is earned only by a correct numerical value equivalent to -0.4 .

4-2-2b



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$x''(6) \approx \frac{x'(8) - x'(4)}{8 - 4} = \frac{6.0 - 7.6}{4} = -0.4$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Other acceptable interpretations include:

- $x''(6)$ is the acceleration of the particle at $t = 6$.
- $x''(6)$ is the rate of change of the particle's velocity at $t = 6$.
- The particle is decelerating at a rate of 0.4 feet per second per second at $t = 6$.
- The velocity of the particle is decreasing at a rate of 0.4 feet per second per second at $t = 6$.

Interpretations that will not earn the point include:

- $x''(6)$ is the rate of change of speed at $t = 6$.
- $x''(6)$ means the particle is slowing down at $t = 6$. (We cannot be certain the particle's velocity is positive at $t = 6$.)
- The particle is decelerating at a rate of -0.4 feet per second per second at $t = 6$. (Decelerating at a negative rate is not correct.)
- The velocity of the particle is decreasing at a rate of -0.4 feet per second per second at $t = 6$. (Decreasing at a negative rate is not correct.)

Units must include the rate of change units, but do not need to include the t (time) units. However, if t units are given, they must be correct.

Units appearing in part (a) do not need to be repeated in part (b).

Incorrect units written in part (b) will override any units presented in part (a).

We may read " $x = 6$ " to mean $t = 6$.

4-2-2b



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Interpretation
- ☐ Units

Solution:

$x''(6)$ is the rate of change of the rate of change of the position of the particle, in feet per second per second, at time $t = 6$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The computations shown must include sum of two explicit products. For example, $7 \cdot 4 + 7.6 \cdot 4$ or $(7 + 7.6) \cdot 4$.

Labeled areas on a graph are not enough without both the sum and explicit products.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$\int_0^8 |x'(t)| dt \approx |x'(0)|(4-0) + |x'(4)|(8-4)$$

$$= 7 \cdot 4 + 7.6 \cdot 4 = 58.4$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The interpretation must clearly indicate the time interval $t = 0$ to $t = 8$. It cannot reference a specific point in time.

Other acceptable ways of indicating the time interval include:

- On the time interval $[0, 8]$

4-2-2b

- $0 \leq t \leq 8$
- Over the first 8 seconds
- During the first 8 seconds
- Over the 8 second time interval
- During the 8 second time interval
- Over the 8 seconds
- During the 8 seconds
- In all of these, “seconds” may be replaced by any unit of time, correct or incorrect. Incorrect time units will cause the second point not to be awarded.

Interpretations that will not earn the point include:

- At time $t = 8$
- After $t = 8$ seconds

The interpretation must include “distance traveled.” “Displacement” does not demonstrate a correct interpretation of the integral given.

Other acceptable ways of indicating distance traveled include:

- The particle traveled a distance of 58.4 feet (or other unit of distance, correct or incorrect) over the time interval $t = 0$ to $t = 8$.

Interpretations that will not earn the point include:

- The particle moved a distance of 58.4 feet over the time interval $t = 0$ to $t = 8$.

Units must include the distance units but do not need to include the t (time) units. However, if the time units are given, they must be correct.

The distance units may have been written in part (c). They do not need to be written again in part (d).

If the distance units in part (d) are incorrect, but distance units are present (e.g., inches instead of feet), and the identical distance unit substitution was already penalized in part (b), the student will not be penalized again in part (d).



0	1	2
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The student response accurately includes **both** of the criteria below.

4-2-2b

- ☐ Interpretation
- ☐ Units

Solution:

$\int_0^8 |x'(t)| dt$ is the total distance traveled by the particle, in feet, from $t = 0$ to $t = 8$ seconds.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for doing the first step of Euler's method with at most one error. This could be done in either equation or tabular form.

If presented in a table and the table is unlabeled, a correct answer will earn both points. An incorrect final answer from an unlabeled table will earn no points.

Some answers that might indicate one error include:

- Uses positive change in time: $12 + 6(4) = 36$, $36 + 7.6(4) = 66.4$
- Uses starting t -value instead of x -value: $8 + 6(-4) = -16$, $-16 + 7.6(-4) = -46.4$
- Uses value from part (c) as starting x -value: $58.4 + 6(-4) = 34.4$, $34.4 + 7.6(-4) = 4$

The second point is earned only by the correct answer. That answer does not need to be simplified.

A response of only $12 + 6(-4) + 7.6(-4)$ would earn both points.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$x(4) \approx x(8) + x'(8)(4 - 8) = 12 + 6 \cdot (-4) = -12$$

$$x(0) \approx x(4) + x'(4)(0 - 4) \approx -12 + 7.6 \cdot (-4) = -42.4$$

Part F

4-2-2b

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for recognition that it is necessary to find the sixth derivative.

This point can be earned by a declared attempt at the sixth derivative, even if the derivative is not correct.

If the derivative is not declared to be the sixth derivative, then it must be correct in order to earn the point.

This point can be earned by the correct derivative only embedded in an incorrect expression for the sixth-degree term of the Taylor polynomial, e.g., $\frac{9(t-4)^8}{6!}(t-2)^6$.

The second point is for using that derivative to write the coefficient of the sixth-degree term. This point would be earned for dividing an evaluation of the declared or correct sixth derivative at $t = 2$ by $6!$. This coefficient does not have to be attached to $(t-2)^6$ in order to earn this point.

The coefficient does not have to be simplified. You may see the coefficient in the answer correctly expressed in any of these forms (among others):

$$\cdot \frac{9(-2)^8}{6!}$$

$$\cdot \frac{9(2-4)^8}{6!}$$

$$\cdot \frac{2304}{6!}$$

$$\cdot \frac{9(256)}{6!}$$

$$\cdot \frac{16}{5}$$

$$\cdot 3.2$$

$$\cdot \frac{2304}{720}$$

The third point is for putting it all together and presenting the correct term. The correct term must have a correct coefficient, attached to $(t-2)^6$. We will not read any presented terms of degree less than six. If there are terms of degree higher than six, the response must clearly indicate that they are not part of the final answer in order to earn this point.

A response of only $\frac{9(2^8)}{6!}(t-2)^6$ (or equivalent) would earn all three points.

If the sixth derivative is never declared and is incorrectly computed, the response can earn at most one point (the second point).

The variable used in the expression need not be t (for example, it might be x).

4-2-2b



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ $s^{(6)}(t)$
- ☐ $\frac{s^{(6)}(2)}{6!}$
- ☐ Term

Solution:

$$s^{(5)}(t) = (t - 4)^9$$

$$s^{(6)}(t) = 9(t - 4)^8$$

$$s^{(6)}(2) = 9(2 - 4)^8 = 9(-2)^8 = 9 \cdot 2^8$$

The sixth-degree term is $\frac{s^{(6)}(2)}{6!}(t - 2)^6 = \frac{9 \cdot 2^8}{6!}(t - 2)^6$.

4-2-2b

52. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	5	10
$x'(t)$ (meters per minute)	6.2	5.6	7.0

The position of a particle moving along a straight line is modeled by a twice-differentiable function $x(t)$, where time t is measured in minutes and $x(t)$ is measured in meters. Selected values of $x'(t)$, the derivative of $x(t)$, are shown in the table above. The position of the particle at time $t = 10$ is $x(10) = 9$.

- Approximate $x''(7.5)$ using the average rate of change of $x'(t)$ over the interval $5 \leq t \leq 10$. Show the computations that lead to your answer.
- Using correct units, explain the meaning of $x''(7.5)$ in the context of the problem.
- Use a right Riemann sum with the two subintervals indicated by the data in the table to approximate $\int_0^{10} |x'(t)| \, dt$. Show the computations that lead to your answer.
- Using correct units, explain the meaning of $\int_0^{10} |x'(t)| \, dt$ in the context of the problem.
- Use Euler's method, starting at $t = 10$ with two steps of equal size, to approximate the position of the particle at time $t = 0$. Show the computations that lead to your answer.
- Let s be a function such that the fifth derivative of s with respect to t is $(t + 3)^8$. Write the sixth-degree term of the sixth-degree Taylor polynomial for s about $t = -1$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The computations shown should include a quotient with at least one subtraction (numerator or denominator).

This point is earned only by a correct numerical value equivalent to 0.28. Note that you may see the answer correctly expressed in other forms, e.g., $\frac{7}{25}$ or $\frac{1.4}{5}$.

4-2-2b



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$x''(7.5) \approx \frac{x'(10) - x'(5)}{10 - 5} = \frac{7.0 - 5.6}{5} = 0.28$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Other acceptable interpretations include:

- $x''(7.5)$ is the acceleration of the particle at $t = 7.5$.
- $x''(7.5)$ is the rate of change of the particle's velocity at $t = 7.5$.
- The particle is accelerating at a rate of 0.28 meters per minute per minute at $t = 7.5$.
- The velocity of the particle is increasing at a rate of 0.28 meters per minute per minute at $t = 7.5$.

Interpretations that will not earn the point include:

- $x''(7.5)$ is the rate of change of speed at $t = 7.5$.
- $x''(7.5)$ means the particle is speeding up at $t = 7.5$. (We cannot be certain the particle's velocity is positive at $t = 7.5$.)

Units must include the rate of change units, but do not need to include the t (time) units. However, if t units are given, they must be correct.

Units appearing in part (a) do not need to be repeated in part (b).

Incorrect units written in part (b) will override any units presented in part (a).

We may read “ $x = 7.5$ ” to mean $t = 7.5$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

4-2-2b

- ☐ Interpretation
- ☐ Units

Solution:

$x''(7.5)$ is the rate of change of the rate of change of the position of the particle, in meters per minute per minute, at time $t = 7.5$ minutes.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The computations shown must include sum of two explicit products. For example, $5.6 \cdot 5 + 7 \cdot 5$ or $(5.6 + 7) \cdot 5$.

Labeled areas on a graph are not enough without both the sum and explicit products.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$\begin{aligned} \int_0^{10} |x'(t)| \, dt &\approx |x'(5)| (5 - 0) + |x'(10)| (10 - 5) \\ &= 5.6 \cdot 5 + 7 \cdot 5 = 63 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The interpretation must clearly indicate the time interval $t = 0$ to $t = 10$. It cannot reference a specific point in time.

Other acceptable ways of indicating the time interval include:

- On the time interval $[0, 10]$
- $0 \leq t \leq 10$
- Over the first 10 minutes
- During the first 10 minutes
- Over the 10 minute time interval

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- During the 10 minute time interval
- Over the 10 minutes
- During the 10 minutes
- In all of these, “minutes” may be replaced by any unit of time, correct or incorrect. Incorrect time units will cause the second point not to be awarded.

Interpretations that will not earn the point include:

- At time $t = 10$
- After $t = 10$ minutes

The interpretation must include “distance traveled.” “Displacement” does not demonstrate a correct interpretation of the integral given.

Other acceptable ways of indicating distance traveled include:

- The particle traveled a distance of 63 meters (or other unit of distance, correct or incorrect) over the time interval $t = 0$ to $t = 10$.

Interpretations that will not earn the point include:

- The particle moved a distance of 63 meters over the time interval $t = 0$ to $t = 10$.

Units must include the distance units but do not need to include the t (time) units. However, if the time units are given, they must be correct.

The distance units may have been written in part (c). They do not need to be written again in part (d).

If the distance units in part (d) are incorrect, but distance units are present (e.g., inches instead of meters), and the identical distance unit substitution was already penalized in part (b), the student will not be penalized again in part (d).



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Interpretation
- ☐ Units

Solution:

$\int_0^{10} |x'(t)| dt$ is the total distance traveled by the particle, in meters, from $t = 0$ to $t = 10$ minutes.

4-2-2b

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for doing the first step of Euler's method with at most one error. This could be done in either equation or tabular form.

If presented in a table and the table is unlabeled, a correct answer will earn both points. An incorrect final answer from an unlabeled table will earn no points.

Some answers that might indicate one error include:

- Uses positive change in time: $9 + 7(5) = 44$, $44 + 5.6(5) = 72$
- Uses starting t -value instead of x -value: $10 + 7(-5) = -25$, $-25 + 5.6(-5) = -53$
- Uses value from part (c) as starting x -value: $63 + 7(-5) = 28$, $28 + 5.6(-5) = 0$

The second point is earned only by the correct answer. That answer does not need to be simplified.

A response of only $9 + 7(-5) + 5.6(-5)$ would earn both points.

✓

0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$x(5) \approx x(10) + x'(10)(5 - 10) = 9 + 7 \cdot (-5) = -26$$

$$x(0) \approx x(5) + x'(5)(0 - 5) \approx -26 + 5.6 \cdot (-5) = -54$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for recognition that it is necessary to find the sixth derivative.

This point can be earned by a declared attempt at the sixth derivative, even if the derivative is not correct.

If the derivative is not declared to be the sixth derivative, then it must be correct in order to earn the point.

4-2-2b

This point can be earned by the correct derivative only embedded in an incorrect expression for the sixth-degree term of the Taylor polynomial, e.g., $\frac{8(t+3)^7}{6!}(t+1)^6$.

The second point is for using that derivative to write the coefficient of the sixth-degree term. This point would be earned for dividing an evaluation of the declared or correct sixth derivative at $t = -1$ by $6!$. This coefficient does not have to be attached to $(t+1)^6$ in order to earn this point.

The coefficient does not have to be simplified. You may see the coefficient in the answer correctly expressed in any of these forms (among others):

$$\cdot \frac{8(-1+3)^7}{6!}$$

$$\cdot \frac{2^{10}}{6!}$$

$$\cdot \frac{1024}{6!}$$

$$\cdot \frac{8(128)}{6!}$$

$$\cdot \frac{1024}{720}$$

$$\cdot \frac{64}{45}$$

· 1.422 (Note: 1.42 would not earn the point.)

The third point is for putting it all together and presenting the correct term. The correct term must have a correct coefficient, attached to $(t+1)^6$. We will not read any presented terms of degree less than six. If there are terms of degree higher than six, the response must clearly indicate that they are not part of the final answer in order to earn this point.

A response of only $\frac{8(2^7)}{6!}(t+1)^6$ (or equivalent) would earn all three points.

If the sixth derivative is never declared and is incorrectly computed, the response can earn at most one point (the second point).

The variable used in the expression need not be t (for example, it might be x).



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

☐ $s^{(6)}(t)$

4-2-2b

☐ $\frac{s^{(6)}(-1)}{6!}$

☐ Term**Solution:**

$$s^{(5)}(t) = (t + 3)^8$$

$$s^{(6)}(t) = 8(t + 3)^7$$

$$s^{(6)}(-1) = 8(-1 + 3)^7 = 8 \cdot 2^7$$

The sixth-degree term is $\frac{s^{(6)}(-1)}{6!}(t + 1)^6 = \frac{8 \cdot 2^7}{6!}(t + 1)^6$.

4-2-2b

53. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	3	6
$x'(t)$ (feet per minute)	7.0	6.2	8.0

The position of a particle moving along a straight line is modeled by a twice-differentiable function $x(t)$, where time t is measured in minutes and $x(t)$ is measured in feet. Selected values of $x'(t)$, the derivative of $x(t)$, are shown in the table above. The position of the particle at time $t = 6$ is $x(6) = 10$.

(a) Approximate $x''(4.5)$ using the average rate of change of $x'(t)$ over the interval $3 \leq t \leq 6$. Show the computations that lead to your answer.

(b) Using correct units, explain the meaning of $x''(4.5)$ in the context of the problem.

(c) Use a left Riemann sum with the two subintervals indicated by the data in the table to approximate $\int_0^6 |x'(t)| \, dt$. Show the computations that lead to your answer.

(d) Using correct units, explain the meaning of $\int_0^6 |x'(t)| \, dt$ in the context of the problem.

(e) Use Euler's method, starting at $t = 6$ with two steps of equal size, to approximate the position of the particle at time $t = 0$. Show the computations that lead to your answer.

(f) Let s be a function such that the third derivative of s with respect to t is $(t + 5)^6$. Write the fourth-degree term of the fourth-degree Taylor polynomial for s about $t = -2$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The computations shown should include a quotient with at least one subtraction (numerator or denominator).

This point is earned only by a correct numerical value equivalent to 0.6.

4-2-2b



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$x''(4.5) \approx \frac{x'(6) - x'(3)}{6 - 3} = \frac{8.0 - 6.2}{3} = 0.6$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Other acceptable interpretations include:

- $x''(4.5)$ is the acceleration of the particle at $t = 4.5$.
- $x''(4.5)$ is the rate of change of the particle's velocity at $t = 4.5$.
- The particle is accelerating at a rate of 0.6 feet per minute per minute at $t = 4.5$.
- The velocity of the particle is increasing at a rate of 0.6 feet per minute per minute at $t = 4.5$.

Interpretations that will not earn the point include:

- $x''(4.5)$ is the rate of change of speed at $t = 4.5$.
- $x''(4.5)$ means the particle is speeding up at $t = 4.5$. (We cannot be certain the particle's velocity is positive at $t = 4.5$.)

Units must include the rate of change units, but do not need to include the t (time) units. However, if t units are given, they must be correct.

Units appearing in part (a) do not need to be repeated in part (b).

Incorrect units written in part (b) will override any units presented in part (a).

We may read " $x = 4.5$ " to mean $t = 4.5$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

4-2-2b

- ☐ Interpretation
- ☐ Units

Solution:

$x''(4.5)$ is the rate of change of the rate of change of the position of the particle, in feet per minute per minute, at time $t = 4.5$ minutes.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The computations shown must include sum of two explicit products. For example, $7 \cdot 3 + 6.2 \cdot 3$ or $(7 + 6.2) \cdot 3$.

Labeled areas on a graph are not enough without both the sum and explicit products.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$\begin{aligned} \int_0^6 |x'(t)| dt &\approx |x'(0)|(3-0) + |x'(3)|(6-3) \\ &= 7 \cdot 3 + 6.2 \cdot 3 = 39.6 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The interpretation must clearly indicate the time interval $t = 0$ to $t = 6$. It cannot reference a specific point in time.

Other acceptable ways of indicating the time interval include:

- On the time interval $[0, 6]$
- $0 \leq t \leq 6$
- Over the first 6 minutes
- During the first 6 minutes
- Over the 6 minute time interval

4-2-2b

- During the 6 minute time interval
- Over the 6 minutes
- During the 6 minutes
- In all of these, “minutes” may be replaced by any unit of time, correct or incorrect. Incorrect time units will cause the second point not to be awarded.

Interpretations that will not earn the point include:

- At time $t = 6$
- After $t = 6$ minutes

The interpretation must include “distance traveled.” “Displacement” does not demonstrate a correct interpretation of the integral given.

Other acceptable ways of indicating distance traveled include:

- The particle traveled a distance of 39.6 feet (or other unit of distance, correct or incorrect) over the time interval $t = 0$ to $t = 6$.

Interpretations that will not earn the point include:

- The particle moved a distance of 39.6 feet over the time interval $t = 0$ to $t = 6$.

Units must include the distance units but do not need to include the t (time) units. However, if the time units are given, they must be correct.

The distance units may have been written in part (c). They do not need to be written again in part (d).

If the distance units in part (d) are incorrect, but distance units are present (e.g., inches instead of feet), and the identical distance unit substitution was already penalized in part (b), the student will not be penalized again in part (d).



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Interpretation
- ☐ Units

Solution:

$\int_0^6 |x'(t)| dt$ is the total distance traveled by the particle, in feet, from $t = 0$ to $t = 6$ minutes.

4-2-2b

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for doing the first step of Euler's method with at most one error. This could be done in either equation or tabular form.

If presented in a table and the table is unlabeled, a correct answer will earn both points. An incorrect final answer from an unlabeled table will earn no points.

Some answers that might indicate one error include:

- Uses positive change in time: $10 + 8(3) = 34$, $34 + 6.2(3) = 52.6$
- Uses starting t -value instead of x -value: $6 + 8(-3) = -18$, $-18 + 6.2(-3) = -36.6$
- Uses value from part (c) as starting x -value: $39.6 + 8(-3) = 15.6$, $15.6 + 6.2(-3) = -3$

The second point is earned only by the correct answer. That answer does not need to be simplified.

A response of only $10 + 8(-3) + 6.2(-3)$ would earn both points.

✓

0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$x(3) \approx x(6) + x'(6)(3 - 6) = 10 + 8 \cdot (-3) = -14$$

$$x(0) \approx x(3) + x'(3)(0 - 3) \approx -14 + 6.2 \cdot (-3) = -32.6$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for recognition that it is necessary to find the fourth derivative.

This point can be earned by a declared attempt at the fourth derivative, even if the derivative is not correct.

If the derivative is not declared to be the fourth derivative, then it must be correct in order to earn the point.

4-2-2b

This point can be earned by the correct derivative only embedded in an incorrect expression for the fourth-degree term of the Taylor polynomial, e.g., $\frac{6(t+5)^5}{4!}(t+2)^4$.

The second point is for using that derivative to write the coefficient of the fourth-degree term. This point would be earned for dividing an evaluation of the declared or correct fourth derivative at $t = -2$ by $4!$. This coefficient does not have to be attached to $(t+2)^4$ in order to earn this point.

The coefficient does not have to be simplified. You may see the coefficient in the answer correctly expressed in any of these forms (among others):

$$\cdot \frac{6(3^5)}{4!}$$

$$\cdot \frac{6(-2+5)^5}{4!}$$

$$\cdot \frac{1458}{4!}$$

$$\cdot \frac{6(243)}{4!}$$

$$\cdot \frac{1458}{24}$$

$$\cdot \frac{243}{4}$$

$$\cdot 60.75$$

The third point is for putting it all together and presenting the correct term. The correct term must have a correct coefficient, attached to $(t+2)^4$. We will not read any presented terms of degree less than four. If there are terms of degree higher than four, the response must clearly indicate that they are not part of the final answer in order to earn this point.

A response of only $\frac{6(3^5)}{4!}(t+2)^4$ (or equivalent) would earn all three points.

If the fourth derivative is never declared and is incorrectly computed, the response can earn at most one point (the second point).

The variable used in the expression need not be t (for example, it might be x).



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

☐ $s^{(4)}(t)$

4-2-2b

- ☐ $\frac{s^{(4)}(-2)}{4!}$
- ☐ Term

Solution:

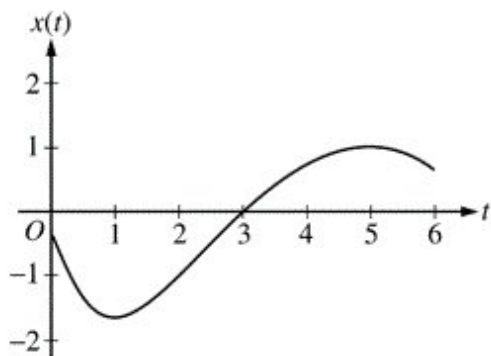
$$s^{(3)}(t) = (t + 5)^6$$

$$s^{(4)}(t) = 6(t + 5)^5$$

$$s^{(4)}(-2) = 6(-2 + 5)^5 = 6 \cdot 3^5$$

The fourth-degree term is $\frac{s^{(4)}(-2)}{4!}(t + 2)^4 = \frac{6 \cdot 3^5}{4!}(t + 2)^4$.

54.



A particle moves along a straight line. The graph of the particle's position $x(t)$ at time t is shown above for $0 < t < 6$. The graph has horizontal tangents at $t = 1$ and $t = 5$ and a point of inflection at $t = 2$. For what values of t is the velocity of the particle increasing?

(A) $0 < t < 2$

(B) $1 < t < 5$

(C) $2 < t < 6$

(D) $3 < t < 5$ only

(E) $1 < t < 2$ only $5 < t < 6$



4-2-2b

55. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	2	5	11
$v(t)$ (meters per second)	3	5.5	7	13

The velocity of a particle, P , moving along the x -axis is modeled by a differentiable function v , where time t is measured in seconds and $v(t)$ is measured in meters per second. Selected values of $v(t)$ are shown in the table above.

(a) Use the data in the table to approximate $v'(8)$ using the average rate of change of $v(t)$ over the interval $5 \leq t \leq 11$. Show the computations that lead to your answer. Indicate units of measure.

(b) Interpret the meaning of $v'(8)$ in the context of the problem.

(c) Justify why there must be a time $t = b$, for $0 \leq b \leq 2$, when the velocity of the particle is 5 meters per second.

(d) Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^{11} v(t) \, dt$. Show the computations that lead to your answer.

(e) Find $\int_0^9 v' \left(\frac{t}{3} + 2 \right) \, dt$. Show the computations that lead to your answer.

(f) Let $h(x) = \int_1^{\frac{5}{2}x} v(2t) \, dt$. Find $h'(1)$. Show the computations that lead to your answer.

(g) The position of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(2) = 5.5$, $g'(2) = -4$, and $g''(2) = -3$. Is the speed of particle Q increasing or decreasing at time $t = 2$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -3x^2y^3$ with initial condition $f(3) = 2$. Write an equation for the line tangent to the graph of f at the point $(3, 2)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-2-2b

First point: Approximation

Need to see both a difference and a quotient as the supporting work. Minimal answers that earn the point include $\frac{13-7}{11-5}$, $\frac{13-7}{6}$, and $\frac{6}{11-5}$.

The answer does not need to be simplified, but any attempt at simplification must be correct.

Numerical values must be pulled from the table. $\frac{v(11)-v(5)}{11-5}$ alone will not earn the point as it does not pull values from the table (yet).

Note: the reciprocal of this quotient gives the same numerical value. This does not earn the point.

Second point: Units

$\frac{m}{s^2}$ or meters per second per second will earn the point.

The units must be tied to a number. Units alone do not earn this point.

Beware of $\left(\frac{m}{s}\right)^2$ or $\frac{m^2}{s}$ (which do not earn the point).

This point must be earned in part (a). If no units are present in part (a) and correct units are shown in part (b), this point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Approximation

☐ Units

Solution:

$$v'(8) \approx \frac{13-7}{11-5} = 1 \text{ meter per second per second}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Interpretation

There are two components to look for:

· “rate of change of velocity” or “acceleration” of the particle

4-2-2b

· at $t = 8$

“The rate at which the velocity is increasing at $t = 8$ ” is acceptable. If the answer to part (a) was reported to be negative, “the rate at which the velocity is decreasing at $t = 8$ ” is acceptable. We will not penalize for the same error twice.

Units are not required in part (b). If units are presented for t , they must be s (seconds).



0	1
---	---

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$v'(8)$ is the rate at which the velocity of the particle is changing, in meters per second per second, at time $t = 8$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: v is continuous

It must be clear that the reason given as to why v is continuous is that v is differentiable.

Neither “ v is differentiable and continuous” nor “ v differentiable, v continuous” are sufficient to earn this point.

“ v is differentiable hence continuous,” by contrast, earns the point.

“ v differentiable $\therefore v$ continuous” earns the point.

Second point: Bounding with table values

Must see the values 3 and 5.5 pulled from the table and correctly compared to 5, in words or using inequality symbols.

The Intermediate Value Theorem (IVT) does not need to be referenced by name.

That said, this point will not be earned if other theorems (MVT, etc.) are invoked.

Responses often “say too much”; especially common are responses that indicate that v is always increasing on the interval $[0, 2]$. This point will not be earned if the response contains such errors in reasoning.

4-2-2b



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ v continuous
- ☐ Bounding 5 with table values

Solution:

v is differentiable. $\Rightarrow v$ is continuous.

$$v(0) = 3 < 5 < 5.5 = v(2)$$

By the Intermediate Value Theorem, there must be a time $t = b$, for $0 \leq b \leq 2$, such that the velocity of the particle is 5 meters per second.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Form of left Riemann sum

Must see a sum of three terms, each of which is a product of two factors: $(b_1 \cdot h_1) + (b_2 \cdot h_2) + (b_3 \cdot h_3)$.

If exactly 5 out of 6 of these numbers ($b_1, b_2, b_3, h_1, h_2, h_3$) are correctly placed in the above expression, we are convinced that the response carries the form of a left Riemann sum. In this case, award the first point. Such a response cannot earn the second point. If fewer than 5 out of 6 values are correct, no points are earned in this problem.

To earn this point, values do not need yet to be imported from the table: the expression

$$v(0) \cdot (2 - 0) + v(2) \cdot (5 - 2) + v(5) \cdot (11 - 5) \text{ is sufficient.}$$

Once the first point is earned, it cannot be lost. Subsequent errors result in the second point not being earned.

Second point: Answer

64.5 or equivalent.

It is possible to earn this point without earning the first point. If there is credible evidence that an attempt was made to find a left Riemann sum but not enough work is shown, this point can be earned in the presence of the correct answer. For example, a sum of three terms with no evidence of products $(6 + 16.5 + 42)$, or a graph with areas labeled and summed to 64.5 or equivalent, can earn this point.

The response of only $3 \cdot 2 + 5.5 \cdot 3 + 7 \cdot 6$ earns both points.

4-2-2b



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of left Riemann sum
- ☐ Answer

Solution:

$$\int_0^{11} v(t) dt \approx v(0)(2-0) + v(2)(5-2) + v(5)(11-5)$$

$$= 3 \cdot 2 + 5.5 \cdot 3 + 7 \cdot 6 = 64.5$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Substitution

There are three elements to look for in a correct substitution. (another variable may be chosen instead of u .)

- $u = \frac{t}{3} + 2$
- $du = \frac{dt}{3}$ or equivalent
- Appropriate limits of integration throughout (2 and 5 for u , 0 and 9 for t).

Substitution can be implicit. A coefficient of 3 accompanying a correct antiderivative convinces us of this:

$$\int_0^9 v' \left(\frac{t}{3} + 2 \right) dt = 3v \left(\frac{t}{3} + 2 \right) \Big|_0^9$$

To elaborate on the third element above: if a response shows an otherwise correct substitution but the limits of integration on a definite integral are unchanged (0 and 9), this point is not earned.

Alternately, the integral can be worked with as an indefinite integral with the variable returned to $\frac{t}{3} + 2$ before evaluating as a means of eliminating the need to change limits. Such a response can earn the first point, if handled correctly.

Second point: Fundamental Theorem of Calculus

We must be convinced that the response has indicated an appropriate connection between $\int v'$ and v .

Expressions/equations that earn this point are:

4-2-2b

· $k(v(5) - v(2))$ or $k(v(\frac{9}{3} + 2) - v(\frac{0}{3} + 2))$, where k is a nonzero constant.

· $kv(\frac{t}{3} + 2)|_0^9$ or $k v(u)|_2^5$, where k is a nonzero constant.

· $\int v'(\frac{t}{3} + 2) dt = kv(\frac{t}{3} + 2)$ or $k \int v'(u) du = kv(u)$, where k is a nonzero constant, with or without appropriate limits of integration displayed on either integral.

· $\int_0^9 v'(\frac{t}{3} + 2) dt = k v(\frac{t^2}{6} + 2t)$ where k is a nonzero constant is a common response. This does not earn the point. Likewise, similar mishandled antiderivatives do not earn the point.

Third point: Answer

4.5 or equivalent, with credible supporting work, earns this point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{3} + 2$. Then $du = \frac{dt}{3}$.

When $t = 0$, $u = 2$.

When $t = 9$, $u = 5$.

$$\int_0^9 v'(\frac{t}{3} + 2) dt = 3 \int_2^5 v'(u) du$$

$$= 3(v(5) - v(2))$$

$$= 3(7 - 5.5) = 4.5$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Fundamental Theorem of Calculus (FTC)

4-2-2b

Substitutes $\frac{5}{2}x$ into $v(2t)$. We must see either $v\left(2 \cdot \frac{5}{2}x\right)$ or $v(5x)$.

The expression $v\left(2 \cdot \frac{5}{2}x\right) - v(2(1))$ or $v\left(2 \cdot \frac{5}{2}x\right) \cdot \frac{5}{2} - v(2(1))$ will earn this point. The error of subtracting $v(2(1))$ will be viewed as an error in applying the chain rule.

Beware of possible substitutions prior to using FTC:

· For example, $u = 2t$, $\frac{1}{2} du = dt$ and rewrite $\int_1^{\frac{5}{2}x} v(2t) dt = \frac{1}{2} \int_2^{5x} v(u) du$.

· Or $u = \frac{5}{2}x$, $\frac{du}{dx} = \frac{5}{2}$ and rewrites: $\frac{d}{dx}[h(x)] = \frac{d}{du}[h(u)] \cdot \frac{du}{dx}$

· Such approaches can earn this point if done properly.

Second point: Chain rule

Multiplies by $\frac{5}{2}$

It is not possible to earn this point without having earned the first point.

Depending on substitutions, this may look more like $\frac{1}{2}$ and 5 multiplied separately.

Third point: Answer

$\frac{35}{2}$ or equivalent

It is not possible to earn this point without having earned both the first and second points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_1^{\frac{5}{2}x} v(2t) dt = v\left(2 \cdot \frac{5}{2}x\right) \cdot \frac{5}{2}$$

$$h'(1) = v(5) \cdot \frac{5}{2} = 7 \cdot \frac{5}{2} = \frac{35}{2}$$

4-2-2b

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Narrows consideration to g' and g''

You must be convinced the response considers only the functions g' and g'' as being critical to the reasoning.

Second point: Answer with reason

A correct response must:

- Conclude “increasing”
- Appeal to the signs of g' and g''

It is not possible to earn this point without having earned the first point.

A response that mentions velocity and acceleration must tie velocity to g' and acceleration to g'' . A response that says only “increasing because velocity and acceleration are both negative” earns neither of the two points. It is not clear that the response is considering g' and g'' . A response that calls out all three given values ($g(2)$, $g'(2)$, and $g''(2)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g' and g'' only, both points can be earned.

The phrase “speeding up” will be interpreted as “increasing.”



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g' and g''
- ☐ Answer with reason

Solution:

Because $g'(2) = -4 < 0$, the velocity of the particle is negative. Because $g''(2) = -3 < 0$, the acceleration of the particle is negative. Therefore, because both the velocity and acceleration are negative, the speed of the particle is increasing at time $t = 2$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Tangent line equation

4-2-2b

This point is earned only for the correct tangent line equation written in any form.

Some common forms:

$$\cdot y - 2 = -3(3)^2(2)^3(x - 3)$$

$$\cdot y - 2 = -216(x - 3)$$

$$\cdot y = -216x + 650$$

$y - 2 = -3x^2y^3(x - 3)$ does not earn the point. This is not the equation of a line. The response can go on to present one of the correct linear forms, and then would earn the point.

$f(x) = -216(x - 3) + 2$ or equivalent does not earn the point. $f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

Some responses have the correct tangent line equation written initially, with subsequent errors in simplification. Such responses do not earn the point.

The point can be earned after pursuing a less fruitful approach (such as solving the differential equation) and then restarting.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(3,2)} = -3 \cdot 3^2 \cdot 2^3 = -216$$

$$y = -216(x - 3) + 2$$

4-2-2b

56. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	2	8	14
$v(t)$ (meters per second)	15	12	6	0

The velocity of a particle, P , moving along the x -axis is modeled by a differentiable function v , where time t is measured in seconds and $v(t)$ is measured in meters per second. Selected values of $v(t)$ are shown in the table above.

- Use the data in the table to approximate $v'(6)$ using the average rate of change of $v(t)$ over the interval $2 \leq t \leq 8$. Show the computations that lead to your answer. Indicate units of measure.
- Interpret the meaning of $v'(6)$ in the context of the problem.
- Justify why there must be a time $t = k$, for $8 \leq k \leq 14$, when the velocity of the particle is 3 meters per second.
- Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^{14} v(t) \, dt$. Show the computations that lead to your answer.
- Find $\int_3^{21} v'\left(\frac{t}{3} + 7\right) \, dt$. Show the computations that lead to your answer.
- Let $p(x) = \int_0^{4x} v(2t) \, dt$. Find $p'(1)$. Show the computations that lead to your answer.
- The position of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(4) = 5$, $g'(4) = 2$, and $g''(4) = -6$. Is the speed of particle Q increasing or decreasing at time $t = 4$? Give a reason for your answer.
- Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -2x + 3y^2$ with initial condition $f(3) = 2$. Write an equation for the line tangent to the graph of f at the point $(3, 2)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-2-2b

First point: Approximation.

Need to see both a difference and a quotient as the supporting work. Minimal answers that earn the point include $\frac{6-12}{8-2}$, $\frac{6-12}{6}$, and $-\frac{6}{8-2}$.

The answer does not need to be simplified, but any attempt at simplification must be correct.

Numerical values must be pulled from the table. $\frac{v(8)-v(2)}{8-2}$ alone will not earn the point as it does not pull values from the table (yet).

Note: the reciprocal of the quotient gives the same numerical answer. This does not earn the point.

Also: $\frac{v(14)-v(8)}{14-8}$ happens to equal -1 as well. A response using an incorrect pair of points does not earn the point.

Second point: Units

$\frac{m}{s^2}$ or meters per second per second will earn the point.

The units must be tied to a number. Units alone do not earn this point.

Beware of $\left(\frac{m}{s}\right)^2$ or $\frac{m^2}{s}$ (which do not earn the point).

This point must be earned in part (a). If no units are present in part (a) and correct units are shown in part (b), this point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$v'(6) \approx \frac{6-12}{8-2} = -1 \text{ meter per second per second}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Interpretation

4-2-2b

There are two components to look for:

- “rate of change of velocity” or “acceleration” of the particle
- at $t = 6$

“The rate at which the velocity is decreasing at $t = 6$ ” is acceptable. If the answer to part (a) was reported to be positive, “the rate at which the velocity is increasing at $t = 6$ is acceptable. We will not penalize for the same error twice.

Responses that say “velocity is decreasing at a rate of $-1 \frac{m}{s^2}$ ” or similar do not earn the point. This is interpreted to mean that the velocity is increasing.

Units are not required in part (b). If units are presented for t , they must be s (seconds).



0	1
---	---

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$v'(6)$ is the rate at which the velocity of the particle is changing, in meters per second per second, at time $t = 6$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: v is continuous

It must be clear that the reason given as to why v is continuous is that v is differentiable.

Neither “ v is differentiable and continuous” nor “ v differentiable, v continuous” are sufficient to earn this point.

“ v is differentiable hence continuous,” by contrast, earns the point.

v differentiable $\therefore v$ continuous” earns the point.

Second point: Bounding with table values

Must see the values 0 and 6 pulled from the table and correctly compared with 3, in words or using inequality symbols.

The Intermediate Value Theorem (IVT) does not need to be referenced by name.

4-2-2b

That said, this point will not be earned if other theorems (MVT, etc.) are invoked.

Responses often “say too much”; especially common are responses that indicate that v is always decreasing on the interval $[8, 14]$. This point will not be earned if the response contains such errors in reasoning.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ v continuous
- ☐ Bounding 3 with table values

Solution:

v is differentiable. $\Rightarrow v$ is continuous.

$$v(14) = 0 < 3 < 6 = v(8)$$

By the Intermediate Value Theorem, there must be a time $t = k$, for $8 < k < 14$, such that the velocity of the particle is 3 meters per second.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Form of left Riemann sum

Must see a sum of three terms, each of which is a product of two factors:

$$(b_1 \cdot h_1) + (b_2 \cdot h_2) + (b_3 \cdot h_3)$$

If exactly 5 out of 6 of these numbers ($b_1, b_2, b_3, h_1, h_2, h_3$) are correctly placed in the above expression, we are convinced that the response carries the form of a left Riemann sum. In this case, award the first point. Such a response cannot earn the second point. If fewer than 5 out of 6 values are correct, no points are earned in this problem.

To earn this point, values do not need yet to be imported from the table: the expression $v(0) \cdot (2 - 0) + v(2) \cdot (8 - 2) + v(8) \cdot (14 - 8)$ is sufficient.

Once the first point is earned, it cannot be lost. Subsequent errors result in the second point not being earned.

Second point: Answer

138 or equivalent.

4-2-2b

It is possible to earn this point without earning the first point. If there is credible evidence that an attempt was made to find a left Riemann sum but not enough work is shown, this point can be earned in the presence of the correct answer. For example, a sum of three terms with no evidence of products $(30 + 72 + 36)$, or a graph with areas labeled and summed to 138 or equivalent, can earn this point.

The response of only $15 \cdot 2 + 12 \cdot 6 + 6 \cdot 6$ earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of left Riemann sum
- ☐ Answer

Solution:

$$\int_0^{14} v(t) dt \approx v(0)(2-0) + v(2)(8-2) + v(8)(14-8)$$

$$= 15 \cdot 2 + 12 \cdot 6 + 6 \cdot 6 = 138$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

1st point: Substitution

There are three elements to look for in a correct substitution. (another variable may be chosen instead of u .)

$$\cdot u = \frac{t}{3} + 7$$

$$\cdot du = \frac{dt}{3} \text{ or equivalent}$$

· Appropriate limits of integration throughout (8 and 14 for u , 3 and 21 for t).

Substitution can be implicit. A coefficient of 3 accompanying a correct antiderivative convinces us of this:

$$\int_3^{21} v' \left(\frac{t}{3} + 7 \right) dt = 3v \left(\frac{t}{3} + 7 \right) \Big|_3^{21}$$

To elaborate on the third element above: if a response shows an otherwise correct substitution but the limits of integration on a definite integral are unchanged (3 and 21), this point is not earned.

Alternately, the integral can be worked with as an indefinite integral with the variable returned to $\frac{t}{3} + 7$ before evaluating as a means of eliminating the need to change limits. Such a response can earn the first point, if handled correctly.

4-2-2b

Second point: Fundamental Theorem of Calculus

We must be convinced that the response has indicated an appropriate connection between $\int v'$ and v .

Expressions/equations that earn this point are:

· $k(v(14) - v(8))$ or $k\left(v\left(\frac{21}{3} + 7\right) - v\left(\frac{3}{3} + 7\right)\right)$, where k is a nonzero constant

· $kv\left(\frac{t}{3} + 7\right)\bigg|_3^{21}$ or $k v(u)\bigg|_8^{14}$, where k is a nonzero constant

· $\int v'\left(\frac{t}{3} + 7\right) dt = kv\left(\frac{t}{3} + 7\right)$ or $k \int v'(u) du = kv(u)$, where k is a nonzero constant, with or without appropriate limits of integration displayed on either integral

· $\int_3^{21} v'\left(\frac{t}{3} + 7\right) dt = k v\left(\frac{t^2}{6} + 7t\right)$ where k is a nonzero constant is a common response. This does not earn the point. Likewise, similar mishandled antiderivatives do not earn the point.

Third point: Answer

–18 or equivalent, with credible supporting work, earns this point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{3} + 7$.

Then $du = \frac{dt}{3}$.

When $t = 3$, $u = 8$.

When $t = 21$, $u = 14$.

$$\begin{aligned} \int_3^{21} v'\left(\frac{t}{3} + 7\right) dt &= 3 \int_8^{14} v'(u) du \\ &= 3(v(14) - v(8)) \end{aligned}$$

4-2-2b

$$= 3(0 - 6) = -18$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Fundamental Theorem of Calculus (FTC)

Substitutes $4x$ into $v(2t)$. We must see either $v(2 \cdot 4x)$ or $v(8x)$.

The expression $v(2 \cdot 4x) - v(2(0))$ or $v(2 \cdot 4x) \cdot 4 - v(2(0))$ will earn this point. The error of subtracting $v(2(0))$ will be viewed as an error in applying the chain rule.

Beware of possible substitutions prior to using FTC:

· For example, $u = 2t$, $\frac{1}{2}du = dt$ and rewrites: $\int_0^{4x} v(2t) dt = \frac{1}{2} \int_0^{8x} v(u) du$.

· Or Alt text: $u = 4x$, $\frac{du}{dx} = 4$ and rewrites: $\frac{d}{dx}[h(x)] = \frac{d}{du}[h(u)] \cdot \frac{du}{dx}$

· Such approaches can earn this point if done properly.

Second point: Chain rule

Multiplies by 4

It is not possible to earn this point without having earned the first point.

Depending on substitutions, this may look more like $\frac{1}{2}$ and 8 multiplied separately.

Third point: Answer

24 or equivalent

It is not possible to earn this point without having earned both the first and second points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

4-2-2b

$$p'(x) = \frac{d}{dx} \int_0^{4x} v(2t) dt = v(2 \cdot 4x) \cdot 4$$

$$p'(1) = v(8) \cdot 4 = 6 \cdot 4 = 24$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Narrows consideration to g' and g''

You must be convinced the response considers only the functions g' and g'' as being critical to the reasoning.

Second point: Answer with reason

A correct response must:

- Conclude “decreasing”
- Appeal to the signs of g' and g''

It is not possible to earn this point without having earned the first point.

A response that mentions velocity and acceleration must tie velocity to g' and acceleration to g'' . A response that says only “decreasing because velocity and acceleration have different signs” earns neither of the two points. It is not clear that the response is considering g' and g'' .

A response that calls out all three given values ($g(4)$, $g'(4)$, and $g''(4)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g' and g'' only, both points can be earned.

The phrase “slowing down” will be interpreted as “decreasing.”



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g' and g''
- ☐ Answer with reason

Solution:

Because $g'(4) = 2 > 0$, the velocity of the particle is positive. Because $g''(4) = -6 < 0$, the acceleration of the particle is negative. Therefore, because the signs of the velocity and acceleration are different, the speed of the particle is decreasing at time $t = 4$.

4-2-2b

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Tangent line equation

This point is earned only for the correct tangent line equation written in any form.

Some common forms:

$$y - 2 = (-2(3) + 3(2)^2)(x - 3)$$

$$y - 2 = 6(x - 3)$$

$$y = 6x - 16$$

$y - 2 = (-2x + 3y^2)(x - 3)$ does not earn the point. This is not the equation of a line. The response can go on to present one of the correct linear forms, and then would earn the point.

$f(x) = 6(x - 3) + 2$ or equivalent does not earn the point. $f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

Some responses have the correct tangent line equation written initially, with subsequent errors in simplification. Such responses do not earn the point.

The point can be earned after pursuing a less fruitful approach (such as solving the differential equation) and then restarting.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(3,2)} = -2 \cdot 3 + 3 \cdot 2^2 = 6$$

$$y = 6(x - 3) + 2$$

4-2-2b

57. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	3	8	11
$v(t)$ (meters per second)	-2	-5	-11	-12

The velocity of a particle, P , moving along the x -axis is modeled by a differentiable function v , where time t is measured in seconds and $v(t)$ is measured in meters per second. Selected values of $v(t)$ are shown in the table above.

(a) Use the data in the table to approximate $v'(5)$ using the average rate of change of $v(t)$ over the interval $3 \leq t \leq 8$. Show the computations that lead to your answer. Indicate units of measure.

(b) Interpret the meaning of $v'(5)$ in the context of the problem.

(c) Justify why there must be a time $t = d$, for $0 \leq d \leq 3$, when the velocity of the particle is -4 meters per second.

(d) Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^{11} v(t) \, dt$. Show the computations that lead to your answer.

(e) Find $\int_0^{10} v'\left(\frac{t}{2} + 3\right) \, dt$. Show the computations that lead to your answer.

(f) Let $r(x) = \int_0^{\frac{3}{2}x} v(2t) \, dt$. Find $r'(1)$. Show the computations that lead to your answer.

(g) The position of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(3) = -4$, $g'(3) = -2$, and $g''(3) = 6$. Is the speed of particle Q increasing or decreasing at time $t = 3$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = 3x^2 - y^3$ with initial condition $f(-1) = 2$. Write an equation for the line tangent to the graph of f at the point $(-1, 2)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-2-2b

First point: Approximation.

Need to see both a difference and a quotient as the supporting work. Minimal answers that earn the point include $\frac{-11-(-5)}{8-3}$, $\frac{-11-(-5)}{5}$, and $\frac{-6}{8-3}$.

The answer does not need to be simplified, but any attempt at simplification must be correct.

Numerical values must be pulled from the table. $\frac{v(8)-v(3)}{8-3}$ alone will not earn the point as it does not pull values from the table (yet).

Second point: Units

$\frac{m}{s^2}$ or meters per second per second will earn the point.

The units must be tied to a number. Units alone do not earn this point.

Beware of $\left(\frac{m}{s}\right)^2$ or $\frac{m^2}{s}$ (which do not earn the point).

This point must be earned in part (a). If no units are present in part (a) and correct units are shown in part (b), this point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$v'(5) \approx \frac{-11-(-5)}{8-3} = -\frac{6}{5} \text{ meters per second per second}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Interpretation

There are two components to look for:

- “rate of change of velocity” or “acceleration” of the particle
- at $t = 5$

4-2-2b

“The rate at which the velocity is decreasing at $t = 5$ ” is acceptable. If the answer to part (a) was reported to be positive, “the rate at which the velocity is increasing at $t = 5$ ” is acceptable. We will not penalize for the same error twice.

Responses that say “velocity is decreasing at a rate of $-\frac{6}{5} \frac{m}{s^2}$ ” or similar do not earn the point. This is interpreted to mean that the velocity is increasing.

Units are not required in part (b). If units are presented for t , they must be s (seconds).



0	1
---	---

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$v'(5)$ is the rate at which the velocity of the particle is changing, in meters per second per second, at time $t = 5$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: v is continuous

It must be clear that the reason given as to why v is continuous is that v is differentiable.

Neither “ v is differentiable and continuous” nor “ v differentiable, v continuous” are sufficient to earn this point.

“ v is differentiable hence continuous,” by contrast, earns the point.

“ v differentiable $\therefore v$ continuous” earns the point.

Second point: Bounding with table values

Must see the values -5 and -2 pulled from the table and correctly compared with -4 , in words or using inequality symbols.

The Intermediate Value Theorem (IVT) does not need to be referenced by name.

That said, this point will not be earned if other theorems (MVT, etc.) are invoked.

Responses often “say too much”; especially common are responses that indicate that v is always decreasing on the interval $[0, 3]$. This point will not be earned if the response contains such errors in reasoning.

4-2-2b



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ v continuous
- ☐ Bounding -4 with table values

Solution:

v is differentiable. $\Rightarrow v$ is continuous.

$$v(3) = -5 < -4 < -2 = v(0)$$

By the Intermediate Value Theorem, there must be a time $t = d$, $0 < d < 3$, such that the velocity of the particle is -4 meters per second.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Form of left Riemann sum

Must see a sum of three terms, each of which is a product of two factors: $(b_1 \cdot h_1) + (b_2 \cdot h_2) + (b_3 \cdot h_3)$

If exactly 5 out of 6 of these numbers (b_1 , b_2 , b_3 , h_1 , h_2 , h_3) are correctly placed in the above expression, we are convinced that the response carries the form of a left Riemann sum. In this case, award the first point. Such a response cannot earn the second point. If fewer than 5 out of 6 values are correct, no points are earned in this problem.

To earn this point, values do not need yet to be imported from the table: the expression

$$v(0) \cdot (3 - 0) + v(3) \cdot (8 - 3) + v(8) \cdot (11 - 8) \text{ is sufficient.}$$

Once the first point is earned, it cannot be lost. Subsequent errors result in the second point not being earned.

Second point: Answer

-64 or equivalent.

It is possible to earn this point without earning the first point. If there is credible evidence that an attempt was made to find a left Riemann sum but not enough work is shown, this point can be earned in the presence of the correct answer. For example, a sum of three terms with no evidence of products ($-6 + -25 + -33$) or a graph with areas labeled and summed to -64 or equivalent, can earn this point.

The response of only $(-2 \cdot 3) + (-5 \cdot 5) + (-11 \cdot 3)$ earns both points.

4-2-2b



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of left Riemann sum
- ☐ Answer

Solution:

$$\int_0^{11} v(t) dt \approx v(0)(3-0) + v(3)(8-3) + v(8)(11-8)$$

$$= -2 \cdot 3 + (-5) \cdot 5 + (-11) \cdot 3 = -64$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Substitution

There are three elements to look for in a correct substitution. (another variable may be chosen instead of u .)

- $u = \frac{t}{2} + 3$
- $du = \frac{dt}{2}$ or equivalent
- Appropriate limits of integration throughout (3 and 8 for u , 0 and 10 for t).

Substitution can be implicit. A coefficient of 2 accompanying a correct antiderivative convinces us of this:

$$\int_0^{10} v' \left(\frac{t}{2} + 3 \right) dt = 2v \left(\frac{t}{2} + 3 \right) \Big|_0^{10}$$

To elaborate on the third element above: if a response shows an otherwise correct substitution but the limits of integration on a definite integral are unchanged (0 and 10), this point is not earned.

Alternately, the integral can be worked with as an indefinite integral with the variable returned to $\frac{t}{2} + 3$ before evaluating as a means of eliminating the need to change limits. Such a response can earn the first point, if handled correctly.

Second point: Fundamental Theorem of Calculus

We must be convinced that the response has indicated an appropriate connection between $\int v'$ and v .

Expressions/equations that earn this point are:

4-2-2b

$$k(v(8) - v(3)) \text{ or } k\left(v\left(\frac{10}{2} + 3\right) - v\left(\frac{0}{2} + 3\right)\right),$$

· where k is a nonzero constant.

$$\cdot kv\left(\frac{t}{2} + 3\right)\Big|_0^{10} \text{ or } kv(u)\Big|_3^8, \text{ where } k \text{ is a nonzero constant.}$$

$$\cdot \int v'\left(\frac{t}{2} + 3\right) dt = kv\left(\frac{t}{2} + 3\right) \text{ or } k \int v'(u) du = kv(u), \text{ where } k \text{ is a nonzero constant, with or without appropriate limits of integration displayed on either integral.}$$

· $\int_0^{10} v'\left(\frac{t}{2} + 3\right) dt = kv\left(\frac{t^2}{4} + 3t\right)$ where k is a nonzero constant is a common response. This does not earn the point. Likewise, similar mishandled antiderivatives do not earn the point.

Third point: Answer

–12 or equivalent, with credible supporting work, earns this point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$\text{Let } u = \frac{t}{2} + 3. \text{ Then } du = \frac{dt}{2}.$$

$$\text{When } t = 0, u = 3.$$

$$\text{When } t = 10, u = 8.$$

$$\int_0^{10} v'\left(\frac{t}{2} + 3\right) dt = 2 \int_3^8 v'(u) du$$

$$= 2(v(8) - v(3))$$

$$= 2(-11 - (-5)) = -12$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-2-2b

First point: Fundamental Theorem of Calculus

Substitutes $\frac{3}{2}x$ into $v(2t)$. We must see either $v\left(2 \cdot \frac{3}{2}x\right)$ or $v(3x)$.

The expression $v\left(2 \cdot \frac{3}{2}x\right) - v(2(0))$ or $v\left(2 \cdot \frac{3}{2}x\right) \cdot \frac{3}{2} - v(2(0))$ will earn this point. The error of subtracting $v(2(0))$ will be viewed as an error in applying the chain rule.

Beware of possible substitutions prior to using FTC:

· For example, $u = 2t$, $\frac{1}{2}du = dt$ and rewrites: $\int_0^{\frac{3}{2}x} v(2t) dt = \frac{1}{2} \int_0^{3x} v(u) du$.

· Or $u = \frac{3}{2}x$, $\frac{du}{dx} = \frac{3}{2}$ and rewrites: $\frac{d}{dx}[h(x)] = \frac{d}{du}[h(u)] \cdot \frac{du}{dx}$

· Such approaches can earn this point if done properly.

Second point: Chain rule

Multiplies by $\frac{3}{2}$.

It is not possible to earn this point without having earned the first point.

Depending on substitutions, this may look more like $\frac{1}{2}$ and 3 multiplied separately.

Third point: Answer

$-\frac{15}{2}$ or equivalent

It is not possible to earn this point without having earned both the first and second points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$r'(x) = \frac{d}{dx} \int_0^{\frac{3}{2}x} v(2t) dt = v\left(2 \cdot \frac{3}{2}x\right) \cdot \frac{3}{2}$$

$$r'(1) = v(3) \cdot \frac{3}{2} = -5 \cdot \frac{3}{2} = -\frac{15}{2}$$

4-2-2b

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Narrows consideration to g' and g''

You must be convinced the response considers only the functions g' and g'' as being critical to the reasoning.

Second point: Answer with reason

A correct response must:

- Conclude “decreasing”
- Appeal to the signs of g' and g''

It is not possible to earn this point without having earned the first point.

A response that mentions velocity and acceleration must tie velocity to g' and acceleration to g'' . A response that says only “decreasing because velocity and acceleration have different signs” earns neither of the two points. It is not clear that the response is considering g' and g''

A response that calls out all three given values ($g(3)$, $g'(3)$, and $g''(3)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g' and g'' only, both points can be earned.

The phrase “slowing down” will be interpreted as “decreasing.”



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g' and g''
- ☐ Answer with reason

Solution:

Because $g'(3) = -2 < 0$, the velocity of the particle is negative. Because $g''(3) = 6 > 0$, the acceleration of the particle is positive. Because the signs of the velocity and acceleration are different, the speed of the particle is decreasing at time $t = 3$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Tangent line equation

4-2-2b

This point is earned only for the correct tangent line equation written in any form.

Some common forms:

$$y - 2 = (3(-1)^2 - 2^3)(x + 1)$$

$$y - 2 = -5(x + 1)$$

$$y = -5x - 3$$

$y - 2 = (3x^2 - y^3)(x + 1)$ does not earn the point. This is not the equation of a line. The response can go on to present one of the correct linear forms, and then would earn the point.

$f(x) = -5(x + 1) + 2$ or equivalent does not earn the point. $f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

Some responses have the correct tangent line equation written initially, with subsequent errors in simplification. Such responses do not earn the point.

The point can be earned after pursuing a less fruitful approach (such as solving the differential equation) and then restarting.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(-1, 2)} = 3 \cdot (-1)^2 - 2^3 = -5$$

$$y = 2 - 5(x + 1)$$

4-2-2b

58. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	3	8	11
$v(t)$ (meters per second)	-15	-12	-8	-3

The velocity of a particle, P , moving along the x -axis is modeled by a differentiable function v , where time t is measured in seconds and $v(t)$ is measured in meters per second. Selected values of $v(t)$ are shown in the table above.

- Use the data in the table to approximate $v'(10)$ using the average rate of change of $v(t)$ over the interval $8 \leq t \leq 11$. Show the computations that lead to your answer. Indicate units of measure.
- Interpret the meaning of $v'(10)$ in the context of the problem.
- Justify why there must be a time $t = k$, for $0 \leq k \leq 3$, when the velocity of the particle is -13 meters per second.
- Use a right Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^{11} v(t) \, dt$. Show the computations that lead to your answer.
- Find $\int_4^{14} v'\left(\frac{t}{2} + 1\right) \, dt$. Show the computations that lead to your answer.
- Let $h(x) = \int_3^{\frac{11}{2}x} v(2t) \, dt$. Find $h'(1)$. Show the computations that lead to your answer.
- The position of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(2) = -5.5$, $g'(2) = 4$, and $g''(2) = 3$. Is the speed of particle Q increasing or decreasing at time $t = 2$? Give a reason for your answer.
- Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = y^2 - xy$ with initial condition $f(3) = 2$. Write an equation for the line tangent to the graph of f at the point $(3, 2)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-2-2b

First point: Approximation.

Need to see both a difference and a quotient as the supporting work. Minimal answers that earn the point include $\frac{-3+8}{11-8}$, $\frac{-3-(-8)}{3}$, and $\frac{5}{11-8}$.

The answer does not need to be simplified, but any attempt at simplification must be correct.

Numerical values must be pulled from the table. $\frac{v(11)-v(8)}{11-8}$ alone will not earn the point as it does not pull values from the table (yet).

Second point: Units

$\frac{m}{s^2}$ end fraction or meters per second per second will earn the point.

The units must be tied to a number. Units alone do not earn this point.

Beware of $\left(\frac{m}{s}\right)^2$ or $\frac{m^2}{s}$ (which do not earn the point).

This point must be earned in part (a). If no units are present in part (a) and correct units are shown in part (b), this point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Approximation

☐ Units

Solution:

$$v'(10) \approx \frac{-3-(-8)}{11-8} = \frac{5}{3} \text{ meters per second per second}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Interpretation

There are two components to look for:

- “rate of change of velocity” or “acceleration” of the particle
- at $t = 10$

4-2-2b

“The rate at which the velocity is increasing at $t = 10$ ” is acceptable. If the answer to part (a) was reported to be negative, “the rate at which the velocity is decreasing at $t = 10$ ” is acceptable. We will not penalize for the same error twice.

Units are not required in part (b). If units are presented for t , they must be s (seconds).



0	1
---	---

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$v'(10)$ is the rate at which the velocity of the particle is changing, in meters per second per second, at time $t = 10$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: v is continuous

It must be clear that the reason given as to why v is continuous is that v is differentiable.

Neither “ v is differentiable and continuous” nor “ v differentiable, v continuous” are sufficient to earn this point.

“ v is differentiable hence continuous,” by contrast, earns the point.

“ v differentiable $\therefore v$ continuous” earns the point.

Second point: Bounding with table values

Must see the values -15 , and -12 pulled from the table and correctly compared to -13 , in words or using inequality symbols.

The Intermediate Value Theorem (IVT) does not need to be referenced by name.

That said, this point will not be earned if other theorems (MVT, etc.) are invoked.

Responses often “say too much”; especially common are responses that indicate that v is always increasing on the interval $[0, 3]$. This point will not be earned if the response contains such errors in reasoning.



0	1	2
---	---	---

4-2-2b

The student response accurately includes **both** of the criteria below.

- ☐ v continuous
- ☐ Bounding -13 with table values

Solution:

v is differentiable. $\Rightarrow v$ is continuous.

$$v(0) = -15 < -13 < -12 = v(3)$$

By the Intermediate Value Theorem, there must be a time $t = k$, for $0 \leq k \leq 3$, such that the velocity of the particle is -13 meters per second.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Form of right Riemann sum

Must see a sum of three terms, each of which is a product of two factors: $(b_1 \cdot h_1) + (b_2 \cdot h_2) + (b_3 \cdot h_3)$.

If exactly 5 out of 6 of these numbers $(b_1, b_2, b_3, h_1, h_2, h_3)$ are correctly placed in the above expression, we are convinced that the response carries the form of a right Riemann sum. In this case, award the first point. Such a response cannot earn the second point. If fewer than 5 out of 6 values are correct, no points are earned in this problem.

To earn this point, values do not need yet to be imported from the table: the expression $v(3) \cdot (3 - 0) + v(8) \cdot (8 - 3) + v(11) \cdot (11 - 8)$ is sufficient.

Once the first point is earned, it cannot be lost. Subsequent errors result in the second point not being earned.

second point: Answer

-85 or equivalent.

It is possible to earn this point without earning the first point. If there is credible evidence that an attempt was made to find a right Riemann sum but not enough work is shown, this point can be earned in the presence of the correct answer. For example, a sum of three terms with no evidence of products $(-36 + (-40) + (-9))$, or a graph with areas labeled and summed to -85 or equivalent, can earn this point.

The response of only $(-12) \cdot 3 + (-8) \cdot 5 + (-3) \cdot 3$ earns both points.



0	1	2
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4-2-2b

The student response accurately includes **both** of the criteria below.

- ☐ Form of right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{11} v(t) dt \approx v(3)(3-0) + v(8)(8-3) + v(11)(11-8)$$

$$= -12 \cdot 3 + (-8) \cdot 5 + (-3) \cdot 3 = -85$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Substitution

There are three elements to look for in a correct substitution. (another variable may be chosen instead of u .)

- $u = \frac{t}{2} + 1$
- $du = \frac{dt}{2}$ or equivalent
- Appropriate limits of integration throughout (3 and 8 for u , 4 and 14 for t).

Substitution can be implicit. A coefficient of 2 accompanying a correct antiderivative convinces us of this:

$$\int_4^{14} v' \left(\frac{t}{2} + 1 \right) dt = 2v \left(\frac{t}{2} + 1 \right) \Big|_4^{14}$$

To elaborate on the third element above: if a response shows an otherwise correct substitution but the limits of integration on a definite integral are unchanged (4 and 14), this point is not earned.

Alternately, the integral can be worked with as an indefinite integral with the variable returned to $\frac{t}{2} + 1$ before evaluating as a means of eliminating the need to change limits. Such a response can earn the first point, if handled correctly.

Second point: Fundamental Theorem of Calculus

We must be convinced that the response has indicated an appropriate connection between $\int v'$ and v .

Expressions/equations that earn this point are:

- $k(v(8) - v(3))$ or $k\left(v\left(\frac{14}{2} + 1\right) - v\left(\frac{4}{2} + 1\right)\right)$, where k is a nonzero constant.
- $kv\left(\frac{t}{2} + 1\right) \Big|_4^{14}$ or $kv(u) \Big|_3^8$, where k is a nonzero constant.

4-2-2b

$\cdot \int v' \left(\frac{t}{2} + 1 \right) dt = kv \left(\frac{t}{2} + 1 \right)$ or $k \int v' (u) du = kv (u)$, where k is a nonzero constant, with or without appropriate limits of integration displayed on either integral.

$\cdot \int_4^{14} v' \left(\frac{t}{2} + 1 \right) dt = k v \left(\frac{t^2}{4} + t \right)$ where k is a nonzero constant is a common response. This does not earn the point. Likewise, similar mishandled antiderivatives do not earn the point.

Third point: Answer

8 or equivalent, with credible supporting work, earns this point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{2} + 1$. Then $du = \frac{dt}{2}$.

When $t = 4$, $u = 3$.

When $t = 14$, $u = 8$.

$$\begin{aligned} \int_4^{14} v' \left(\frac{t}{2} + 1 \right) dt &= 2 \int_3^8 v' (u) du \\ &= 2 (v (8) - v (3)) \\ &= 2 (-8 - (-12)) = 8 \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Fundamental Theorem of Calculus (FTC)

Substitutes $\frac{11}{2}x$ into $v(2t)$. We must see either $v \left(2 \cdot \frac{11}{2}x \right)$ or $v(11x)$.

The expression $v \left(2 \cdot \frac{11}{2}x \right) - v (2 (3))$ or $v \left(2 \cdot \frac{11}{2}x \right) \cdot \frac{11}{2} - v (2 (3))$ will earn this point. The error of subtracting $v (2 (3))$ will be viewed as an error in applying the chain rule.

4-2-2b

Beware of possible substitutions prior to using FTC:

· For example, $u = 2t$, $\frac{1}{2}du = dt$ and rewrites: $\int_3^{\frac{11}{2}x} v(2t)dt = \frac{1}{2} \int_6^{11x} v(u)du$.

· Or $u = \frac{11}{2}x$, $\frac{du}{dx} = \frac{11}{2}$ and rewrites: $\frac{d}{dx}[h(x)] = \frac{d}{du}[h(u)] \cdot \frac{du}{dx}$

· Such approaches can earn this point if done properly.

Second point: Chain rule

Multiplies by $\frac{11}{2}$

It is not possible to earn this point without having earned the first point.

Depending on substitutions, this may look more like $\frac{1}{2}$ and 11 multiplied separately.

Third point: Answer

$-\frac{33}{2}$ or equivalent

It is not possible to earn this point without having earned both the first and second points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_3^{\frac{11}{2}x} v(2t) dt = v\left(2 \cdot \frac{11}{2}x\right) \cdot \frac{11}{2}$$

$$h'(1) = v(11) \cdot \frac{11}{2} = -3 \cdot \frac{11}{2} = -\frac{33}{2}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Narrows consideration to g' and g''

4-2-2b

You must be convinced the response considers only the functions g' and g'' as being critical to the reasoning.

Second point: Answer with reason

A correct response must:

- Conclude “increasing”
- Appeal to the signs of g' and g''

It is not possible to earn this point without having earned the first point.

A response that mentions velocity and acceleration must tie velocity to g' and acceleration to g'' . A response that says only “increasing because velocity and acceleration are both negative” earns neither of the two points. It is not clear that the response is considering g' and g'' . A response that calls out all three given values ($g(2)$, $g'(2)$, and $g''(2)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g' and g'' only, both points can be earned.

The phrase “speeding up” will be interpreted as “increasing.”



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g' and g''
- ☐ Answer with reason

Solution:

Because $g'(2) = 4 > 0$, the velocity of the particle is positive. Because $g''(2) = 3 > 0$, the acceleration of the particle is positive. Therefore, because both the velocity and acceleration are positive, the speed of the particle is increasing at time $t = 2$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Tangent line equation

This point is earned only for the correct tangent line equation written in any form.

Some common forms:

· $y - 2 = ((2)^2 - 3 \cdot 2)(x - 3)$

· $y - 2 = -2(x - 3)$

4-2-2b

$$y = -2x + 8$$

$y - 2 = (y^2 - xy)(x - 3)$ does not earn the point. This is not the equation of a line. The response can go on to present one of the correct linear forms, and then would earn the point.

$f(x) = -2(x - 3) + 2$ or equivalent does not earn the point. $f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

Some responses have the correct tangent line equation written initially, with subsequent errors in simplification. Such responses do not earn the point.

The point can be earned after pursuing a less fruitful approach (such as solving the differential equation) and then restarting.



0	1
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The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(3,2)} = 2^2 - 3 \cdot 2 = -2$$

$$y = -2(x - 3) + 2$$

4-2-2b

59. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	4	10	12
$v(t)$ (meters per second)	2	6	15	20

The velocity of a particle, P , moving along the x -axis is modeled by a differentiable function v , where time t is measured in seconds and $v(t)$ is measured in meters per second. Selected values of $v(t)$ are shown in the table above.

- Use the data in the table to approximate $v'(11)$ using the average rate of change of $v(t)$ over the interval $10 \leq t \leq 12$. Show the computations that lead to your answer. Indicate units of measure.
- Interpret the meaning of $v'(11)$ in the context of the problem.
- Justify why there must be a time $t = c$, for $4 \leq c \leq 10$, when the velocity of the particle is 8 meters per second.
- Use a right Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^{12} v(t) \, dt$. Show the computations that lead to your answer.
- Find $\int_4^{12} v' \left(\frac{t}{2} - 2 \right) \, dt$. Show the computations that lead to your answer.
- Let $h(x) = \int_2^{4x} v(3t) \, dt$. Find $h'(1)$. Show the computations that lead to your answer.
- The position of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(4) = 7$, $g'(4) = -3$, and $g''(4) = -2$. Is the speed of particle Q increasing or decreasing at time $t = 4$? Give a reason for your answer.
- Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = 3x^2 - y^3$ with initial condition $f(2) = 1$. Write an equation for the line tangent to the graph of f at the point $(2, 1)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-2-2b

First point: Approximation

Need to see both a difference and a quotient as the supporting work. Minimal answers that earn the point include $\frac{20-15}{12-10}$, $\frac{20-15}{2}$, and $\frac{5}{12-10}$.

The answer does not need to be simplified, but any attempt at simplification must be correct.

Numerical values must be pulled from the table. $\frac{v(12)-v(10)}{12-10}$ alone is not sufficient as it does not pull values from the table (yet).

Second point: Units

$\frac{m}{s^2}$ or meters per second per second will earn the point.

The units must be tied to a number. Units alone do not earn this point.

Beware of $\left(\frac{m}{s}\right)^2$ or $\frac{m^2}{s}$ (which do not earn the point).

This point must be earned in part (a). If no units are present in part (a) and correct units are shown in part (b), this point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$v'(11) \approx \frac{20-15}{12-10} = \frac{5}{2} \text{ meters per second per second}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Interpretation

There are two components to look for:

- “rate of change of velocity” or “acceleration” of the particle
- at $t = 11$

4-2-2b

“The rate at which the velocity is increasing at $t = 11$ ” is acceptable. If the answer to part (a) was reported to be negative, “the rate at which the velocity is decreasing at $t = 11$ ” is acceptable. We will not penalize for the same error twice.

Units are not required in part (b). If units are presented for t , they must be s (seconds).



0	1
---	---

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$v'(11)$ is the rate at which the velocity of the particle is changing, in meters per second per second, at time $t = 11$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: v is continuous

It must be clear that the reason given as to why v is continuous is that v is differentiable.

Neither “ v is differentiable and continuous” nor “ v differentiable, v continuous” are sufficient to earn this point.

“ v is differentiable hence continuous,” by contrast, earns the point.

“ v differentiable $\therefore v$ continuous” earns the point.

Second point: Bounding with table values

Must see the values 6 and 15 pulled from the table and correctly compared with 8, in words or using inequality symbols.

The Intermediate Value Theorem (IVT) does not need to be referenced by name.

That said, this point will not be earned if other theorems (MVT, etc.) are invoked.

Responses often “say too much”; especially common are responses that indicate that v is always increasing on the interval $[6, 15]$. This point will not be earned if the response contains such errors in reasoning.



0	1	2
---	---	---

4-2-2b

The student response accurately includes **both** of the criteria below.

- ☐ v continuous
- ☐ Bounding 8 with table values

Solution:

v is differentiable. $\Rightarrow v$ is continuous.

$$v(4) = 6 < 8 < 15 = v(10)$$

By the Intermediate Value Theorem, there must be a time $t = c$, for $4 < c < 10$, such that the velocity of the particle is 8 meters per second.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Form of right Riemann sum

Must see a sum of three terms, each of which is a product of two factors: $(b_1 \cdot h_1) + (b_2 \cdot h_2) + (b_3 \cdot h_3)$.

If exactly 5 out of 6 of these numbers (b_1 , b_2 , b_3 , h_1 , h_2 , h_3) are correctly placed in the above expression, we are convinced that the response carries the form of a right Riemann sum. In this case, award the first point. Such a response cannot earn the second point. If fewer than 5 out of 6 values are correct, no points are earned in this problem.

To earn this point, values do not need yet to be imported from the table: the expression $v(4) \cdot (4 - 0) + v(10) \cdot (10 - 4) + v(12) \cdot (12 - 10)$ is sufficient.

Once the first point is earned, it cannot be lost. Subsequent errors result in the second point not being earned.

Second point: Answer

154 or equivalent.

It is possible to earn this point without earning the first point. If there is credible evidence that an attempt was made to find a right Riemann sum but not enough work is shown, this point can be earned in the presence of the correct answer. For example, a sum of three terms with no evidence of products $(24 + 90 + 40)$, or a graph with areas labeled and summed to 154 or equivalent, can earn this point.

The response of only $6 \cdot 4 + 15 \cdot 6 + 20 \cdot 2$ earns both points.



0	1	2
---	---	---

4-2-2b

The student response accurately includes **both** of the criteria below.

- ☐ Form of right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{12} v(t) dt \approx v(4)(4-0) + v(10)(10-4) + v(12)(12-10)$$

$$= 6 \cdot 4 + 15 \cdot 6 + 20 \cdot 2 = 154$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Substitution

There are three elements to look for in a correct substitution. (another variable may be chosen instead of u .)

- $u = \frac{t}{2} - 2$
- $du = \frac{dt}{2}$ or equivalent
- Appropriate limits of integration throughout (0 and 4 for u , 4 and 12 for t).

Substitution can be implicit. A coefficient of 2 accompanying a correct antiderivative convinces us of this:

$$\int_4^{12} v' \left(\frac{t}{2} - 2 \right) dt = 2v \left(\frac{t}{2} - 2 \right) \Big|_4^{12}$$

To elaborate on the third element above: if a response shows an otherwise correct substitution but the limits of integration on a definite integral are unchanged (4 and 12), this point is not earned.

Alternately, the integral can be worked with as an indefinite integral with the variable returned to $\frac{t}{2} - 2$ before evaluating as a means of eliminating the need to change limits. Such a response can earn the first point, if handled correctly.

Second point: Fundamental Theorem of Calculus

We must be convinced that the response has indicated an appropriate connection between $\int v'$ and v .

Expressions/equations that earn this point are:

- $k(v(4) - v(0))$ or $k\left(v\left(\frac{12}{2} - 2\right) - v\left(\frac{4}{2} - 2\right)\right)$, where k is a nonzero constant.
- $kv\left(\frac{t}{2} - 2\right) \Big|_4^{12}$ or $kv(u) \Big|_0^4$, where k is a nonzero constant.

4-2-2b

$\cdot \int v' \left(\frac{t}{2} - 2 \right) dt = kv \left(\frac{t}{2} - 2 \right)$ or $k \int v' (u) du = kv (u)$, where k is a nonzero constant, with or without appropriate limits of integration displayed on either integral.

$\cdot \int_4^{12} v' \left(\frac{t}{2} - 2 \right) dt = k v \left(\frac{t^2}{4} - 2t \right)$ where k is a nonzero constant is a common response. This does not earn the point. Likewise, similar mishandled antiderivatives do not earn the point.

Third point: Answer

8 or equivalent, with credible supporting work, earns this point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{2} - 2$. Then $du = \frac{dt}{2}$.

When $t = 4$, $u = 0$.

When $t = 12$, $u = 4$.

$$\begin{aligned} \int_4^{12} v' \left(\frac{t}{2} - 2 \right) dt &= 2 \int_0^4 v' (u) du \\ &= 2 (v (4) - v (0)) \\ &= 2(6 - 2) = 8 \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Fundamental Theorem of Calculus

Substitutes $4x$ into $v (3t)$. We must see either $v (3 \cdot 4x)$ or $v (12x)$.

The expression $v (3 \cdot 4x) - v (3 (2))$ or $v (3 \cdot 4x) \cdot 4 - v (3 (2))$ will earn this point. The error of subtracting $v (3 (2))$ will be viewed as an error in applying the chain rule.

4-2-2b

Beware of possible substitutions prior to using FTC:

· For example, $u = 3t$, $\frac{1}{3}du = dt$ and rewrites: $\int_2^{4x} v(3t) dt = \frac{1}{3} \int_6^{12x} v(u) du$.

· Or $u = 4x$, $\frac{du}{dx} = 4$ and rewrites: $\frac{d}{dx}[h(x)] = \frac{d}{du}[h(u)] \cdot \frac{du}{dx}$

· Such approaches can earn this point if done properly.

Second point: Chain rule

Multiplies by 4

It is not possible to earn this point without having earned the first point.

Depending on substitutions, this may look more like $\frac{1}{3}$ and 12 multiplied separately.

Third point: Answer

80 or equivalent

It is not possible to earn this point without having earned both the first and second points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_2^{4x} v(3t) dt = v(3 \cdot 4x) \cdot 4$$

$$h'(1) = v(12) \cdot 4 = 20 \cdot 4 = 80$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Narrows consideration to g' and g''

You must be convinced the response considers only the functions g' and g'' as being critical to the reasoning.

4-2-2b

Second point: Answer with reason

A correct response must:

- Conclude “increasing”
- Appeal to the signs of g' and g''

It is not possible to earn this point without having earned the first point.

A response that mentions velocity and acceleration must tie velocity to g' and acceleration to g'' . A response that says only “increasing because velocity and acceleration are both negative” earns neither of the two points. It is not clear that the response is considering g' and g'' . A response that calls out all three given values ($g(4)$, $g'(4)$, and $g''(4)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g' and g'' only, both points can be earned.

The phrase “speeding up” will be interpreted as “increasing.”



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g' and g''
- ☐ Answer with reason

Solution:

Because $g'(4) = -3 < 0$, the velocity of the particle is negative. Because $g''(4) = -2 < 0$, the acceleration of the particle is negative. Therefore, because both the velocity and acceleration are negative, the speed of the particle is increasing at time $t = 4$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Tangent line equation

This point is earned only for the correct tangent line equation written in any form.

Some common forms:

- $y - 1 = (3(2)^2 - (1)^3)(x - 2)$
- $y - 1 = 11(x - 2)$
- $y = 11x - 21$

4-2-2b

f does not earn the point. This is not the equation of a line. The response can go on present one of the correct linear forms, and then would earn the point.

$f(x) = 11(x - 2) + 1$ or equivalent does not earn the point. $f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

Some responses have the correct tangent line equation written initially, with subsequent errors in simplification. Such responses do not earn the point.

The point can be earned after pursuing a less fruitful approach (such as solving the differential equation) and then restarting.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(2,1)} = 3 \cdot 2^2 - 1^3 = 11$$

$$y = 11(x - 2) + 1$$

4-2-2b

60. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	3	6	9	12
$p(t)$ (feet)	3	5.5	7	13	17

The position of a particle, A , moving along the x -axis is modeled by a twice-differentiable function p , where time t is measured in seconds and $p(t)$ is measured in feet. Selected values of $p(t)$ are shown in the table above.

(a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate the value of $\int_0^{12} p(t) \, dt$. Show the computations that lead to your answer.

(b) Use the data in the table to approximate $p'(10)$ using the average rate of change of $p(t)$ over the interval $9 \leq t \leq 12$. Show the computations that lead to your answer. Indicate units of measure.

(c) Interpret the meaning of $p'(10)$ in the context of the problem.

(d) Must there exist a value of c , for $0 < c < 12$, such that $p'(c)$ is equal to the average rate of change of p over the interval $0 \leq x \leq 12$? Justify your answer.

(e) Find $\int_0^3 p'(2t + 3) \, dt$. Show the computations that lead to your answer.

(f) Let $h(x) = \int_2^{3x} p(3t) \, dt$. Find $h'(1)$. Show the computations that lead to your answer.

(g) The velocity of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(2) = -6$, $g'(2) = 4$, and $g''(2) = -3$. Is the speed of particle Q increasing or decreasing at time $t = 2$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = 4x^2y^3$ with initial condition $f\left(\frac{1}{2}\right) = -2$. Write an equation for the line tangent to the graph of f at the point $\left(\frac{1}{2}, -2\right)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-2-2b

First Point

Philosophy: Earned for correct midpoint Riemann sum, with appropriate values taken from the table.

Need to see sum of two terms, each as a product of two factors, i.e. $\text{base}_1 \cdot \text{height}_1 + \text{base}_2 \cdot \text{height}_2$

$p(3) \cdot (6 - 0) + p(9) \cdot (12 - 6)$ or $p(3) \cdot (6) + p(9) \cdot (6)$ earns first point.

If at least three of the four base/height values are correct in the sum of products, the expression earns the first point.

Second Point

111 or equivalent

Can be earned without first point with Riemann sum setup demonstration. With credible evidence that a midpoint Riemann sum attempt was made without not enough work shown, this point can be earned in the presence of the correct answer. For example, a sum of two terms with no evidence of products ($33 + 78$) or a graph with areas labeled and summed to 111 or equivalent, can earn this point.

Examples:

- $111 = 33 + 78$ by itself earns second point, not first point
- 111 declared, with $5.5 \cdot 6$ and $13 \cdot 6$ identified (perhaps graphically) earns second point, not first point
- $k(5.5 \cdot 6 + 13 \cdot 6)$, nonzero constant $k \neq 1$ by itself earns first point, not second point
- $p(3) \cdot (6) + p(9) \cdot (12 - 6)$ by itself earns first point, not second point



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint Riemann sum
- ☐ Answer

Solution:

$$\int_0^{12} p(t) dt \approx p(3)(6 - 0) + p(9)(12 - 6)$$

$$= 5.5 \cdot 6 + 13 \cdot 6 = 111$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-2-2b

First Point

Philosophy: Earned for correct difference quotient, with appropriate values from the table.

Need to see both a difference and a quotient

$\frac{p(12)-p(9)}{12-9}$ by itself doesn't earn first point; however $\frac{17-13}{12-9}$ or $\frac{17-13}{3}$ by itself earns first point

Second Point

Earned for correct units of ft per sec or ft/sec or f/s.

Correct units must be tied to a number to earn second point.

Correct units must appear in part (b); correct units in part (c) only doesn't earn second point.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$p'(10) \approx \frac{17-13}{12-9} = \frac{4}{3} \text{ feet per second}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct $p'(10)$ interpretation.

Need two components in response:

- “rate of change of position” or “velocity” of the particle
- at $t = 10$

Correct ways to describe rate at time at $t = 10$ include:

- Rate at which position is changing/increasing
- Rate at which particle is moving right
- Instantaneous velocity or velocity of particle

4-2-2b

Ambiguous/incorrect ways to describe rate include:

- Rate at which particle is changing/increasing
- Rate at which particle is moving

Units are not required in part (c). If units are presented for t , they must be in seconds.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Interpretation

Solution:

$p'(10)$ is the rate at which the position of the particle is changing, in feet per second, at time $t = 10$ seconds.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for both Mean Value Theorem (MVT) conditions with the appropriate implication that p is continuous because we know that p is differentiable.

Ways to earn first point include:

- p differentiable $\Rightarrow p$ continuous
- p differentiable, hence p continuous
- p differentiable, so p continuous

p differentiable and p continuous does not earn first point.

Second Point

Philosophy: Earned for concluding “yes” and invoking Mean Value Theorem or its equivalent formulation, with consideration of the continuity of p .

MVT title does not have to be stated to earn second point.

This point is not earned if other theorems (IVT, etc.) are invoked.

If presented, the value for $p'(c) = \frac{7}{6}$ must be correct to earn second point.

4-2-2b

Examples:

- “By MVT, yes since p differentiable” earns neither point
- “By MVT, yes since p continuous” earns second point only
- “By MVT, yes since p differentiable and continuous” earns second point only
- “By MVT, yes since p differentiable hence continuous” earns both points
- “By MVT, yes since p twice differentiable hence continuous” earns both points
- “By MVT, yes since function differentiable hence continuous” earns both points



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

p is differentiable. $\Rightarrow p$ is continuous.

Therefore, the Mean Value Theorem can be applied to p on the interval $0 \leq x \leq 12$ to guarantee that there exists a value of c , for $0 < c < 12$, such that $p'(c)$ equals the average rate of change of p over the interval $0 \leq x \leq 12$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct substitution application.

Need three elements in response (with consistent use of u or another substitution variable):

- $u = 2t + 3$ (can be explicit or implicit)
- $du = 2 dt$ (yielding integral coefficient of $\frac{1}{2}$)
- Correct integral limits throughout, with t from 0 to 3 and u from 3 to 9

Omitting the coefficient of $\frac{1}{2}$ or using some other value will not earn first point; response could earn second point, but not third point.

4-2-2b

In the third element above, if response shows an otherwise correct substitution with unchanged limits of integration on a definite integral from 0 to 3, then point is not earned.

Alternately, the integral can be worked with as an indefinite integral with u returned to $2t + 3$ in the antiderivative before evaluating as a means to eliminate the need to change limits. Such a response can earn the first point if handled correctly.

For example: $\int_0^3 p'(2t + 3) dt = \frac{1}{2}p(2t + 3) \Big|_0^3$ earns the substitution point.

Second Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, with evidence of connection between $\int p'(u)$ and $p(u)$.

Ways to earn second point include:

· $k(p(9) - p(3))$ or $k(p(2 \cdot 3 + 3) - p(2 \cdot 0 + 3))$ for k a nonzero constant

· $k p(2t + 3) \Big|_0^3$ or $k p(u) \Big|_3^9$ for k a nonzero constant

· $\int p'(2t + 3) dt = k p(2t + 3)$ or $\int p'(u) du = k p(u)$ for any nonzero constant k , with or without appropriate limits of integration on either integral

First and second points can be earned simultaneously by responses such as $\int_0^3 p'(2t + 3) dt = \frac{1}{2}p(2t + 3) \Big|_0^3$

Third Point

Earned only for an answer of 3.75 or equivalent value.

Value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring Examples:

· $\frac{1}{2} \int p'(u) du = \frac{1}{2}p(u)$ earns 0 – 1 – 0

· $\frac{1}{2} \int p'(u) du = \frac{1}{2}p(u) \Big|_0^3 = \frac{1}{2}(p(3) - p(0))$ earns 0 – 1 – 0

· $\int p'(u) du = p(u) \Big|_3^9 = (p(9) - p(3)) = 7.5$ earns 0 – 1 – 0

· $2 \int p'(u) du = 2p(u) \Big|_0^3 = 2[13 - 5.5]$ earns 0 – 1 – 0

· $\int_0^3 p'(2t + 3) dt = p(t^2 + 3t) \Big|_0^3 = p(18) - p(0)$ earns 0 – 0 – 0

4-2-2b

$$\cdot \frac{1}{2} \int_0^3 p'(u) \, du = \frac{1}{2}(p(9) - p(3)) = 3.75 \text{ earns } 0 - 1 - 1$$

$$\cdot \int_0^3 p'(2t + 3) \, dt = \frac{1}{2}(p(9) - p(3)) = 3.75 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^3 p'(2t + 3) \, dt = \frac{1}{2}p(u) \Big|_3^9 = \frac{1}{2}(p(9) - p(3)) = 3.75 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^3 p'(2t + 3) \, dt = \frac{1}{2}p(2t + 3) \Big|_0^3 = \frac{1}{2}(p(9) - p(3)) \text{ earns } 1 - 1 - 0$$

$$\cdot \frac{1}{2} \int p'(u) = \frac{1}{2}p(u); \frac{1}{2}p(2t + 3) \Big|_0^3 = \frac{1}{2}(13 - 5.5) \text{ earns } 1 - 1 - 1$$



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t + 3$. Then $du = 2 \, dt$.

When $t = 0$, $u = 3$.

When $t = 3$, $u = 9$.

$$\int_0^3 p'(2t + 3) \, dt = \frac{1}{2} \int_3^9 p'(u) \, du$$

$$= \frac{1}{2}(p(9) - p(3))$$

$$= \frac{1}{2}(13 - 5.5) = 3.75$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

4-2-2b

Philosophy: Earned for correct Fundamental Theorem of Calculus application, resulting in $a \cdot p(9x)$ for some nonzero constant a .

Ways to earn first point include responses of $p(3 \cdot 3x)$ or $p(9x)$ (Note $p(9t)$ would also earn this point.)

The expression $p(3 \cdot 3x) - p(3 \cdot 2)$ or $(p(3 \cdot 3x) - p(3 \cdot 2)) \cdot 3$ earns this point. The $p(3 \cdot 2)$ is viewed as a chain rule error.

Any response correctly presenting and evaluating the derivative of the integral in terms of the substitution $u = 3t$ earns first point. Part of this response may include $\int_2^{3x} p(3t) dt = \frac{1}{3} \int_6^{9x} p(u) du$.

Second Point

Philosophy: This point is earned for correctly applying the chain rule to find $h'(x)$ or $h'(1)$. That is accomplished by multiplying the term involving $p(x)$ by 3.

Can only be earned if response earned first (FTC) point.

The first two points can be, and often are, earned simultaneously.

The point is not earned if the presented derivative is $a \cdot p(9x)$ for any $a \neq 3$.

The point is not earned for responses of $3p(3t) \big|_2^{3x}$ or $3(p(9x) - p(b))$, for any value of b .

Third Point

Can only be earned if response earned first (FTC) and second (chain rule) points.

Earned only for answer of 39 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring examples:

$\cdot p(3 \cdot 3x) - p(6); p(9) - p(6) = 13 - 7$ earns 1 – 0 – 0

$\cdot p(3 \cdot 3x) \cdot 3 - p(6) \cdot 0; 39$ earns 1 – 1 – 1

$\cdot 3p(3t) \big|_2^{3x} = 3(p(9) - p(6))$ earns 1 – 0 – 0

$\cdot p(3x) \cdot 3; p(3) \cdot 3 = 5.5 \cdot 3$ earns 0 – 0 – 0

$\cdot p(9x) \cdot 3; p(9) \cdot 3$ earns 1 – 1 – 0



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

4-2-2b

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_2^{3x} p(3t) dt = p(3 \cdot 3x) \cdot 3$$

$$h'(1) = p(9) \cdot 3 = 13 \cdot 3 = 39$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for narrowing consideration of given information to g and g' .

g and g' need not be evaluated to earn the first point.

Considering velocity and acceleration not sufficient to earn first point, unless acceleration is linked to g' .

If response considers g and g' and g'' , then eventually g'' must be eliminated to earn first point.

Second Point

Philosophy: Earned for “decreasing” with reason, using the opposite signs of g and g' .

Ways to earn second point include:

- “decreasing since $g < 0$ and $g' > 0$ ”
- “particle slowing down since g and g' have opposite signs”

Can only be earned if response earned first point.

Examples:

- “decreasing since velocity and acceleration have opposite signs” earns neither point
- “decreasing since velocity < 0 and acceleration > 0 ” earns neither point
- “decreasing since $(-6) \cdot 4 = -24 < 0$ ” earns both points



0	1	2
---	---	---

4-2-2b

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g and g'
- ☐ Answer with reason

Solution:

Because $g(2) = -6 < 0$, the velocity of the particle is negative. Because $g'(2) = 4 > 0$, the acceleration of the particle is positive. Because the velocity and acceleration have opposite signs, the speed of the particle is decreasing at time $t = 2$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct tangent line equation.

Correct responses include:

· $y = -8\left(x - \frac{1}{2}\right) - 2$ or $y + 2 = -8\left(x - \frac{1}{2}\right)$

· $y = -8x + 2$

· $y = \left(4 \cdot \left(\frac{1}{2}\right)^2 \cdot (-2)^3\right)\left(x - \frac{1}{2}\right) - 2$

Incorrect responses include:

· $y = (4x^2y^3)\left(x - \frac{1}{2}\right) - 2$

· $y = 8x + 2$

· $f(x) = -8x + 2$



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Tangent line equation

Solution:

$$\frac{dy}{dx} \Big|_{\left(\frac{1}{2}, -2\right)} = 4 \cdot \left(\frac{1}{2}\right)^2 \cdot (-2)^3 = -8$$

4-2-2b

$$y = -8 \left(x - \frac{1}{2} \right) - 2$$

4-2-2b

61. A particle is traveling along the x -axis. For time $0 \leq t \leq 5$, the velocity of the particle is defined by $v(t) = 3^{-t^2} - \sqrt{t + 0.4}$. At time $t = 5$, the particle is at position $x = -4$.

**Part A**

Find $a(2)$, the acceleration of the particle at time $t = 2$. Show the setup for your calculations. Is the speed of the particle increasing or decreasing at time $t = 2$? Give a reason for your answer.

Part B

At what time in the interval $0 \leq t \leq 5$ does the particle change direction? Explain your reasoning.

Part C

What is the position of the particle at time $t = 0$? Show the work that leads to your answer.

Part D

Find the total distance traveled by the particle over the time interval $0 \leq t \leq 5$. Show the setup for your calculations.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P1** is earned for the correct answer with a setup demonstrating the relationship $v' = a$. This could be shown by “ $v'(2) = -0.377$ ” or “ $v' = a$ and $a(2) = -0.377$.” A response of just “ $a(2) = -0.377$ ” is not sufficient to earn **P1**.
- For **P1**, a reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn the point.
- A response with an unsupported value of -0.377 does not earn **P1**.

Model Solution:

$$a(2) = v'(2) = -0.377001$$

The acceleration of the particle at time $t = 2$ is -0.377 .



0	1
---	---

The response accurately includes the criteria below.

- Acceleration with setup

4-2-2b

Part A Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To be eligible for **P2**, a response must declare a value of $a(2)$, declare a value of $v'(2)$, or present the value -0.377 . A response with an unsupported value of -0.377 is eligible for **P2**.
- P2** can only be earned for a response that is consistent with a negative velocity at time $t = 2$ and the presented value of $a(2)$.
- Any presented value of $v(2)$ must be correct to the number of decimal places presented (up to three decimal places) in order to be eligible for **P2**.
- To earn **P2**, the response does not need to present a value of $v(2)$; an implied sign is sufficient. Any statement equivalent to “The speed of the particle is increasing because $a(2)$ and $v(2)$ have the same sign” will earn **P2**, provided the presented value of $a(2)$ is negative.
- Alternate solutions for **P2**:
 - Speed $= |v(t)|$ and its derivative is positive when $t = 2$, therefore the speed of the particle is increasing.
 - $a(2)v(2) > 0$, therefore the speed of the particle is increasing.

Model Solution:

$$v(2) = -1.536848 < 0$$

Because $v(2)$ and $a(2) = v'(2)$ have the same sign, the speed of the particle is increasing at time $t = 2$.



0	1
---	---

The response accurately includes the criteria below.

- Answer with reason

Part B Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- P3** can be earned by explicitly presenting $v(t) = 0$.
- P3** can also be earned by identifying $t = 0.356$.

Model Solution:

$$v(t) = 0 \Rightarrow t = 0.356432$$

4-2-2b



0	1
---	---

The response accurately includes the criteria below.

- Considers $v(t) = 0$

Part B Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P4**, a response does not need to say $v(t)$ changes sign “from positive to negative,” but it cannot say $v(t)$ changes sign “from negative to positive.”
- For **P4**, a reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn the point, unless an earlier point was not earned due to inappropriate rounding.

Model Solution:

The particle changes direction at time $t = 0.356$ because $v(t)$ changes sign there (from positive to negative).



0	1
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The response accurately includes the criteria below.

- Answer with reason

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P5** is earned with a definite integral with correct lower and upper limits in either order, with or without the differential.
 - Correct handling of the sign of a reversal in the limits of integration is not assessed in **P5**.
 - Writing an indefinite integral and never presenting a definite integral does not earn **P5**.

Model Solution:

4-2-2b

$$x(0) = x(5) + \int_5^0 v(t)dt = -4 + \int_5^0 v(t)dt$$



0

1

The response accurately includes the criteria below.

- Integral

Part C Point 6

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P6** is earned for adding the initial condition to a definite or indefinite integral. This addition may be presented explicitly (e.g., $-4 \pm \int_5^0 v(t)dt$) or implied by the combination of a correct integral and answer (e.g., “ $\int_5^0 v(t)dt = 7.351$, so 3.351”). A response of $x(5) \pm \int_5^0 v(t)dt$ alone has not yet earned **P6**, but it will earn **P6** if the response also explicitly identifies “ $x(5) = -4$ ” or presents the correct answer.
- If the differential dt is missing:
 - Writing $-4 \pm \int_5^0 v(t)dt$ or $-4 \pm \int_0^5 v(t)$ earns **P6**.
 - Because $\pm \int_5^0 v(t) - 4$ may mean $\pm \int_5^0 v(t)dt - 4$ or $\pm \int_5^0 (v(t) - 4)dt$:
 - An integral of $\int_5^0 v(t) - 4$ or $-\int_0^5 v(t) - 4$ in the presence of the correct answer of 3.351 provides sufficient evidence to infer that the correct placement of the differential was used. Such a response earns **P6**.
 - The equation $\int_0^5 v(t) - 4 = -11.351$ or $-\int_5^0 v(t) - 4 = -11.351$ earns **P6**.
 - An integral of $\int_5^0 v(t) - 4$ or $-\int_0^5 v(t) - 4$ in the presence of a correct but inappropriately rounded answer of 3, 3.3, 3.4 or 3.35 earns **P6**.
 - Writing $\int_5^0 v(t) - 4$ or $-\int_0^5 v(t) - 4$ in the presence of no answer or an incorrect answer does not provide sufficient evidence that the correct placement of the differential was used. Such a response does not earn **P6**.

Model Solution:

4-2-2b

$$x(0) = x(5) + \int_5^0 v(t) dt = -4 + \int_5^0 v(t) dt$$



0	1
---	---

The response accurately includes the criteria below.

- Uses initial condition

Part C Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P7** is earned for the correct answer, with or without supporting work. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn the point, unless an earlier point was not earned due to inappropriate rounding.
- Incorrect or unclear communication between the correct integral and the correct answer is treated as scratch work and is not considered in scoring. The following examples earn **P7** for the correct answer:

- $-4 - \int_5^0 v(t) dt = 3.351$
- $-4 + \int_0^5 v(t) dt = 3.351$
- $\int_5^0 v(t) dt = 7.351 - 4 = 3.351$
- $\int_5^0 v(t) dt = 3.351$

Model Solution:

The position of the particle at time $t = 0$ is 3.351.



0	1
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The response accurately includes the criteria below.

- Answer

Part D Point 8

4-2-2b

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P8** is earned for any of the following, with or without the differential dt :
 - $\int_0^5 |v(t)| dt$
 - $\int_0^a v(t) dt - \int_a^5 v(t) dt$ for $0.35 \leq a \leq 0.36$, or equivalent
 - $-\int_0^a v(t) dt + \int_a^5 v(t) dt$ for $0.35 \leq a \leq 0.36$ (This will produce a negative answer of -7.493 , which can be resolved to give the correct answer.)
- Alternate solution:

$$\int_0^{0.356432} v(t) dt - \int_{0.356432}^5 v(t) dt$$

Model Solution:

$$\int_0^5 |v(t)| dt$$



0	1
---	---

The response accurately include the criteria below.

- Setup

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P9** is earned for the correct answer, with or without supporting work. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn the point, unless an earlier point was not earned due to inappropriate rounding.

Model Solution:

The total distance traveled by the particle is 7.493 (or 7.492).

4-2-2b



0	1
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The response accurately includes the criteria below.

- Answer

A particle moves along the y -axis with velocity given by $v(t) = t \sin(t^2)$ for $t \geq 0$.

62.  In what direction (up or down) is the particle moving at time $t = 1.5$?

Part A

1 point is earned for the answer and reason

$$v(1.5) = 1.5 \sin(1.5^2) = 1.167$$

Up, because $v(1.5) > 0$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer and reason

$$v(1.5) = 1.5 \sin(1.5^2) = 1.167$$

Up, because $v(1.5) > 0$

4-2-2b

63.  Find the acceleration of the particle at time $t = 1.5$. Is the velocity of the particle increasing at $t = 1.5$? Why or why not?

Part B

1 point is earned for: $a(1.5)$

1 point is earned for the conclusion and reasons

$$a(t) = v'(t) = \sin t^2 + 2t^2 \cos t^2$$

$$a(1.5) = v'(1.5) = -2.048 \text{ or } -2.049$$

No; v is decreasing at 1.5 because $v'(1.5) < 0$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $a(1.5)$

1 point is earned for the conclusion and reasons

$$a(t) = v'(t) = \sin t^2 + 2t^2 \cos t^2$$

$$a(1.5) = v'(1.5) = -2.048 \text{ or } -2.049$$

No; v is decreasing at 1.5 because $v'(1.5) < 0$

4-2-2b

64.  A particle moves along the y -axis with velocity given by $v(t) = t \sin(t^2)$ for $t \geq 0$.
- (a) In which direction (up or down) is the particle moving at time $t = 1.5$? Why?
- (b) Find the acceleration of the particle at time $t = 1.5$. Is the velocity of the particle increasing at $t = 1.5$? Why or why not?
- (c) Given that $y(t)$ is the position of the particle at time t and that $y(0) = 3$, find $y(2)$.
- (d) Find the total distance traveled by the particle from $t = 0$ to $t = 2$.

Part A

1 point is earned for the answer and reason

$$v(1.5) = 1.5 \sin(1.5^2) = 1.167$$

Up, because $v(1.5) > 0$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer and reason

$$v(1.5) = 1.5 \sin(1.5^2) = 1.167$$

Up, because $v(1.5) > 0$

Part B

1 point is earned for: $a(1.5)$

1 point is earned for the conclusion and reason

$$a(t) = v'(t) = \sin t^2 + 2t^2 \cos t^2$$

$$a(1.5) = v'(1.5) = -2.048 \text{ or } -2.049$$

No; v is decreasing at 1.5 because $v'(1.5) < 0$



0	1	2
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The student response earns all of the following points:

1 point is earned for: $a(1.5)$

4-2-2b

1 point is earned for the conclusion and reason

$$a(t) = v'(t) = \sin t^2 + 2t^2 \cos t^2$$

$$a(1.5) = v'(1.5) = -2.048 \text{ or } -2.049$$

No; v is decreasing at 1.5 because $v'(1.5) < 0$

Part C

1 point is earned for: $y(t) = \int v(t) dt$

1 point is earned for: $y(t) = -\frac{1}{2} \cos t^2 + C$

$$y(t) = \int v(t) dt$$

$$= \int t \sin t^2 dt = -\frac{\cos t^2}{2} + C$$

1 point is earned for:

$$y(0) = 3 = -\frac{1}{2} + C \Rightarrow C = \frac{7}{2}$$

ed for: $y(2)$

$$y(t) = -\frac{1}{2} \cos t^2 + \frac{7}{2}$$

$$y(2) = -\frac{1}{2} \cos 4 + \frac{7}{2} = 3.826 \text{ or } 3.827$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for: $y(t) = \int v(t) dt$

1 point is earned for: $y(t) = -\frac{1}{2} \cos t^2 + C$

1 point is earned for: $y(2)$

4-2-2b

$$y(t) = \int v(t) dt$$

$$= \int t \sin t^2 dt = -\frac{\cos t^2}{2} + C$$

$$y(0) = 3 = -\frac{1}{2} + C \Rightarrow C = \frac{7}{2}$$

$$y(t) = -\frac{1}{2}\cos t^2 + \frac{7}{2}$$

$$y(2) = -\frac{1}{2}\cos 4 + \frac{7}{2} = 3.826 \text{ or } 3.827$$

Part D

1 point is earned for the limits of 0 and 2 on an integral of $v(t)$ or $|v(t)|$

or

uses $y(0)$ and $y(2)$ to compute distance

1 point is earned for the handles change of direction at student's turning point

1 point is earned for the answer

0/1 if incorrect turning point

$$\text{distance} = \int_0^2 |v(t)| dt = 1.173$$

or

$$v(t) = t \sin t^2 = 0$$

$$t = 0 \text{ or } t = \sqrt{\pi} \approx 1.772$$

$$y(0) = 3; y(\sqrt{\pi}) = 4; y(2) = 3.826 \text{ or } 3.827$$

$$[y(\sqrt{\pi}) - y(0)] + [y(2) - y(\sqrt{\pi})]$$

$$= 1.173 \text{ or } 1.174$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the limits of 0 and 2 on an integral of $v(t)$ or $|v(t)|$

or

uses $y(0)$ and $y(2)$ to compute distance

1 point is earned for the handles change of direction at student's turning point

4-2-2b

1 point is earned for the answer

0/1 if incorrect turning point

$$\text{distance} = \int_0^2 |v(t)| dt = 1.173$$

or

$$v(t) = t \sin t^2 = 0$$

$$t = 0 \text{ or } t = \sqrt{\pi} \approx 1.772$$

$$y(0) = 3; y(\sqrt{\pi}) = 4; y(2) = 3.826 \text{ or } 3.827$$

$$[y(\sqrt{\pi}) - y(0)] + [y(\sqrt{\pi}) - y(2)]$$

$$= 1.173 \text{ or } 1.174$$

65.  A particle moves along a straight line with velocity given by $v(t) = t^2 - 3t + e^{-t/2}$ and acceleration given by $a(t) = 2t - 3 - \frac{1}{2}e^{-t/2}$ for time $0 \leq t \leq 4$. On which open intervals is the speed of the particle increasing?

- (A) $(0, 0.318)$ and $(2.921, 4)$
 (B) $(1.612, 4)$ only
 (C) $(0, 0.318)$ and $(1.612, 2.921)$
 (D) $(0.318, 1.612)$ and $(2.921, 4)$



Answer D

Correct. The speed of the particle is increasing on intervals where the velocity $v(t)$ and the acceleration $a(t)$ have the same sign. This will happen where $v(t) \cdot a(t) > 0$. The graphing calculator is used to solve $v(t)a(t) = 0$ and examine the graph of $v(t)a(t)$. Since $v(t)a(t) = 0$ at $t = 0.318$, $t = 1.612$, and $t = 2.921$, the graph of $v(t)a(t)$ indicates that $v(t)a(t)$ is positive for $0.318 < t < 1.612$ and $2.921 < t < 4$.

4-2-2b

66. A toy train moves along a straight set up on a table. The position $x(t)$ of the train at the time t seconds is measured in centimeters from the center of the track. At time $t=1$, the train is 6 centimeters to the left of the center, so $x(1)=-6$. For $0 \leq t \leq 4$, the velocity of the train at the time t is given by $v(t)=3t^2-12$, where $v(t)$ is measured in centimeters per second.

Find the total distance traveled by the train during the time interval $0 \leq t \leq 4$.

Part B

The response can earn up to 3 points:

1 point: For identification of $t=2$

$$v(t) = 3(t^2 - 4) = 0 \Rightarrow t = -2, t = 2$$

1 point: For consideration of $x(0)$, $x(2)$ and $x(4)$

1 point: For the correct answer

$$= |x(2) - x(0)| + |x(4) - x(2)|$$

$$\text{Distance} = |-11 - 5| + |21 - (-11)|$$

$$= 16 + 32 = 48$$



0	1	2	3
---	---	---	---

The response earns all three of the following points:

1 point: For identification of $t=2$

$$v(t) = 3(t^2 - 4) = 0 \Rightarrow t = -2, t = 2$$

1 point: For consideration of $x(0)$, $x(2)$ and $x(4)$

1 point: For the correct answer

$$= |x(2) - x(0)| + |x(4) - x(2)|$$

$$\text{Distance} = |-11 - 5| + |21 - (-11)|$$

$$= 16 + 32 = 48$$

4-2-2b

67. A toy train moves along a straight track set up on a table. The position $x(t)$ of the train at time t seconds is measured in centimeters from the center of the track. At time $t = 1$, the train is 6 centimeters to the left of the center, so $x(1) = -6$. For $0 \leq t \leq 4$, the velocity of the train at time t is given by $v(t) = 3t^2 - 12$, where $v(t)$ is measured in centimeters per second.
- (a) For $0 \leq t \leq 4$, find $x(t)$.
- (b) Find the total distance traveled by the train during the time interval $0 \leq t \leq 4$.
- (c) A toy bus moving on the same table has position given by $(x(t), y(t))$. Here, $x(t)$ is the function found in part (a), and $y(t) = 2 + 12 \sin(t/4)$ is the distance from the bus to the train track, in centimeters. Write, but do not evaluate, an integral expression that gives the total distance traveled by the bus during the time interval $0 \leq t \leq 4$.

Part A

The response can earn up to 3 points:

1 point is earned for the correct integral

1 point is earned for the correct antiderivative

1 point is earned for the correct answer

$$\begin{aligned}
 x(t) &= -6 + \int_1^t (3u^2 - 12) du \\
 &= -6 + [u^3 - 12u]_{u=1}^{u=t} \\
 &= t^3 - 12t + 5
 \end{aligned}$$



0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point is earned for the correct integral

1 point is earned for the correct antiderivative

1 point is earned for the correct answer

$$\begin{aligned}
 x(t) &= -6 + \int_1^t (3u^2 - 12) du \\
 &= -6 + [u^3 - 12u]_{u=1}^{u=t} \\
 &= t^3 - 12t + 5
 \end{aligned}$$

Part B

The response can earn up to 3 points:

1 point is earned for identification of $t = 2$

4-2-2b

$$v(t) = 3(t^2 - 4) = 0 \Rightarrow t = -2, t = 2$$

1 point is earned for consideration of $x(0)$, $x(2)$ and $x(4)$

1 point is earned for the correct answer

$$\begin{aligned}\text{Distance} &= |x(2) - x(0)| + |x(4) - x(2)| \\ &= |-11 - 5| + |21 - (-11)| \\ &= 16 + 32 = 48\end{aligned}$$



0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point is earned for identification of $t = 2$

$$v(t) = 3(t^2 - 4) = 0 \Rightarrow t = -2, t = 2$$

1 point is earned for consideration of $x(0)$, $x(2)$ and $x(4)$

1 point is earned for the correct answer

$$\begin{aligned}\text{Distance} &= |x(2) - x(0)| + |x(4) - x(2)| \\ &= |-11 - 5| + |21 - (-11)| \\ &= 16 + 32 = 48\end{aligned}$$

Part C

The response can earn up to 3 points:

1 point is earned for $y'(t)$

$$y'(t) = 3 \cos\left(\frac{t}{4}\right)$$

2 points are earned for the integral

$$\text{Distance} = \int_0^4 \sqrt{(3t^2 - 12)^2 + \left(3 \cos\left(\frac{t}{4}\right)\right)^2} dt$$



0	1	2	3
---	---	---	---

4-2-2b

The response earns all of the following points:

1 point is earned for $y'(t)$

$$y'(t) = 3 \cos\left(\frac{t}{4}\right)$$

2 points are earned for the integral

$$\text{Distance} = \int_0^4 \sqrt{(3t^2 - 12)^2 + \left(3 \cos\left(\frac{t}{4}\right)\right)^2} dt$$

68.

t (min)	0	5	10	15	20	25	30	35	40
$v(t)$ (mpm)	7.0	9.2	9.5	7.0	4.5	2.4	2.4	4.3	7.3

A test plane flies in a straight line with positive velocity $v(t)$, in miles per minute at time t minutes, where v is a differentiable function of t . Selected values of $v(t)$ for $0 \leq t \leq 40$ are shown in the table above.

 The function f , defined by $f(t) = 6 + \cos\left(\frac{t}{10}\right) + 3 \sin\left(\frac{7t}{40}\right)$, is used to model the velocity of the plane, in miles per minute, for $0 \leq t \leq 40$. According to this model, what is the acceleration of the plane at $t=23$? Indicates units of measure.

Part C

1 point is earned for the answer with units

$$f'(23) = -0.407 \text{ or } -0.408 \text{ miles per minute}^2$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for the answer with units

$$f'(23) = -0.407 \text{ or } -0.408 \text{ miles per minute}^2$$

4-2-2b

69.



t (min)	0	5	10	15	20	25	30	35	40
$v(t)$ (mpm)	7.0	9.2	9.5	7.0	4.5	2.4	2.4	4.3	7.3

A test plane flies in a straight line with positive velocity $v(t)$, in miles per minute at time t minutes, where v is a differentiable function of t . Selected values of $v(t)$ for $0 \leq t \leq 40$ are shown in the table above.

(a) Use a midpoint Riemann sum with four subintervals of equal length and values from the table to approximate $\int_0^{40} v(t) dt$. Show the computations that lead to your answer. Using correct units, explain the meaning of $\int_0^{40} v(t) dt$ in terms of the plane's flight.

(b) Based on the values in the table, what is the smallest number of instances at which the acceleration of the plane could equal zero on the open interval $0 < t < 40$? Justify your answer.

(c) The function f , defined by $f(t) = 6 + \cos\left(\frac{t}{10}\right) + 3 \sin\left(\frac{7t}{40}\right)$, is used to model the velocity of the plane, in miles per minute, for $0 \leq t \leq 40$. According to this model, what is the acceleration of the plane at $t = 23$? Indicate units of measure.

(d) According to the model f , given in part (c), what is the average velocity of the plane, in miles per minute, over the time interval $0 \leq t \leq 40$?

Part A

1 point is earned for $v(5) + v(15) + v(25) + v(35)$

1 point is earned for the answer

1 point is earned for the meaning with units

Midpoint Riemann sum is $10 \cdot [v(5) + v(15) + v(25) + v(35)] = 10 \cdot [9.2 + 7.0 + 2.4 + 4.3] = 229$

The integral gives the total distance in miles that the plane flies during the 40 minutes.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $v(5) + v(15) + v(25) + v(35)$

1 point is earned for the answer

1 point is earned for the meaning with units

Midpoint Riemann sum is

4-2-2b

$$10 \cdot [v(5) + v(15) + v(25) + v(35)] = 10 \cdot [9.2 + 7.0 + 2.4 + 4.3] = 229$$

The integral gives the total distance in miles that the plane flies during the 40 minutes.

Part B

1 point is earned for the two instances

1 point is earned for the justification

By the Mean Value Theorem, $v'(t) = 0$ somewhere in the interval $(0,15)$ and somewhere in the interval $(25, 30)$. Therefore the acceleration will equal 0 for at least two values of t .



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the two instances

1 point is earned for the justification

By the Mean Value Theorem, $v'(t) = 0$ somewhere in the interval $(0,15)$ and somewhere in the interval $(25, 30)$. Therefore the acceleration will equal 0 for at least two values of t .

Part C

1 point is earned for the answer with units

$$f'(23) = -0.407 \text{ or } -0.408 \text{ miles per minute}^2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer with units

$$f'(23) = -0.407 \text{ or } -0.408 \text{ miles per minute}^2$$

Part D

4-2-2b

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Average velocity} = \frac{1}{40} \int_0^{40} f(t) dt$$

= 5.916 miles per minute



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Average velocity} = \frac{1}{40} \int_0^{40} f(t) dt$$

= 5.916 miles per minute

4-2-2b

70. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Stephen swims back and forth along a straight path in a 50-meter-long pool for 90 seconds. Stephen's velocity is modeled by $v(t) = 2.38e^{-0.02t} \sin\left(\frac{\pi}{56}t\right)$, where t is measured in seconds and $v(t)$ is measured in meters per second.

- (a) Find all times t in the interval $0 < t < 90$ at which Stephen changes direction. Give a reason for your answer.
- (b) Find Stephen's acceleration at time $t = 60$ seconds. Show the setup for your calculations, and indicate units of measure. Is Stephen speeding up or slowing down at time $t = 60$ seconds? Give a reason for your answer.
- (c) Find the distance between Stephen's position at time $t = 20$ seconds and his position at time $t = 80$ seconds. Show the setup for your calculations.
- (d) Find the total distance Stephen swims over the time interval $0 \leq t \leq 90$ seconds. Show the setup for your calculations.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that considers $v(t) = 0$ earns the first point.

A response of "Stephen changes direction when his velocity changes sign" earns the first point for considering the sign of $v(t)$ but must include the answer of $t = 56$ in order to earn the second point.

A response of $t = 56$ with no supporting work does not earn either point.

Any presented values of outside the interval $0 < t < 90$ will not affect scoring.

4-2-2b



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers sign of $v(t)$
- ☐ Answer with reason

Solution:

For $0 < t < 90$, $v(t) = 0 \Rightarrow t = 56$.

Stephen changes direction when his velocity changes sign. This occurs at $t = 56$ seconds.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The minimum work needed to earn the first point is $v'(60) = -0.036$.

- $a(60) = -0.0360162$ is not sufficient to earn the first point. The connection $v'(t) = a(t)$ or $v'(60) = a(60)$ must be explicitly shown.

A response must declare a value for $a(60)$ to be eligible for the second point.

In order to earn the third point the presented conclusion must be consistent with a negative velocity at time $t = 60$ and the presented value of $a(60)$.

A response does not need to find the value of $v(60)$; an implied sign is sufficient.

- The statement “Stephen is speeding up because $a(60)$ and $v(60)$ have the same sign” (or equivalent) earns the third point, provided a negative value is presented for $a(60)$.

A response that reports an incorrect sign or value of $v(60)$ does not earn the third point. Any presented value of $v(60)$ must be correct for the number of digits presented, from one up to three decimal places in order to earn the third point.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode).

- In degree mode, there are two possible values for $v'(60)$. A response that declares $v'(60) = -0.000141$ does not earn the first point but would earn the third point in the presence of $v(60) = 0.042089$ or $v(60) > 0$ and the conclusion that Stephen is slowing down.

4-2-2b

- Similarly, a response that declares $v'(60) = 0.039304$ does not earn the first point but would earn the third point in the presence of $v(60) = 0.042089$ or $v(60) > 0$ and the conclusion that Stephen is speeding up.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ $a(60)$ with setup
- ☐ Acceleration units
- ☐ Speeding up with reason

Solution:

$$v'(60) = a(60) = -0.0360162$$

Stephen's acceleration at time $t = 60$ seconds is -0.036 meter per second per second.

$$v(60) = -0.1595124 < 0$$

Stephen is speeding up at time $t = 60$ seconds because Stephen's velocity and acceleration are both negative at that time.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned only for $\int_{20}^{80} v(t)dt$ (or the equivalent) with or without the differential.

The second point is earned only for an answer of 23.384 (or 23.383) regardless of whether the first point was earned.

Degree mode: In degree mode, $\int_{20}^{80} v(t)dt = 2.407982$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Integral
- ☐ Answer

4-2-2b

Solution:

$$\int_{20}^{80} v(t) dt = 23.383997$$

The distance between Stephen's positions at $t = 20$ seconds and $t = 80$ seconds is 23.384 (or 23.383) meters.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned only for $\int_0^{90} |v(t)| dt$ or $\int_0^{56} v(t) dt - \int_{56}^{90} v(t) dt$ (or the equivalent), with or without the differential(s).

The second point is earned only for an answer of 62.164 regardless of whether the first point was earned.

Degree mode: In degree mode, $\int_0^{90} |v(t)| dt = 3.127892$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Integral

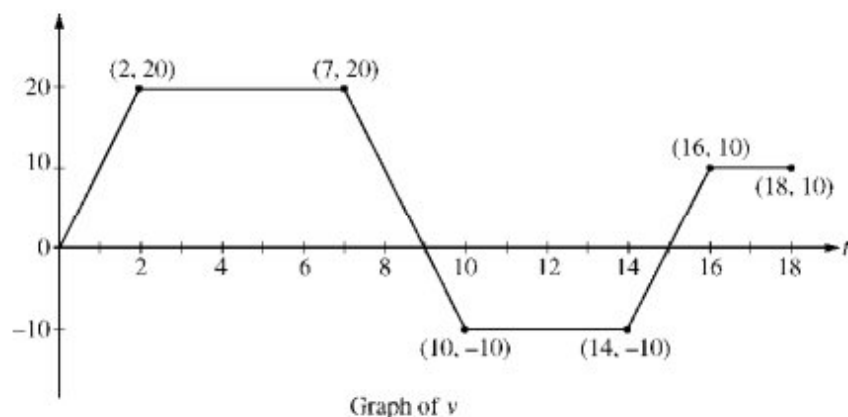
☐ Answer

Solution:

$$\int_0^{90} |v(t)| dt = 62.164216$$

The total distance Stephen swims over the time interval $0 \leq t \leq 90$ seconds is 62.164 meters.

4-2-2b



A squirrel starts at building A at time $t=0$ and travels along a straight, horizontal wire connected to building B . For $0 \leq t \leq 18$, the squirrel's velocity is modeled by the piecewise-linear function defined by the graph above.

71. Write expressions for the squirrel's acceleration $a(t)$, velocity $v(t)$, and distance $x(t)$ from building A that are valid for the time interval $7 < t < 10$.

Part D

1 point is earned for $a(t)$

1 point is earned for $v(t)$

2 points are earned for $x(t)$

$$\text{For } 7 < t < 10, a(t) = \frac{20 - (-10)}{7 - 10} = -10$$

$$v(t) = 20 - 10(t - 7) = -10t + 90$$

$$x(7) = \frac{7+5}{2} \cdot 20 = 120$$

$$x(t) = x(7) + \int_7^t (-10u + 90) du$$

$$= 120 + (-5u^2 + 90u) \Big|_{u=7}^{u=t}$$

$$= -5t^2 + 90t - 265$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $a(t)$

1 point is earned for $v(t)$

4-2-2b

2 points are earned for $x(t)$

$$\text{For } 7 < t < 10, a(t) = \frac{20 - (-10)}{7 - 10} = -10$$

$$v(t) = 20 - 10(t - 7) = -10t + 90$$

$$x(7) = \frac{7+5}{2} \cdot 20 = 120$$

$$x(t) = x(7) + \int_7^t (-10u + 90) du$$

$$= 120 + (-5u^2 + 90u) \Big|_{u=7}^{u=t}$$

$$= -5t^2 + 90t - 265$$

- 72.** At what times in the interval $0 < t < 18$, if any, does the squirrel change direction? Give a reason for your answer.

Part A

1 point is earned for the t -values

1 point is earned for the explanation

The squirrel changes direction whenever its velocity changes sign. This occurs at $t=9$ and $t=15$.



0	1	2
---	---	---

The student response earns all of the following points:

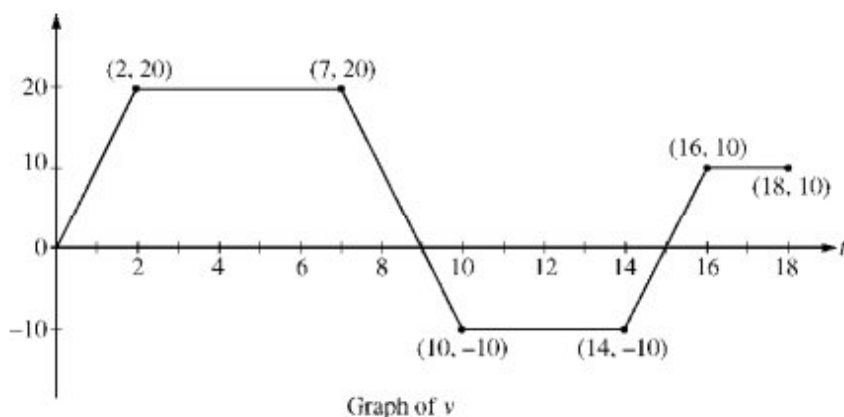
1 point is earned for the t -values

1 point is earned for the explanation

The squirrel changes direction whenever its velocity changes sign. This occurs at $t=9$ and $t=15$.

4-2-2b

73.



A squirrel starts at building A at time $t = 0$ and travels along a straight, horizontal wire connected to building B . For $0 \leq t \leq 18$, the squirrel's velocity is modeled by the piecewise-linear function defined by the graph above.

- At what times in the interval $0 < t < 18$, if any, does the squirrel change direction? Give a reason for your answer.
- At what time in the interval $0 \leq t \leq 18$ is the squirrel farthest from building A ? How far from building A is the squirrel at that time?
- Find the total distance the squirrel travels during the time interval $0 \leq t \leq 18$.
- Write expressions for the squirrel's acceleration $a(t)$, velocity $v(t)$, and distance $x(t)$ from building A that are valid for the time interval $7 < t < 10$.

Part A

1 point is earned for the t -values

1 point is earned for the explanation

The squirrel changes direction whenever its velocity changes sign. This occurs at $t = 9$ and $t = 15$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the t -values

1 point is earned for the explanation

The squirrel changes direction whenever its velocity changes sign. This occurs at $t = 9$ and $t = 15$.

Part B

1 point is earned for identifying candidates

4-2-2b

1 point is earned for answers

Velocity is 0 at $t = 0$, $t = 9$, and $t = 15$.

t	position at time t
0	0
9	$\frac{9+5}{2} \cdot 20 = 140$
15	$140 - \frac{6+4}{2} \cdot 10 = 90$
18	$90 + \frac{3+2}{2} \cdot 10 = 115$

The squirrel is farthest from building A at time $t = 9$;

its greatest distance from the building is 140.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for identifying candidates

1 point is earned for answers

Velocity is 0 at $t = 0$, $t = 9$, and $t = 15$.

t	position at time t
0	0
9	$\frac{9+5}{2} \cdot 20 = 140$
15	$140 - \frac{6+4}{2} \cdot 10 = 90$
18	$90 + \frac{3+2}{2} \cdot 10 = 115$

The squirrel is farthest from building A at time $t = 9$;

its greatest distance from the building is 140.

Part C

1 point is earned for the answer

4-2-2b

The total distance traveled is $\int_0^{18} |v(t)| dt = 140 + 50 + 25 = 215$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

The total distance traveled is $\int_0^{18} |v(t)| dt = 140 + 50 + 25 = 215$.

Part D

1 point is earned for $a(t)$

1 point is earned for $v(t)$

2 points are earned for $x(t)$

$$\text{For } 7 < t < 10, a(t) = \frac{20 - (-10)}{7 - 10} = -10$$

$$v(t) = 20 - 10(t - 7) = -10t + 90$$

$$x(7) = \frac{7+5}{2} \cdot 20 = 120$$

$$x(t) = x(7) + \int_7^t (-10u + 90) du$$

$$= 120 + (-5u^2 + 90u) \Big|_{u=7}^{u=t}$$

$$= -5t^2 + 90t - 265$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $a(t)$

1 point is earned for $v(t)$

2 points are earned for $x(t)$

4-2-2b

For $7 < t < 10$, $a(t) = \frac{20 - (-10)}{7 - 10} = -10$

$$v(t) = 20 - 10(t - 7) = -10t + 90$$

$$x(7) = \frac{7+5}{2} \cdot 20 = 120$$

$$\begin{aligned} x(t) &= x(7) + \int_7^t (-10u + 90) du \\ &= 120 + (-5u^2 + 90u) \Big|_{u=7}^{u=t} \\ &= -5t^2 + 90t - 265 \end{aligned}$$

74. Let $v(t)$ be the velocity, in feet per second, of a skydiver at time t seconds, $t \geq 0$. After her parachute opens, her velocity satisfies the differential equation $\frac{dv}{dt} = -2v - 32$, with initial condition $v(0) = -50$. It is safe to land when her speed is 20 feet per second. At what time t does she reach this speed?

Part C

1 point is earned for correctly setting $v(t) = -20$

$$v(t) = -34e^{-2t} - 16 = -20$$

1 point is earned for the correct solution

must be exponential $v(t)$

$$e^{-2t} = \frac{2}{17}; t = -\frac{1}{2} \ln\left(\frac{2}{17}\right) = 1.070$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for correctly setting $v(t) = -20$

$$v(t) = -34e^{-2t} - 16 = -20$$

1 point is earned for the correct solution

must be exponential $v(t)$

$$e^{-2t} = \frac{2}{17}; t = -\frac{1}{2} \ln\left(\frac{2}{17}\right) = 1.070$$

4-2-2b

75. Let $v(t)$ be the velocity, in feet per second, of a skydiver at time t seconds, $t \geq 0$. After her parachute opens, her velocity satisfies the differential equation $\frac{dv}{dt} = -2v - 32$, with initial condition $v(0) = -50$.
- (a) Use separation of variables to find an expression for v in terms of t , where t is measured in seconds.
- (b) Terminal velocity is defined as $\lim_{t \rightarrow \infty} v(t)$. Find the terminal velocity of the skydiver to the nearest foot per second.
- (c) It is safe to land when her speed is 20 feet per second. At what time t does she reach this speed?

Part A

Solution: Max 6/6

1 point is earned for correctly separating variables

$$\frac{dv}{dt} = -2v - 32 = -2(v + 16)$$

1 point is earned for the correct antiderivative of dv side

$$0/1 \text{ if not } \int \frac{dv}{av + b}, a \neq 0$$

$$\frac{dv}{v + 16} = -2dt$$

1 point is earned for the correct antiderivative of dt side

$$\ln |v + 16| = -2t + A$$

1 point is earned for the correct constant of integration

$$|v + 16| = e^{-2t+A} = e^A e^{-2t}$$

1 point is earned for correctly using initial condition $v(0) = -50$

$$v + 16 = Ce^{-2t}$$

$$-50 + 16 = Ce^0; C = -34$$

1 point is earned for correctly solving for $v(t)$

4-2-2b

0/1 if not solving $\frac{dv}{av+b} = k dt$ where a, b, k nonzero

0/1 if no constant of integration

0/6 if variables not separated

$$v = -34e^{-2t} - 16$$

OR

Integrating Factor Solution: Max 5/6

1 point is earned for correct use of integrating factor

$$v'(t) = -2v(t) - 32$$

$$e^{2t}(v'(t) + 2v(t)) = -32e^{2t}$$

$$\frac{d}{dt}(v(t)e^{2t}) = -32e^{2t}$$

$$v(t)e^{2t} = \int -32e^{2t} dt = -16e^{2t} + C$$

$$v(t) = -16 + Ce^{-2t}$$

$$-50 = -16 + C; \quad C = -34$$

4 points are earned for the correct solution

$$v(t) = -16 - 34e^{-2t}$$



0	1	2	3	4	5	6
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The student response earns all of the following points:

Solution: Max 6/6

1 point is earned for correctly separating variables

$$\frac{dv}{dt} = -2v - 32 = -2(v + 16)$$

1 point is earned for the correct antiderivative of dv side

4-2-2b

0/1 if not $\int \frac{dv}{av+b}, a \neq 0$

$$\frac{dv}{v+16} = -2dt$$

1 point is earned for the correct antiderivative of dt side

$$\ln|v+16| = -2t + A$$

1 point is earned for the correct constant of integration

$$|v+16| = e^{-2t+A} = e^A e^{-2t}$$

1 point is earned for correctly using initial condition $v(0) = -50$

$$v+16 = Ce^{-2t}$$

$$-50+16 = Ce^0; C = -34$$

1 point is earned for correctly solving for $v(t)$

0/1 if not solving $\frac{dv}{av+b} = k dt$ where a, b, k nonzero

0/1 if no constant of integration

0/6 if variables not separated

$$v = -34e^{-2t} - 16$$

OR

Integrating Factor Solution: Max 5/6

1 point is earned for correct use of integrating factor

4-2-2b

$$v'(t) = -2v(t) - 32$$

$$e^{2t}(v'(t) + 2v(t)) = -32e^{2t}$$

$$\frac{d}{dt}(v(t)e^{2t}) = -32e^{2t}$$

$$v(t)e^{2t} = \int -32e^{2t} dt = -16e^{2t} + C$$

$$v(t) = -16 + Ce^{-2t}$$

$$-50 = -16 + C; \quad C = -34$$

4 points are earned for the correct solution

$$v(t) = -16 - 34e^{-2t}$$

Part B

1 point is earned for the correct limit value must be exponential $v(t)$ with finite limit

$$\lim_{t \rightarrow \infty} v(t) = \lim_{t \rightarrow \infty} (-34e^{-2t} - 16) = -16$$



0

1

The student response earns all of the following points:

1 point is earned for the correct limit value must be exponential $v(t)$ with finite limit

$$\lim_{t \rightarrow \infty} v(t) = \lim_{t \rightarrow \infty} (-34e^{-2t} - 16) = -16$$

Part C

1 point is earned for correctly setting $v(t) = -20$

$$v(t) = -34e^{-2t} - 16 = -20$$

1 point is earned for the correct solution must be exponential $v(t)$

4-2-2b

$$e^{-2t} = \frac{2}{17}; \quad t = -\frac{1}{2} \ln \left(\frac{2}{17} \right) = 1.070$$



0	1	2
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The student response earns all of the following points:

1 point is earned for correctly setting $v(t) = -20$

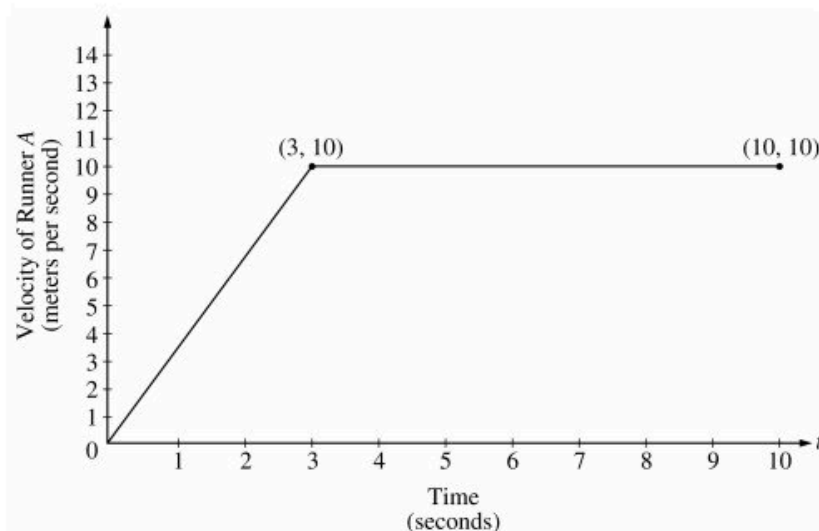
$$v(t) = -34e^{-2t} - 16 = -20$$

1 point is earned for the correct solution must be exponential $v(t)$

$$e^{-2t} = \frac{2}{17}; \quad t = -\frac{1}{2} \ln \left(\frac{2}{17} \right) = 1.070$$

4-2-2b

76.



Two runners, A and B , run on a straight racetrack for $0 \leq t \leq 10$ seconds. The graph above, which consists of two line segments, shows the velocity, in meters per second, of Runner A . The velocity, in meters per second, of Runner B is given by the function v defined by $v(t) = \frac{24t}{2t+3}$.

Find the acceleration of Runner A and the acceleration of Runner B at time $t = 2$ seconds. Indicate the units of measure.

Part B

1 point is earned for the acceleration for Runner A

1 point is earned for the acceleration for Runner B

Runner A : $acceleration = \frac{10}{3} = 3.333meters/sec^2$

Runner B : $a(2) = v'(2) = \frac{72}{(2t+3)^2} \Big|_{t=2}$
 $= \frac{72}{49} = 1.469meters/sec^2$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the acceleration for Runner A

1 point is earned for the acceleration for Runner B

Runner A : $acceleration = \frac{10}{3} = 3.333meters/sec^2$

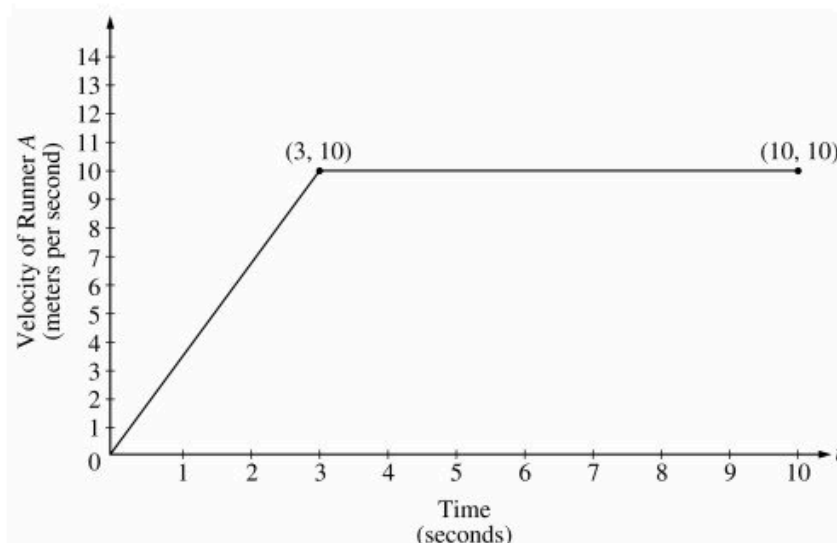
4-2-2b

Runner B:

$$a(2) = v'(2) = \frac{72}{(2t+3)^2} \Big|_{t=2}$$
$$= \frac{72}{49} = 1.469 \text{ meters/sec}^2$$

4-2-2b

77.



Two runners, A and B , run on a straight racetrack for $0 \leq t \leq 10$ seconds. The graph above, which consists of two line segments, shows the velocity, in meters per second, of Runner A . The velocity, in meters per second, of Runner B is given by the function v defined by $v(t) = \frac{24t}{2t+3}$.

- Find the velocity of Runner A and the velocity of Runner B at time $t = 2$ seconds. Indicate units of measure.
- Find the acceleration of Runner A and the acceleration of Runner B at time $t = 2$ seconds. Indicate units of measure.
- Find the total distance run by Runner A and the total distance run by Runner B over the time interval $0 \leq t \leq 10$ seconds. Indicate units of measure.

Part A

1 point is earned for the velocity for Runner A

1 point is earned for the velocity for Runner B

$$\begin{aligned} \text{Runner } A: \text{ velocity} &= \frac{10}{3} \cdot 2 = \frac{20}{3} \\ &= 6.666 \text{ or } 6.667 \text{ meters/sec} \end{aligned}$$

$$\text{Runner } B: v(2) = \frac{48}{7} = 6.857 \text{ meters/sec}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the velocity for Runner A

4-2-2b

1 point is earned for the velocity for Runner *B*

$$\begin{aligned}\text{Runner } A: \text{ velocity} &= \frac{10}{3} \cdot 2 = \frac{20}{3} \\ &= 6.666 \text{ or } 6.667 \text{ meters/sec}\end{aligned}$$

$$\text{Runner } B: v(2) = \frac{48}{7} = 6.857 \text{ meters/sec}$$

Part B

1 point is earned for the acceleration for Runner *A*

1 point is earned for the acceleration for Runner *B*

$$\text{Runner } A: \text{ acceleration} = \frac{10}{3} = 3.333 \text{ meters/sec}^2$$

$$\begin{aligned}\text{Runner } B: a(2) = v'(2) &= \frac{72}{(2t+3)^2} \Big|_{t=2} \\ &= \frac{72}{49} = 1.469 \text{ meters/sec}^2\end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the acceleration for Runner *A*

1 point is earned for the acceleration for Runner *B*

$$\text{Runner } A: \text{ acceleration} = \frac{10}{3} = 3.333 \text{ meters/sec}^2$$

$$\begin{aligned}\text{Runner } B: a(2) = v'(2) &= \frac{72}{(2t+3)^2} \Big|_{t=2} \\ &= \frac{72}{49} = 1.469 \text{ meters/sec}^2\end{aligned}$$

Part C

2 points are earned for distance for Runner *A*

1 : method

1 : answer

4-2-2b

2 points are earned for distance for Runner *B*

1 : integral

1 : answer

$$\text{Runner } A: \text{ distance} = \frac{1}{2}(3)(10) + 7(10) = 85 \text{ meters}$$

$$\text{Runner } B: \text{ distance} = \int_0^{10} \frac{24t}{2t+3} dt = 83.336 \text{ meters}$$

1 point is earned for the units

(units) meters/sec in part (a), meters/sec² in part (b), and meters in part (c), or equivalent.



0	1	2	3	4	5
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The student response earns all of the following points:

2 points are earned for distance for Runner *A*

1 : method

1 : answer

2 points are earned for distance for Runner *B*

1 : integral

1 : answer

$$\text{Runner } A: \text{ distance} = \frac{1}{2}(3)(10) + 7(10) = 85 \text{ meters}$$

$$\text{Runner } B: \text{ distance} = \int_0^{10} \frac{24t}{2t+3} dt = 83.336 \text{ meters}$$

1 point is earned for the units

(units) meters/sec in part (a), meters/sec² in part (b), and meters in part (c), or equivalent.

4-2-2b

78.

t (seconds)	0	10	20	30	40	50	60	70	80
$v(t)$ (feet per second)	5	14	22	29	35	40	44	47	49

Rocket A has positive velocity $v(t)$ after being launched upward from an initial height of 0 feet at time $t = 0$ seconds. The velocity of the rocket is recorded for selected values of t over the interval $0 \leq t \leq 80$ seconds, as show in the table above.

Find the average acceleration of rocket A over the time interval $0 \leq t \leq 80$ seconds. Indicate units of measure.

Part A

1 point is earned for the answer

Average acceleration of rocket A is

$$\frac{v(80)-v(0)}{80-0} = \frac{49-5}{80} = \frac{11}{20} ft/sec^2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

Average acceleration of rocket A is

$$\frac{v(80)-v(0)}{80-0} = \frac{49-5}{80} = \frac{11}{20} ft/sec^2$$

4-2-2b

79.

t (seconds)	0	10	20	30	40	50	60	70	80
$v(t)$ (feet per second)	5	14	22	29	35	40	44	47	49

Rocket A has positive velocity $v(t)$ after being launched upward from an initial height of 0 feet at time $t = 0$ seconds. The velocity of the rocket is recorded for selected values of t over the interval $0 \leq t \leq 80$ seconds, as shown in the table above.

(a) Find the average acceleration of rocket A over the time interval $0 \leq t \leq 80$ seconds. Indicate units of measure.

(b) Using correct units, explain the meaning of $\int_{10}^{70} v(t) dt$ in terms of the rocket's flight. Use a midpoint Riemann sum with 3 subintervals of equal length to approximate $\int_{10}^{70} v(t) dt$.

(c) Rocket B is launched upward with an acceleration of $a(t) = \frac{3}{\sqrt{t+1}}$ feet per second per second. At time $t = 0$ seconds, the initial height of the rocket is 0 feet, and the initial velocity is 2 feet per second. Which of the two rockets is traveling faster at time $t = 80$ seconds? Explain your answer.

Part A

1 point is earned for the answer

Average acceleration of rocket A is

$$\frac{v(80) - v(0)}{80 - 0} = \frac{49 - 5}{80} = \frac{11}{20} \text{ ft/sec}^2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

Average acceleration of rocket A is

$$\frac{v(80) - v(0)}{80 - 0} = \frac{49 - 5}{80} = \frac{11}{20} \text{ ft/sec}^2$$

Part B

4-2-2b

1 point is earned for the explanation

1 point is earned if uses $v(20)$, $v(40)$, $v(60)$

1 point is earned for the value

Since the velocity is positive, $\int_{10}^{70} v(t) dt$ represents the distance, in feet, traveled by rocket A from $t = 10$ seconds to $t = 70$ seconds.

A midpoint Riemann sum is

$$20[v(20) + v(40) + v(60)]$$

$$= 20[22 + 35 + 44] = 2020 \text{ ft}$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned if uses $v(20)$, $v(40)$, $v(60)$

1 point is earned for the value

Since the velocity is positive, $\int_{10}^{70} v(t) dt$ represents the distance, in feet, traveled by rocket A from $t = 10$ seconds to $t = 70$ seconds.

A midpoint Riemann sum is

$$20[v(20) + v(40) + v(60)]$$

$$= 20[22 + 35 + 44] = 2020 \text{ ft}$$

Part C

1 point is earned for $6\sqrt{t+1}$

1 point is earned for constant of integration

1 point is earned for uses initial condition

1 point is earned for $v_B(80)$, compares to $v(80)$, and draws a conclusion

Let $v_B(t)$ be the velocity of rocket B at time t .

4-2-2b

$$v_B(t) = \int \frac{3}{\sqrt{t+1}} dt = 6\sqrt{t+1} + C$$

$$2 = v_B(0) = 6 + C$$

$$v_B(t) = 6\sqrt{t+1} - 4$$

$$v_B(80) = 50 > 49 = v_B(80)$$

Rocket B is traveling faster at time $t = 80$ seconds.



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for $6\sqrt{t+1}$

1 point is earned for constant of integration

1 point is earned for uses initial condition

1 point is earned for $v_B(80)$, compares to $v(80)$, and draws a conclusion

Let be the velocity of rocket B at time t .

$$v_B(t) = \int \frac{3}{\sqrt{t+1}} dt = 6\sqrt{t+1} + C$$

$$2 = v_B(0) = 6 + C$$

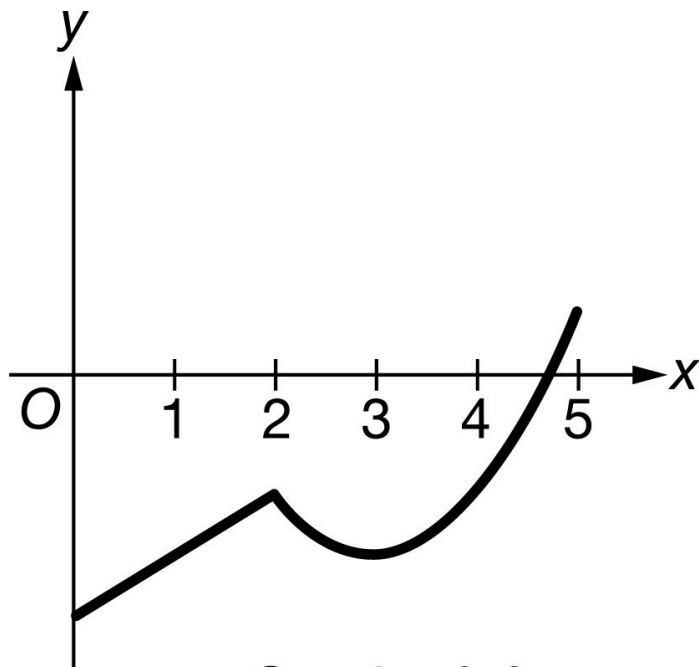
$$v_B(t) = 6\sqrt{t+1} - 4$$

$$v_B(80) = 50 > 49 = v_B(80)$$

Rocket B is traveling faster at time $t = 80$ seconds.

4-2-2b

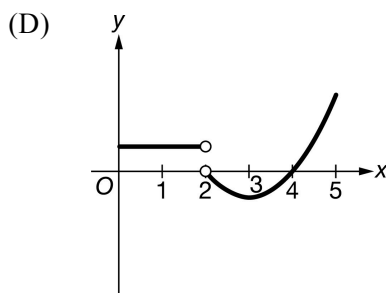
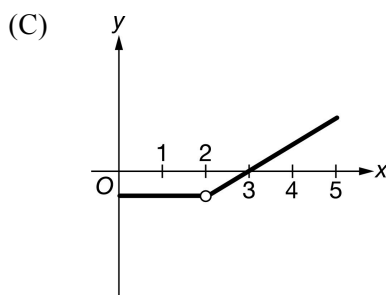
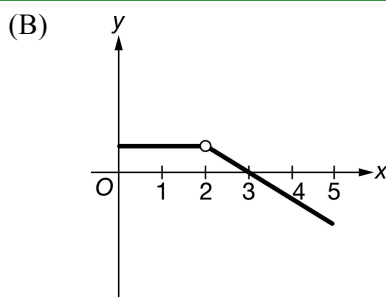
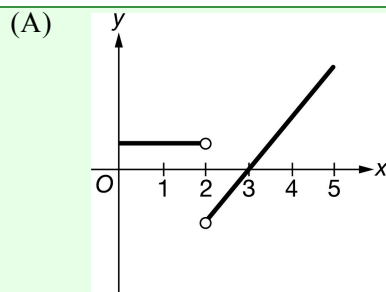
80.

Graph of f

The graph of $y = f(x)$ on the interval $0 < x < 5$ is shown above.

Which of the following could be the graph of $y = f'(x)$?

4-2-2b



Answer A

Correct. The graph of f indicates that f is increasing from $x = 0$ to $x = 2$, then decreasing from $x = 2$ to $x = 3$, and then increasing from $x = 3$ to $x = 5$. Therefore, the graph of f' should be positive from $x = 0$ to $x = 2$, negative from $x = 2$ to $x = 3$, and positive from $x = 3$ to $x = 5$. This graph is the only one that has this behavior, so it could be the graph of f' . Some other features of the graph of f support this conclusion. Since f is not differentiable at $x = 2$, the graph of f' should not be defined at $x = 2$. Since f has a local minimum at $x = 3$ and is differentiable there, $f'(3)$ should equal 0. This graph is consistent with those observations.

4-2-2b

81.  The height h , in meters, of an object at time t is given by $h(t) = 24t + 24t^{3/2} - 16t^2$. What is the height of the object at the instant when it reaches its maximum upward velocity?
- (A) 2.545 meters
- (B) 10.263 meters ✓
- (C) 34.125 meters
- (D) 54.889 meters
- (E) 89.005 meters
82.  The velocity of a particle moving along the x -axis is given by $v(t) = \frac{1-t^3}{1+t}$ for time $t \geq 0$. Which of the following statements is true about the particle at time $t = 1.5$?
- (A) The acceleration is negative, and the speed is decreasing.
- (B) The acceleration is negative, and the speed is increasing. ✓
- (C) The acceleration is positive, and the speed is decreasing.
- (D) The acceleration is positive, and the speed is increasing.

Answer B

Correct. At time $t = 1.5$, $v(1.5) = -0.95$ and $a(1.5) = v'(1.5) = -2.32$. Therefore, the acceleration is negative, and the speed is increasing because the velocity and acceleration have the same sign. An alternative solution is to graph the velocity function on the interval $[0, 3]$ and observe that at $t = 1.5$ (in the middle of the window), the graph is below the t -axis and decreasing. This means that the acceleration is negative (since the velocity is decreasing) and that the graph of the speed, the absolute value of the velocity, would be above the t -axis and increasing.

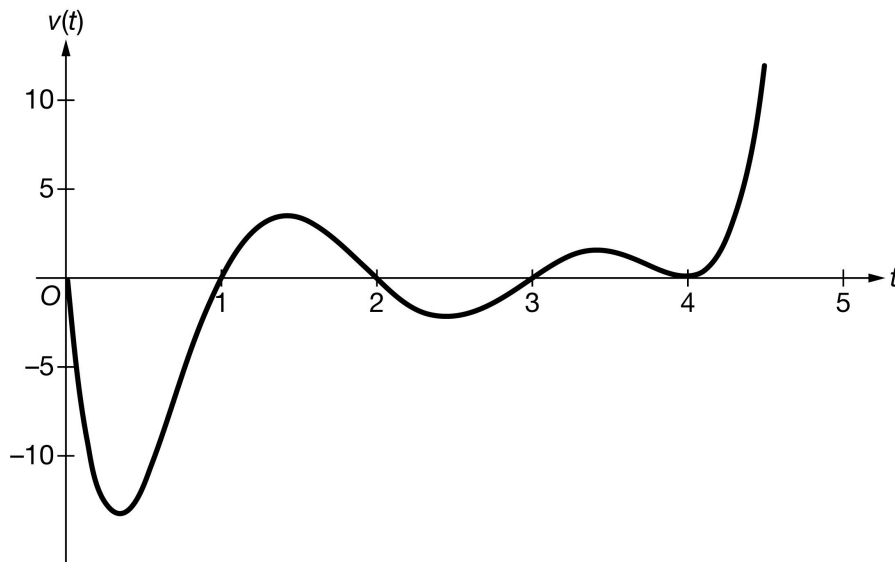
83. For time $t > 0$, the position of a particle moving along the x -axis is given by $x(t) = t^4 - 4t^3$. What is the acceleration of the particle when the velocity of the particle is 0?
- (A) -16
- (B) 0
- (C) 36 ✓
- (D) 96

Answer C

Correct. The velocity of the particle is $v(t) = x'(t) = 4t^3 - 12t^2 = 4t^2(t - 3)$. The velocity is 0 when $t = 3$.
 $a(t) = v'(t) = 12t^2 - 24t \Rightarrow a(3) = 12(9) - 24(3) = 36$

4-2-2b

84.



A particle moves along the y -axis. The graph of the particle's velocity $v(t)$ at time t is shown above for $0 < t < 4.5$. How many times does the particle change direction over the time interval $0 < t < 4.5$?

- (A) Three
(B) Four
(C) Five
(D) Eight

**Answer A**

Correct. The particle's velocity $v(t)$ changes from negative to positive at $t = 1$ and $t = 3$ and from positive to negative at $t = 2$. Therefore, the particle changes direction from downwards to upwards at $t = 1$ and $t = 3$ and from upwards to downwards at $t = 2$. The particle does not change direction at $t = 4$ because there is no sign change in the velocity.

4-2-2b

85. A particle moves along the x -axis so that at time $t \geq 0$ its position is given by $x(t) = 2t^3 + 3t^2 - 36t + 50$. What is the total distance traveled by the particle over the time interval $0 \leq t \leq 5$?
- (A) 145
(B) 180
(C) 195
(D) 233

Answer D

Correct. The velocity of the particle is given by $x'(t) = 6t^2 + 6t - 36 = 6(t + 3)(t - 2)$. This is negative for $0 \leq t < 2$ and positive for $2 < t \leq 5$. So the particle is moving left over the time interval $0 \leq t < 2$ and moving right over the time interval $2 < t \leq 5$. Therefore, the total distance traveled by the particle over the time interval $0 \leq t \leq 5$ is given by $(x(0) - x(2)) + (x(5) - x(2)) = (50 - 6) + (195 - 6) = 233$.

86. A particle moves along the y -axis so that at time $t \geq 0$ its position is given by $y(t) = t^3 - 6t^2 + 9t$. Over the time interval $0 < t < 4$, for what values of t is the speed of the particle increasing?
- (A) $2 < t < 4$
(B) $3 < t < 4$ only
(C) $0 < t < 1$ and $3 < t < 4$
(D) $1 < t < 2$ and $3 < t < 4$

Answer D

Correct. The velocity of the particle is given by $y'(t) = 3t^2 - 12t + 9 = 3(t - 1)(t - 3)$, and the acceleration of the particle is given by $y''(t) = 6t - 12 = 6(t - 2)$. The speed of the particle is increasing when the velocity and acceleration have the same sign, which occurs for $1 < t < 2$ and $3 < t < 4$.

4-2-2b

87. An object moves along a straight line so that at any time t , $0 \leq t \leq 9$, its position is given by $x(t) = 7 + 6t - t^2$. For what value of t is the object at rest?

(A) $t = 3$

(B) $t = 6$

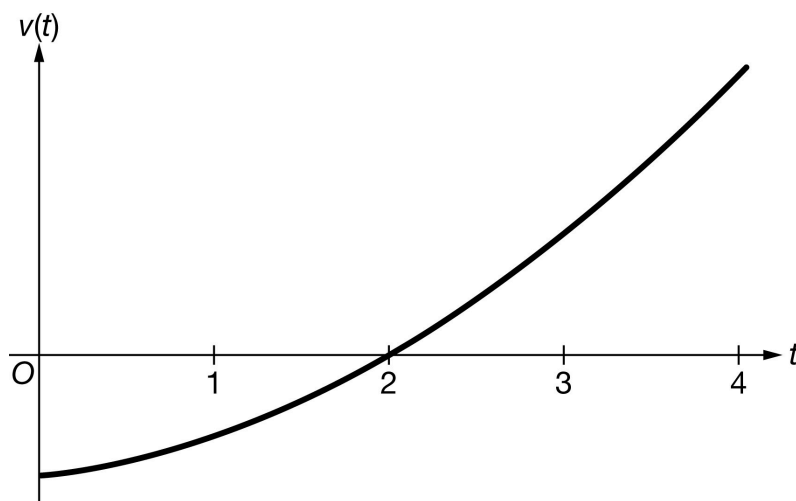
(C) $t = \frac{13}{2}$

(D) $t = 7$

**Answer A**

Correct. The object is at rest when its velocity equals 0. Because the object's velocity is given by $x'(t) = 6 - 2t$, it is at rest at $t = 3$.

88.



A particle traveling on the x -axis has position $x(t)$ at time t . The graph of the particle's velocity $v(t)$ is shown above for $0 \leq t \leq 4$. Which of the following expressions gives the total distance traveled by the particle over the time interval $0 \leq t \leq 4$?

(A) $x(0) - x(4)$

(B) $x(4) - x(0)$

(C) $(x(0) - x(2)) + (x(4) - x(2))$

(D) $(x(2) - x(0)) + (x(2) - x(4))$

**Answer C**

Correct. Since the particle's velocity is negative for $0 \leq t < 2$ and positive for $2 < t \leq 4$, the particle moves to the left for $0 \leq t < 2$ and to the right for $2 < t \leq 4$. Therefore, the particle moves $x(0) - x(2)$ units to the left and then $x(4) - x(2)$ units to the right.

4-2-2b

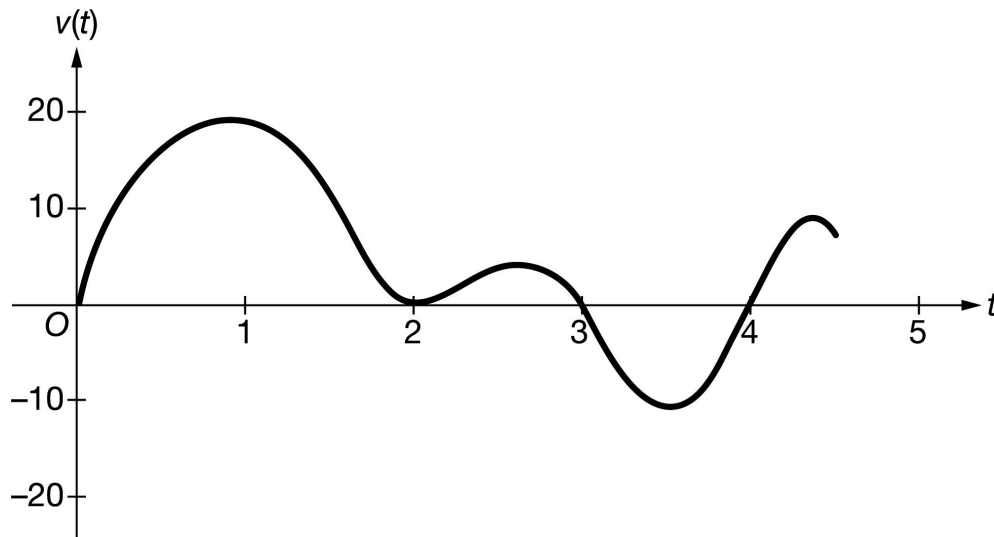
89. A particle moves along the x -axis so that at any time $t \geq 0$ its position is given by $x(t) = te^{-at}$, where a is a positive constant. At what time t is the particle's position farthest to the right?
- (A) $t = 0$
- (B) $t = \frac{1}{a}$ ✓
- (C) $t = a$
- (D) There is no such value of a .

Answer B

Correct. The velocity of the particle is given by $x'(t) = e^{-at} + te^{-at}(-a) = e^{-at}(1 - at)$. Since $x'(t)$ is positive for $0 \leq t < \frac{1}{a}$ and negative for $t > \frac{1}{a}$, the particle is moving to the right for $0 \leq t < \frac{1}{a}$ and moving to the left for $t > \frac{1}{a}$. Therefore, the particle is farthest to the right at $t = \frac{1}{a}$.

4-2-2b

90.



A particle moves along the x -axis. The graph of the particle's velocity $v(t)$ at time t is shown above for $0 < t < 4.5$. How many times does the particle change direction over the time interval $0 < t < 4.5$?

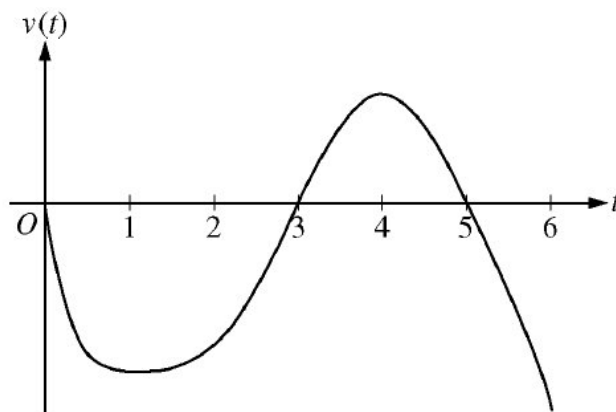
- (A) Two
- (B) Three
- (C) Five
- (D) Seven

**Answer A**

Correct. The particle's velocity $v(t)$ changes from positive to negative at $t = 3$ and from negative to positive at $t = 4$. Therefore, the particle changes direction from left to right at $t = 4$ and from right to left at $t = 3$. The particle does not change direction at $t = 2$ because there is no sign change in the velocity.

4-2-2b

91.

Graph of v

A particle moves along the x -axis so that its velocity at time t , for $0 \leq t \leq 6$, is given by a differentiable function v whose graph is shown above. The velocity is 0 at $t = 0$, $t = 3$, and $t = 5$, and the graph has horizontal tangents at $t = 1$ and $t = 4$. The areas of the regions bounded by the t -axis and the graph of v on the intervals $[0, 3]$, $[3, 5]$, and $[5, 6]$ are 8, 3, and 2, respectively. At time $t = 0$, the particle is at $x = -2$.

- (a) For $0 \leq t \leq 6$, find both the time and the position of the particle when the particle is farthest to the left. Justify your answer.
- (b) For how many values of t , where $0 \leq t \leq 6$, is the particle at $x = -8$? Explain your reasoning.
- (c) On the interval $2 < t < 3$, is the speed of the particle increasing or decreasing? Give a reason for your answer.
- (d) During what time intervals, if any, is the acceleration of the particle negative? Justify your answer.

Part A

The response can earn up to 3 points:

1 point is earned for identifying $t = 3$ as a candidate

Since $v(t) < 0$ for $0 < t < 3$ and $5 < t < 6$, and $v(t) > 0$ for $3 < t < 5$, we consider $t = 3$ and $t = 6$.

1 point is earned for consideration of $\int_0^6 v(t) dt$

$$x(3) = -2 + \int_0^3 v(t) dt = -2 - 8 = -10 \quad x(6) = -2 + \int_0^6 v(t) dt = -2 - 8 + 3 - 2 = -9$$

1 point is earned for the correct conclusion

Therefore, the particle is farthest left at time $t = 3$ when its position is $x(3) = -10$.



0	1	2	3
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The response earns all of the following points:

4-2-2b

1 point is earned for identifying $t = 3$ as a candidate

Since $v(t) < 0$ for $0 < t < 3$ and $5 < t < 6$, and $v(t) > 0$ for $3 < t < 5$, we consider $t = 3$ and $t = 6$.

1 point is earned for consideration of $\int_0^6 v(t) dt$

$$x(3) = -2 + \int_0^3 v(t) dt = -2 - 8 = -10 \quad x(6) = -2 + \int_0^6 v(t) dt = -2 - 8 + 3 - 2 = -9$$

1 point is earned for the correct conclusion

Therefore, the particle is farthest left at time $t = 3$ when its position is $x(3) = -10$.

Part B

The response can earn up to 3 points:

1 point is earned for the positions at $t = 3$, $t = 5$, and $t = 6$

1 point is earned for the description of motion

1 point is earned for the correctly conclusion

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.

By the Intermediate Value Theorem, there are three values of t for which the particle is at $x(t) = -8$.



0	1	2	3
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The response earns all of the following points:

1 point is earned for the positions at $t = 3$, $t = 5$, and $t = 6$

1 point is earned for the description of motion

1 point is earned for the correctly conclusion

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.

By the Intermediate Value Theorem, there are three values of t for which the particle is at $x(t) = -8$.

Part C

The response can earn up to 1 point:

4-2-2b

1 point is earned for the correct answer with reason

The speed is decreasing on the interval $2 < t < 3$ since on this interval $v < 0$ and v is increasing.



0	1
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The response earns all of the following point:

1 point is earned for the correct answer with reason

The speed is decreasing on the interval $2 < t < 3$ since on this interval $v < 0$ and v is increasing.

Part D

The response can earn up to 2 points:

1 point is earned for the correct answer

1 point is earned for correct justification

The acceleration is negative on the intervals $0 < t < 1$ and $4 < t < 6$ since velocity is decreasing on these intervals.



0	1	2
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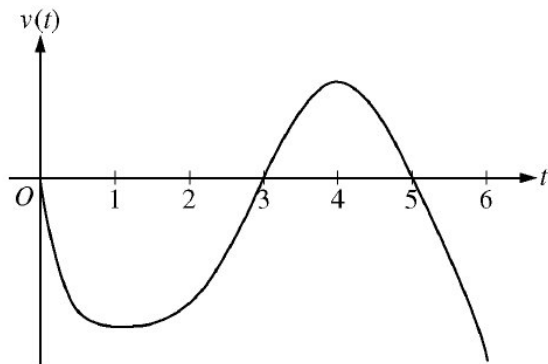
The response earns all of the following points:

1 point is earned for the correct answer

1 point is earned for correct justification

The acceleration is negative on the intervals $0 < t < 1$ and $4 < t < 6$ since velocity is decreasing on these intervals.

4-2-2b

Graph of v

A particle moves along the x -axis so that its velocity at time t , for $0 \leq t \leq 6$, is given by a differentiable function v whose graph is shown above. The velocity is 0 at $t = 0$, $t = 3$, and $t = 5$, and the graph has horizontal tangents at $t = 1$ and $t = 4$. The areas of the regions bounded by the t -axis and the graph of v on the intervals $[0, 3]$, $[3, 5]$, and $[5, 6]$ are 8, 3, and 2, respectively. At time $t = 0$, the particle is at $x = -2$.

92. On the interval $2 < t < 3$, is the speed of the particle increasing decreasing? Give a reason for your answer.

Part C

The response can earn up to 1 point:

1 point: For the correct answer with reason

The speed is decreasing on the interval $2 < t < 3$ since on this interval $v < 0$ and v is increasing.



0	1
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The response earns the following point:

1 point: For the correct answer with reason

The speed is decreasing on the interval $2 < t < 3$ since on this interval $v < 0$ and v is increasing.

4-2-2b

93. During what time intervals, if any, is the acceleration of the particle negative? Justify your answer.
-

Part D

The response can earn up to 2 points:

1 point: For the correct answer

1 point: For correct justification

The acceleration is negative on the intervals $0 < t < 1$ and $4 < t < 6$ since velocity is decreasing on these intervals.



0	1	2
---	---	---

The response earns two of the following points:

1 point: For the correct answer

1 point: For correct justification

The acceleration is negative on the intervals $0 < t < 1$ and $4 < t < 6$ since velocity is decreasing on these intervals.

94.  The position of an object attached to a spring is given by $y(t) = \frac{1}{6}\cos(5t) - \frac{1}{4}\sin(5t)$, where t is time in seconds. In the first 4 seconds, how many times is the velocity of the object equal to 0?

- (A) Zero
(B) Three
(C) Five
(D) Six
(E) Seven

